SHORT-TERM SOIL-MOISTURE FORECASTING IN PATHUM THANI: A QUALITY-FILTERED SMAP L3 BENCHMARK

by

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Abstract

This study benchmarked baseline models for short-term forecasting of SMAP enhanced Level 3 (L3) surface soil moisture at a selected cell of its 9 km grid in Pathum Thani, Thailand, using recommended-quality retrievals, 2016–2024; a cell's value represents the radiometer's footprint of about 36 km. A province-wide quality scan found no recommended-quality retrieval in the 33 urban-classed cells of 53, covering 76.2% of the province. The benchmark was therefore run at one non-urban cell at 2, 3, 5 and 8 days. Filtering removed mostly wet-season retrievals (usable on 10% of October days), under-representing the wettest states.

Ridge regression, random forest and XGBoost, on L3 history alone or with Sentinel-1 backscatter or reanalysis weather, were evaluated walk-forward against persistence, climatology, damped persistence and the mean of the last three retrievals. At 3, 5 and 8 days every learned model on L3 history beat persistence, but so did that mean alone (skill, the fractional reduction in RMSE, $+0.069$ to $+0.204$), so the meaningful test is against that mean. At 8 days ridge regression added $+0.060$ over it ($p = 0.002$), surviving adjustment for twelve comparisons, also without the few training pairs whose target falls in the test year, but not with a wider penalty range (adjusted $p = 0.051$); the random forest's $+0.081$ depended on its tuning. Ridge's $+0.042$ at 3 days ($p = 0.009$) did not survive; no gain was measurable at 2 or 5 days. A rule defined afterwards, that mean damped towards the seasonal normal, was beaten by no learned model except, unadjusted, under the benchmark's own climatology. At 3 days much of the gain came from filtering short-lived departures, a share partly built into their definition; wetting after heavy reanalysis rainfall went unforecast. A two-pass target gave the same broad pattern, and ridge's 8-day gain was positive at all 20 non-urban cells. Sentinel-1 and weather added no measurable additional information under the present data, quality filtering and evaluation protocol. Errors (best unbiased RMSE 0.031–0.035 m^3/m^3) are against the retrieval, unverified in situ. On SMAP Level 4, no model beat persistence.

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List of Terms

Cell	One square of SMAP's 9 km grid. Its value describes the area the radiometer senses, about 36 km across, not the square alone.
L3, L4	SMAP Level 3, the satellite retrieval itself; SMAP Level 4, a land-surface model with SMAP observations assimilated into it.
Recommended quality	A retrieval whose quality flag is 0 or 8. Only these are used in the L3 benchmark.
Morning-first target	The morning retrieval of a day, or the evening one where the morning one is missing. The target of the benchmark.
Two-pass target	The mean of the morning and evening retrievals of a day, or the one available. Used in one robustness run.
Horizon	The number of days between the retrieval a forecast starts from and the retrieval it predicts.
Persistence	The forecast that the value at the horizon equals the latest retrieval.
Climatology	The forecast that the value equals the average for that time of year in the training years.
Mean of three	The mean of the latest three retrievals. Also written three-retrieval mean, and mean3 in tables.
Damped persistence	The latest retrieval relaxed towards the training-period mean by a fitted factor.
DAM3	The mean of three with its departure from the seasonal normal multiplied by one fitted fraction. A diagnostic defined after the main results.
Skill	One minus the ratio of a forecast's RMSE to that of a reference forecast on the same samples. Positive when the forecast is the more accurate.
RMSE, ubRMSE	Root-mean-square error; the same after the mean error is removed (unbiased RMSE).
Walk-forward evaluation	Training on all years before a test year and testing on that year, for each test year in turn.
Diebold–Mariano test	A test of whether two forecasts differ in accuracy on the same samples.
Holm adjustment	A correction of p-values for the number of comparisons made together: here the twelve comparisons of three learned models with the mean of three at four horizons.
Significant, suggestive	Significant: one of those twelve comparisons that survives the Holm adjustment. Suggestive: a p-value below 0.05 that does not survive it.
Main run, purged run	The benchmark at cell (3,6); the same without the few training pairs whose target falls in the test year.
Seed check	Refits of the random forest and XGBoost with 21 random seeds.
Product intercomparison	A comparison of products with one another, none of which is a ground truth. It is not a validation.

Chapter 1

Introduction

1.1 Background

Soil moisture is the water held in the unsaturated soil column. Although it represents only a small fraction of the global freshwater store, it exerts control over processes far larger than its volume suggests: it partitions rainfall into infiltration and runoff, governs the exchange of water and energy between the land and the atmosphere, and modulates droughts, heatwaves, and convective precipitation (Seneviratne et al., 2010). It is formally recognised as an Essential Climate Variable, and it is a decisive input to agriculture, irrigation scheduling, drought monitoring, and flood early warning.

A property of soil moisture that is central to this study is its temporal *memory*. Because water drains and evaporates slowly relative to the pace of daily weather, a wet or dry anomaly persists for days to weeks (Orth & Seneviratne, 2013; Seneviratne et al., 2010). This memory is what makes soil moisture *predictable* on short horizons, and it also sets the reference every forecasting method must beat—the *persistence* forecast, which simply assumes that conditions tomorrow equal conditions today.

Satellite microwave remote sensing has made soil moisture observable at near-global scale. The Soil Moisture Active Passive (SMAP) mission retrieves surface soil moisture every two to three days (Entekhabi et al., 2010), and long, harmonised records such as the European Space Agency Climate Change Initiative (ESA CCI) product now underpin Earth-system research (Dorigo et al., 2017). These products, however, describe the *past and present* state of the land surface; they do not, by themselves, tell a decision-maker what the soil will be like in the coming days. Converting the observed record into a short-term *forecast* is the problem this study addresses.

SMAP soil moisture is distributed at more than one processing level, and the distinction matters throughout this report. Level 3 (L3) carries the radiometer retrieval itself, which exists only on days when the satellite passes over and the retrieval succeeds. Level 4 (L4) assimilates SMAP observations into a land-surface model (Reichle et al., 2019) and is therefore complete in space and time. The study forecast both: L3 as its target, and L4 as a characterisation of the assimilated product.

1.2 Motivation and Study Context

Pathum Thani province, immediately north of Bangkok, covers approximately 1,514 km^2 (the geoBoundaries outline used throughout, Section 3.3) of strongly mixed land use: dense communities and industrial estates along the transport corridors interleaved with the paddy fields and canal (*khlong*) network of the Rangsit agricultural area. Two recurring hazards make short-term soil-moisture information valuable here. First, in the dry season the province's rice-based agriculture is exposed to agricultural drought and irrigation stress, and knowing a few days ahead how wet the root zone will be is the kind of information an irrigation decision would draw on. Second, the province was among

the areas most severely affected by the 2011 Chao Phraya River flood, when prolonged saturation reached major industrial estates and caused losses of national significance (Komori et al., 2012), and the wetness of the soil governs how much heavy rain becomes runoff (Seneviratne et al., 2010). A short-horizon soil-moisture forecast whose skill has been measured locally would therefore be of practical value to agriculture, water management and disaster preparedness in the province. This study takes the first step towards it: a baseline for forecasting the surface layer at one SMAP grid cell. It is not an operational forecast for irrigation or flood warning, and the root zone, the wettest part of the year and the urban-flagged area remain outside it (Section 1.6).

1.3 Statement of the Problem

Despite decades of research, short-term soil-moisture forecasting remains scientifically difficult. The difficulty is not a single obstacle but a combination of properties of the variable, the data, and the modelling task. This section makes each explicit, because each motivated a specific design choice in the study.

1.3.1 The soil-moisture response is nonlinear and regime-dependent

Soil moisture responds to its drivers through thresholds and saturation effects: the same rainfall produces very different wetting on dry versus already-saturated soil, and drydown rates depend nonlinearly on soil texture, vegetation, and atmospheric demand. Linear statistical models and simple interpolation cannot represent these interactions, which is why data-driven methods that learn nonlinear predictor–response relationships have become dominant (Lamichhane et al., 2025). *Why it matters:* the learned baselines included two tree ensembles, which capture nonlinearity and feature interactions without hand-specified physics, and, on L3, ridge regression, a linear model that checks whether that capacity is needed.

1.3.2 Predictability decays with lead time

Because forecasting skill rides on soil-moisture memory, it is high at a one-day horizon and decays as the horizon lengthens and the persistence reference weakens (Orth & Seneviratne, 2013). Studies that report skill only at a single horizon obscure this decay. Two recent works report that the *advantage* of an enhanced deep model over a standard deep baseline—an LSTM in one case, a ConvLSTM in the other—*widens* with lead time (Q. Li et al., 2022; Wu et al., 2025), but neither measures skill against persistence (Table 2.2), so how much of their skill memory alone would supply is not known. *Why it matters:* the study evaluated several horizons and reported skill *relative to persistence* at each, not absolute error alone.

1.3.3 Spatial heterogeneity and observation-scale mismatch

Passive-microwave soil moisture is retrieved at coarse resolution (tens of kilometres), averaging together hydrologically distinct surfaces, whereas decisions are made at field-to-sub-district scale. Fine-resolution ancillary data—radar backscatter, vegetation and temperature indices, terrain, and soil—carry the sub-pixel information, but exploiting them requires fusing many heterogeneous sources (Das et al., 2019; Liu et al., 2020). In a peri-urban province the averaging has a further consequence: over 2016–2024, no cell that SMAP classes as urban carried a single recommended-quality retrieval. *Why it matters:* the study checked retrieval quality cell by cell across the province before choosing where to forecast, and aligned every other source to the chosen SMAP cell.

1.3.4 Sparse ground truth

Thailand has no dense operational in-situ soil-moisture network, the International Soil Moisture Network (Dorigo et al., 2011) holds no station in the country, and no SMAP core or candidate validation site lies anywhere in Mainland Southeast Asia (Ganepola et al., 2026), so neither purely statistical interpolation nor supervised training against abundant local stations is straightforward. A workable approach must lean on spatially complete satellite and reanalysis products and on physically meaningful predictors rather than on dense point measurements (Carranza et al., 2021). *Why it matters:* the study took its target from a satellite retrieval and, in place of validation, compared SMAP L4 with an ECMWF blend of operational analyses and reanalysis and with MERRA-2 (Gelaro et al., 2017) as a product intercomparison.

1.3.5 Fragmented data and the absence of a regional benchmark

The variables needed for forecasting are scattered across providers, formats, resolutions, and revisit frequencies, with no automated pipeline that assembles them for Pathum Thani. More fundamentally, *no published baseline exists for how well standard models forecast soil moisture in this province*. Without such a benchmark—including the deceptively strong persistence forecast—any future claim that an advanced model "improves" results would be unanchored. *Why it matters:* establishing a reproducible, persistence-referenced benchmark was itself a necessary contribution.

1.3.6 Opacity of data-driven models

High accuracy does not guarantee that a model has learned physically sensible relationships rather than dataset artefacts, and unexplained forecasts are poorly trusted by the agencies that would use them (Ren et al., 2023; Y. Wang & Zha, 2024). *Why it matters:* the study paired the benchmark with an explainability analysis (SHAP, applied to the L4 random forest) and with an input ladder that measures what each data source adds on L3.

1.4 Research Questions

The study was organised around five research questions. They were revised from those of the proposal once the data had been examined, so that each can be answered by the work actually carried out:

RQ1. At the selected SMAP L3 cell in Pathum Thani, how much skill do standard machine-learning models (ridge regression, random forest and XGBoost) achieve in forecasting surface soil moisture at 2, 3, 5 and 8 days, measured against persistence, and do they add information beyond a simple smoothing baseline, the mean of the last three retrievals? How do they compare with damped persistence, a further reference forecast? As a separate characterisation, how do random forest and XGBoost compare with persistence on SMAP L4 at 1, 3 and 7 days? (Ridge regression was not run on L4.)

RQ2. How do forecast skill and persistence error change with lead time, and could part of that change reflect the orbit's sampling—8-day pairs taken mostly from the same track against 2-, 3- and 5-day pairs taken from different tracks—rather than the loss of soil-moisture memory?

RQ3. What information does each input add, when inputs are added in the order proposed by the advisor: the L3 history alone, then Sentinel-1 backscatter, then weather (rainfall, temperature and shortwave radiation)? For the SMAP L4 random forest, which features carry the largest attributions according to SHAP?

RQ4. Where do the baselines fail—by season on L3, by antecedent rainfall and current wetness on L4, and by retrieval quality, namely the SMAP quality flag, the urban classification of most of the province's cells, and the loss of wet-season retrievals under the quality rule—and what does this reveal about the capabilities the subsequent thesis must add?

RQ5. How do SMAP L4, SMAP L3, the ECMWF blend and MERRA-2 differ as forecast targets—in completeness, footprint, autocorrelation and, where a benchmark was run, forecast skill—and how closely do SMAP L4, the ECMWF blend and MERRA-2 agree in a product intercomparison?

1.5 Research Objectives

Corresponding to the research questions, the objectives were as follows. Each is stated with its technical contribution and the outcome delivered.

1. **Review** the literature on AI-based soil-moisture forecasting, land-surface modelling, and Earth observation, and synthesise it as a method-evolution narrative from statistical models to foundation models, identifying the limitations that motivate the subsequent thesis. *Outcome:* a comparative review of prior work and an explicit statement of research gaps (Chapter 2).
2. **Construct** a reproducible dataset for Pathum Thani over 2016–2024. *Outcome:* a province-wide scan of SMAP L3 retrieval quality over a box of 56 SMAP 9 km cells enclosing the province, 53 of which returned at least one retrieval, and a single-cell benchmark dataset at

the chosen cell that combines recommended-quality L3 retrievals with Sentinel-1 backscatter and daily weather; the SMAP L4 dataset around the original study point was retained for the L4 benchmark. Both are regenerated by scripts. In place of the separate exploratory figure set planned in the proposal, the data were characterised through the quality and sampling profile reported in Chapter 4.

3. **Formulate and implement** short-term forecasting as supervised regression, at 2, 3, 5 and 8 days on SMAP L3 at the selected cell, with features defined over the sequence of retrievals rather than the calendar, and at 1, 3 and 7 days on SMAP L4. The baselines were persistence, seasonal climatology, random forest and XGBoost on both products, with ridge regression added as a third learner on L3 only, and the mean of the last three retrievals and damped persistence added as smoothing references on L3. After the main results, a one-parameter rule, the three-retrieval mean damped towards the seasonal normal, was added as a diagnostic benchmark; it is not a proposed forecasting model. *Outcome:* a working, scripted forecasting pipeline for both products (RQ1).

4. **Evaluate** forecast skill with expanding-window walk-forward validation and tuning inside each training fold. RMSE, unbiased RMSE (ubRMSE), bias and R were computed for every forecast and kept in the result files; the headline is a skill score against persistence and, on L3, against the three-retrieval mean, tested with Diebold–Mariano and moving-block bootstrap intervals, with a per-fold Wilcoxon test as a consistency check across folds. On L4 the tabulated skill is the mean of the per-fold skills, whereas on L3 it is computed once over the pooled test samples. The L3 benchmark was repeated at two neighbouring non-urban cells and at cell (3,6) with the mean of the morning and evening retrievals as the target, as robustness checks, and, with all retrievals, at the urban cell of the original study point, as a contrast. *Outcome:* significance-tested benchmark tables for SMAP L4 and SMAP L3, and an analysis of how skill and persistence error vary with horizon and orbit geometry (RQ1, RQ2).

5. **Measure information gain** by adding inputs to the L3 history in the order proposed by the advisor—Sentinel-1 backscatter, then weather—and by SHAP attribution for the SMAP L4 random forest. *Outcome:* an input ladder evaluated under the benchmark protocol, and a SHAP driver analysis on L4 (RQ3).

6. **Analyse failure modes and retrieval quality**: on L3 by season, through the behaviour of the forecasts on the real record, and through the quality structure of the record—the quality flag, the urban classification and the loss of wet-season retrievals—and by comparing forecast error with SMAP's $0.04\,\mathrm{m}^3/\mathrm{m}^3$ accuracy requirement; on L4 by antecedent rainfall and current wetness, conditions known at forecast time. *Outcome:* an error and quality analysis that motivates the thesis design (RQ4). The land-cover failure analysis planned in the proposal was not carried out, because ESA WorldCover was not joined.

7. **Compare soil-moisture products**—SMAP L4, SMAP L3, the ECMWF blend and MERRA-2—as forecast targets and by product intercomparison. *Outcome:* a comparison of the products as targets, and an intercomparison of the daily series of SMAP L4, the ECMWF blend and MERRA-2 by raw and anomaly correlation and by response to rain (RQ5). GLDAS, named in the proposal, could not be obtained.

1.6 Scope and Limitations

The study was bounded as follows.

Area and target. Two scopes are kept apart throughout this report. The quality and data-availability scan covers the whole of Pathum Thani province, all seven districts. The forecasting benchmark does not: it refers to one selected SMAP L3 cell in Pathum Thani, with two neighbouring cells as robustness checks, and its results are not results for the province. How forecast error varies across the province, which the advisor asked to be examined, is answered only in part. As a check made afterwards, the reference forecasts and ridge regression were scored at the 20 non-urban cells of the scan (Section 4.3.3); the urban-flagged cells, 76.2% of the province, cannot be scored on recommended retrievals, and the twelve eastern cells, within 26 km of cell (3,6), sense largely the same ground as it. The study concerns the **surface** soil-moisture layer (top $\sim$5 cm); the root zone, on which irrigation decisions depend, is outside it. The target is the SMAP enhanced Level 3 product, SPL3SMP_E version 006 (`https://nsidc.org/data/spl3smp_e/versions/6`), restricted to recommended-quality retrievals, `retrieval_qual_flag` 0 or 8, as advised by Chinthaka Ganepola on 25 September 2026 and documented in the product's user guide (National Snow and Ice Data Center [NSIDC], 2023). It is forecast at a single cell of the 9 km grid, (3,6), centred at 14.0833° N, 100.8766° E in the east of Thanyaburi district (district boundaries from geoBoundaries, `https://www.geoboundaries.org`); 43% of the cell lies in Nong Suea, 38% in Thanyaburi and 10% in Lam Luk Ka. The 9 km is the spacing of the grid: the value posted at a cell represents the radiometer's footprint, about 36 km across (Chan et al., 2018), so it describes a wider area than the cell, and neighbouring cells share most of it (Section 3.3.5). The cell was chosen because 76.2% of the province's area lies in cells that SMAP classes as urban, and those cells carry no recommended-quality retrieval over 2016–2024; a further 1.9% lies in a cell that never returned a retrieval, which leaves 21.9% in non-urban cells. Of the 12 cells at least 90% inside the province only 2 are not urban, and cell (3,6) has the more recommended-quality days of the two (981 against 946). The original study point, in Mueang Pathum Thani district, lies in an urban cell whose retrievals are all not recommended; with all its retrievals, it serves as a contrast. The L3 target is the value at one cell of the 9 km grid at one instant: the morning overpass, or the evening overpass where the morning one is missing. One robustness run replaced it with the mean of the two passes; an evening-only target was not tested.

Period and record. The period is 2016-01-01 to 2024-12-31, 3,288 days, of which 981 (29.8%) carry a recommended-quality retrieval at the chosen cell. The quality rule removes the retrievals stored at the cell's maximum value, which SMAP marks as unsuccessful, and 77% of the retrievals it rejects fall in May–October and follow more rain than the accepted ones, so the filtered record under-represents the wettest conditions.

Horizons. L3 is forecast at **2, 3, 5 and 8 days**. These horizons were chosen from the separations with enough retrieval pairs and approved by the advisor on 6 October 2026. One and 7 days were excluded at the chosen cell because exact 1- and 7-day retrieval pairs number only 41 each over nine years, too few for a walk-forward protocol that needs at least 120 training and 20 test samples per fold; 4-day pairs do not occur. Six-day pairs are numerous enough (326), but 6 days was not made a benchmark horizon, to keep the four horizons consistent with the analysis presented on 23 September;

it was run afterwards as a check (Section 4.7.1). Over all exact-gap pairs, persistence error at 6 days ($0.0430\,\mathrm{m^3/m^3}$) is higher than at 5 or 8 days, so the 6-day result cannot be read off the evaluated horizons.

The L4 benchmark. The SMAP L4 benchmark at **1, 3 and 7 days** is kept as a characterisation. Its series is a box mean ($\pm 0.15^\circ$ around the original point, 57% inside the province) of daily-averaged 3-hourly values, so L4 differs from L3 in product level, footprint, time sampling and location. The two benchmarks also differ in horizons and in how skill is aggregated—on L4 the mean of the per-fold skills, on L3 skill over the pooled test samples—and in their folds: eight testing 2017–2024 on L4, against six (2 days, 2019–2024) or seven (2018–2024) on L3. A contrast between the L4 and L3 results mixes all of these, which this study cannot separate; the two experiments are reported separately and are not a direct product-against-product comparison.

Inputs. The L3 models draw on the retrieval history, on Sentinel-1 radiometrically terrain-corrected backscatter from Microsoft Planetary Computer (`https://planetarycomputer.microsoft.com/dataset/sentinel-1-rtc`, CC-BY-4.0), paired backward only—the latest scene on or before the issue date—and on daily rainfall, temperature and shortwave radiation from an ECMWF reanalysis (ERA5/ERA5-Land, Open-Meteo ERA5-Seamless, `https://open-meteo.com`). The earlier L4 benchmark drew its weather from the ECMWF blend (Open-Meteo best match of the IFS, ERA5 and ERA5-Land). Land cover and GLDAS were not used.

Models. Only **baseline** forecasters were used: simple references such as persistence, climatology and, on L3, smoothing of recent retrievals, and the standard learners random forest and XGBoost, with ridge regression added on L3 only. Deep-learning, Transformer, graph-based, physics-informed and hybrid models are *deliberately left to the thesis* (Section 1.8).

Evaluation. The work is retrospective, with no real-time operational deployment. Because the International Soil Moisture Network has no station in Thailand and no in-situ data were obtained, forecasts are scored against the satellite retrieval itself, and comparisons with other products are reported as product intercomparison, not validation.

Some of these boundaries were choices and others were imposed by the data. Together they isolate a self-contained problem that could be benchmarked rigorously.

1.7 Contributions

The study delivered the following.

1. **A province-wide quality scan of SMAP L3.** It showed that recommended-quality retrievals are absent from the urban-classed cells that cover 76.2% of Pathum Thani, cells whose sur-

roundings are more than a quarter urban, including the cell of the original study point, and that the recurring maximum value earlier read as a saturation ceiling is SMAP's marker of an unsuccessful retrieval, which the quality rule removes.

2. **A reproducible single-cell benchmark dataset** at a non-urban cell, combining recommended-quality L3 retrievals, Sentinel-1 backscatter paired without access to later scenes, and daily reanalysis weather.

3. **A persistence-referenced, significance-tested benchmark on SMAP L3** at one selected non-urban cell, at 2, 3, 5 and 8 days, in which each learned model is scored also against the mean of the last three retrievals, so that it is credited only for what simple smoothing of recent retrievals cannot supply. Much of the apparent gain of the learned models over persistence came from that smoothing; beyond it, only ridge regression's small gain at 8 days is established, within the twelve comparisons made, and its gain at 3 days is suggestive (Section 5.5). Two neighbouring cells that share most of the main cell's footprint, a target averaging the two passes and the original urban cell with all its retrievals serve as checks, with ridge regression and the reference forecasts scored at all 20 non-urban cells of the scan, and the earlier SMAP L4 benchmark, on which no learned model beat persistence, is reported beside it.

4. **A measurement of what Sentinel-1 and weather add** to the L3 history, following the advisor's input ladder. At cell (3,6) neither added measurable additional information under the present data, quality filtering and evaluation protocol.

5. **An account of what the skill consists of on the real record.** Isolated, short-lived changes in the retrievals are distinguished from wetting that followed heavy reanalysis rainfall: much of the 3-day gain comes from the former, a share partly built into how such changes are identified, and the latter is not forecast. A one-parameter rule defined after the main results serves as a diagnostic benchmark, which no learned model clearly beat.

6. **A measurement of persistence error at every separation that occurs up to 8 days, and at 11, 16 and 24 days.** At the chosen cell persistence error does not increase from 2 to 8 days (the differences are small and were not tested), so a change in skill across horizons cannot be read as a change in predictability alone. The study found no consistent evidence that orbit geometry affects it: the lower pooled 8-day error comes partly from pairs that join the morning and evening passes, which are not same-track, and within each pass the ordering is mixed or reversed.

7. **A product intercomparison** of SMAP L4, the ECMWF blend and MERRA-2 over the study period.

Because the quality scan, the dataset and the benchmark are useful whatever accuracy the models achieved, these contributions do not depend on any model beating persistence.

1.8 Relation to the AI Air Project and the Proposed Thesis

This special study is the *baseline* stage of a larger programme. It executed Objective 4 of the AI Air environmental AIoT platform (AIT Fundamental Fund, FY2569) by producing the platform's first

soil-moisture dataset and its first forecasting benchmark. The Master's thesis that follows will use this dataset and these baselines as its point of comparison, and any model it proposes will be judged by whether it improves on the benchmark established here. Following the order of Figure 3.1, that model is to be chosen only after the information gain of candidate inputs has been measured; the results here point to inputs of a different kind rather than to a larger model on the same retrieval history (Chapter 5). This benchmark-then-advance structure mirrors the parent project and ensures that the present study stands on its own.

1.9 Organisation of the Report

Chapter 2 reviews the evolution of soil-moisture forecasting methods, compares representative studies, and sets out the research gaps—those drawn from the literature and those surfaced by this study—in Table 2.3. Chapter 3 describes the method as executed, including the departures from the proposal and the reasons for them: the move from SMAP L4 to quality-filtered SMAP L3 at a non-urban cell, the feature construction required by a gapped record, and the choice of horizons within the limits the satellite's orbit sets; it ends with a table of every design decision, its setting and its reason (Table 3.4). Chapter 4 reports the data and their quality, the SMAP L4 benchmark, the SMAP L3 benchmark with its smoothing references, its season split, its robustness runs and its comparison with the L4 benchmark, the input ladder, the behaviour of the forecasts on the real record, the same forecasts under other yardsticks, the behaviour of skill across horizons and its relation to the orbit, retrieval quality and the former saturation ceiling, accuracy relative to SMAP's requirement, and the product intercomparison, and closes with a summary of the findings against the five research questions (Table 4.24). Chapter 5 interprets these findings, states the limitations and the conclusions, and returns to the research gaps to derive specific topics for future research (Table 5.3).

Chapter 2

Literature Review and Technical Background

2.1 Overview

Research on soil-moisture estimation and forecasting has evolved through several overlapping generations of method, from statistical and geostatistical techniques, through conventional machine learning, to deep learning and, most recently, attention/Transformer models, graph neural networks, physics-informed and hybrid models, and geospatial foundation models. Figure 2.1 summarises this progression, which is broadly one of increasing model capacity, richer spatial and temporal representation, and—as a trade-off—greater data and computational demand. This chapter reviews each generation, compares representative studies (Table 2.1), and distils the research gaps that motivate both this special study and the subsequent thesis.



Figure 2.1: Evolution of methods for soil-moisture estimation and forecasting. This special study establishes the statistical/conventional-ML baseline; the thesis may draw on the later generations once an input has shown information gain.

2.2 Soil-Moisture Memory and Predictability

The physical basis for short-term forecasting is temporal persistence. The soil stores water, so a wet or dry anomaly outlasts the weather that caused it, and this soil-moisture memory makes the recent state a strong predictor of the near-future state (Seneviratne et al., 2010). Orth and Seneviratne (2013) quantified this in a model study of Swiss catchments: information on the initial soil moisture alone gave significantly better forecasts than climatology up to about one week ahead for soil moisture, and two to three days ahead for streamflow. Two consequences follow. First, the problem is tractable: much of tomorrow's soil moisture is explained by today's. Second, persistence is a demanding baseline. Climatology and persistence are the standard references for skill scores (Murphy, 1992), and this study reports skill *relative to persistence* as well as absolute error.

These studies concern the soil column. Work on the layer that SMAP senses points to a shorter memory. From one year of SMAP observations, McColl, Alemohammad, et al. (2017) found that the surface layer retains a median of 14% of the precipitation falling on land after three days, and McColl, Wang, et al. (2017) characterised surface drydowns worldwide by an exponential decay time scale that is shorter over sandier soils and in more arid regions. Shellito et al. (2016) compared SMAP with in-situ probes after 63 rain events: the median drying time scale fitted to SMAP was 4.08 days, against 7.33 days for the probes, a difference the authors attribute to SMAP sensing a shallower layer than the

probes, especially after rainfall. A single surface retrieval may therefore be a weaker predictor of the next than the memory of the soil column suggests, which bears on the choice of reference forecasts in Chapter 3.

2.3 Earth Observation, Land-Surface Modelling, and Reanalysis Data Sources

Data-driven forecasting requires a consistent observed record and spatially complete predictors, which come from three complementary families. *Satellite* passive-microwave radiometry gives direct soil-moisture retrievals: SMAP with a two-to-three-day revisit (Entekhabi et al., 2010), the earlier SMOS L-band mission (Kerr et al., 2010), and the multi-sensor ESA CCI record (Dorigo et al., 2017); where finer detail is needed, the SMAP/Sentinel-1 active–passive product disaggregates SMAP to one-to-three kilometres (Das et al., 2019). *Land-surface models and reanalysis* give spatially complete, gap-free fields: ERA5-Land supplies soil moisture together with the precipitation and temperature that force it and the evapotranspiration its land model computes (Muñoz-Sabater et al., 2021), and the Global Land Data Assimilation System (GLDAS) supplies soil moisture from land-surface models forced by observation-based meteorological data (Rodell et al., 2004). Because satellite, land-surface-model and reanalysis products can disagree, this study compared SMAP with an ECMWF blend of operational analyses and reanalysis and with MERRA-2, which took the place of GLDAS when GLDAS could not be obtained; the comparison is a product intercomparison, not a validation (Chapter 3). *Ground truth* for validation comes from the International Soil Moisture Network (ISMN), a hosting facility for global in-situ measurements (Dorigo et al., 2011), although its coverage of Southeast Asia is limited to two networks, in Malaysia and Myanmar, both since discontinued (Ganepola et al., 2026). Static descriptors such as SoilGrids soil texture (Hengl et al., 2017) add time-invariant context. Cloud platforms—most notably Google Earth Engine (Gorelick et al., 2017)—host these archives with a processing back-end, making multi-source feature assembly feasible without bulk downloading.

2.4 Evolution of Forecasting Methods

2.4.1 Statistical and geostatistical methods

The earliest and still-indispensable methods are statistical. Persistence and seasonal climatology are the reference forecasts (Murphy, 1992; Orth & Seneviratne, 2013); autoregressive models such as (S)ARIMA capture temporal structure, as in the SARIMA-plus-water- balance model of Fu et al. (2023) for multi-depth grassland soil moisture; and geostatistical interpolation (ordinary/regression kriging, inverse-distance weighting) maps soil moisture in space, as compared by Yao et al. (2013), who found regression kriging most accurate in complex terrain. Linear regression, a weighted sum of predictors such as recent values, is the simplest learned model. When the predictors are correlated with one another, weights fitted by least squares alone can be unreliable; ridge regression (Hoerl & Kennard, 1970) penalises their size, accepting a small bias in exchange for a smaller mean square error. Kang et al. (2018) used it to upscale sparse in-situ soil moisture to the pixel of a remote-sensing

product, because correlations among the station measurements otherwise led to failed predictions; with six or with nine stations it was more accurate than both ordinary least squares and the arithmetic mean of the stations. This study uses ridge regression as the step between a fixed-weight mean of recent retrievals and the tree ensembles (Section 3.7.2). These methods are transparent and data-light but assume linearity or spatial stationarity and cannot exploit rich, nonlinear predictor sets.

2.4.2 Conventional machine learning

Conventional machine learning replaced fixed statistical forms with flexible nonlinear regressors. Support vector machines were an early success (Ahmad et al., 2010); tree ensembles then became dominant. Carranza et al. (2021) showed random forests match process-based estimation of root-zone soil moisture; Liu et al. (2020) compared six algorithms for downscaling and found random forest best; and P.-Y. Chen et al. (2025) used a random forest driven by hourly and cumulative rainfall to predict soil water content one to two days ahead. Gradient boosting, especially XGBoost (T. Chen & Guestrin, 2016), is a particularly strong tabular learner; Ren et al. (2023) paired XGBoost with SHAP for interpretable provincial-scale prediction. A recent review of 144 studies found random forest the most-used and most-often-best algorithm, with support-vector methods next (Lamichhane et al., 2025)—evidence that tree ensembles are the appropriate *baseline* class. Their limitation is that they treat each sample independently and do not natively model temporal sequence or spatial structure.

2.4.3 Deep learning

Deep learning introduced explicit temporal and spatial modelling. The Long Short-Term Memory (LSTM) network (Hochreiter & Schmidhuber, 1997) learns long temporal dependencies and became the workhorse of data-driven hydrology: Kratzert et al. (2018) modelled rainfall–runoff across hundreds of catchments, and Fang et al. (2017) used a deep network to produce spatiotemporally seamless SMAP soil moisture. Q. Li et al. (2022) added residual learning to an encoder–decoder LSTM, with gains over a plain LSTM that grow with lead time. The closest earlier work to this study forecasts the SMAP L3 product itself: Fang and Shen (2020) trained an LSTM that takes in the most recent SMAP observations together with atmospheric forcing and static attributes, and reported a median RMSE of 0.022 and a median correlation of 0.92 against SMAP at one day, and 0.024 and 0.90 at two and three days, with the lead time counted from the latest SMAP observation. Its comparisons are with an earlier published method and with the same network without the recent observations; persistence is not among its references. It is not one of the 21 studies compared in Table 2.1. Convolutional and convolutional-recurrent models add spatial structure: L. Li et al. (2022) combined encoder–decoder ConvLSTM with attention for multistep forecasting; Wu et al. (2025) injected causal structure into ConvLSTM for 1–14-day root-zone forecasting; and Hosseinizadeh et al. (2025) trained a ConvLSTM on dense Sentinel-1-derived labels for high-resolution mapping. A systematic benchmark by Y. Wang et al. (2024) compared CNN, LSTM, Transformer, and hybrid models with random forest, SVR, and extreme learning machines across five depths and 1-, 3-, and 7-day horizons. LSTM was the most accurate of the three basic networks and outperformed the three machine-learning models, and two LSTM hybrids, one with feature attention and one trained adversarially, improved on it. Deep

models improve accuracy but need more data and compute and are less transparent.

2.4.4 Attention and Transformer models

Attention mechanisms and Transformers weight the most informative past states directly. Zheng et al. (2024) combined a GRU with a Transformer encoder for root-zone forecasting, and Y. Wang and Zha (2024) compared Transformer and LSTM models with an interpretability analysis, finding the Transformer slightly more accurate at every lag tested and identifying groundwater level as a key predictor. Transformers can capture long-range dependencies but are data-hungry and prone to overfitting on short records.

2.4.5 Graph neural networks

Graph neural networks (GNNs) represent locations as nodes and their spatial or hydrological relationships as edges, letting information propagate between neighbouring sites. Pan et al. (2025) combined a graph-convolutional LSTM with ConvLSTM for multistep gridded forecasting, reducing error against a ConvLSTM baseline at 1–7-day horizons, and W. Wang et al. (2025) used a spatiotemporal graph-convolutional network over a wireless-sensor topology for long-horizon forecasting. Vyas and Bandyopadhyay (2020) learned the graph structure itself for missing-label-robust forecasting. The same template transfers to related spatiotemporal geoscience problems such as streamflow (Akkala et al., 2025). GNNs model spatial coupling explicitly but require a graph construction choice and more data.

2.4.6 Physics-informed and hybrid models

Physics-informed and hybrid models constrain learning with domain knowledge. Chavoshi et al. (2025) embedded the Richardson–Richards equation in a physics-informed network for vadose-zone profiles; Xi et al. (2025) coupled a process-based water-balance branch with an LSTM branch, beating both pure-physics and pure-deep-learning baselines; and L. Li et al. (2024) fused numerical-weather-prediction soil-moisture forecasts into a ConvLSTM for 1–16-day lead. These models improve robustness and extrapolation at the cost of added complexity and coupling assumptions.

2.4.7 Foundation and self-supervised models

The newest generation pretrains large models on unlabelled Earth-observation archives and fine-tunes them for downstream tasks. Geospatial foundation models such as Prithvi-EO (Jakubik et al., 2023) and SatMAE (Cong et al., 2022) learn transferable representations from massive satellite corpora, and L. Wang et al. (2025) applied self-supervised pretraining with a physics-guided spatiotemporal

Transformer to improve soil-moisture generalisation to unseen conditions. These approaches promise strong data efficiency but are, so far, rarely soil-moisture-specific and are computationally heavy.

2.5 Explainable Machine Learning

Because accuracy alone does not establish that a model is physically sound, explainability is increasingly required. SHapley Additive exPlanations (SHAP) assign each feature a contribution to each prediction on the principled basis of cooperative-game Shapley values (Lundberg & Lee, 2017); averaging the magnitudes of these contributions over many predictions gives a global ranking of the features. In soil-moisture work SHAP has been used to confirm that predictions respond sensibly to drivers and to rank predictors (Ren et al., 2023); attention weights have been read similarly (Y. Wang & Zha, 2024). This study applied SHAP to its SMAP L4 random forest for the same purpose, in line with the explainability requirement of the AI Air platform.

2.6 Comparative Analysis of Prior Work

Table 2.1 compares representative studies across the method generations by approach, data and study area, forecast horizon, reported performance (as stated by the authors), and principal limitation. The studies were chosen to represent each method generation and were not drawn by a systematic search, so a statement below that none of them does something describes this set of 21 and is not a claim about the whole literature.

Table 2.1: Comparison of representative soil-moisture estimation and forecasting studies across method generations. Performance is quoted as each paper states it, in the paper's own units.

Study	Method	Data / area	Horizon	Reported performance	Key limitation
Yao et al. (2013)	Kriging, IDW, reg. kriging	In-situ + terrain; 2 km^2 Loess catchment	Mapping	Reg. kriging RMSE 2.72 vs OK 4.04 (10–20 cm)	Geostatistics; weak where autocorrelation low
Orth and Seneviratne (2013)	Initial soil-moisture information	Water-balance model, observed forcing; Swiss catchments	days–weeks	Skill over climatology to about 1 week	Not an ML model; case study
Ahmad et al. (2010)	SVM vs ANN/MLR	TRMM backscatter, NDVI; 10 sites, Lower Colorado basin	Retrieval	SVM best of three; R 0.34–0.77, RMSE $<2\%$ against VIC soil moisture	Retrieval, not forecast; site-specific

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Table 2.1 (continued)

Study	Method	Data / area	Horizon	Reported performance	Key limitation
Fu et al. (2023)	SARIMA + water balance	Multi-depth soil moisture, 2012–2022; Inner Mongolia	Time series	Beats plain SARIMA at 40/100/200 cm	Seasonal signal weakens with depth; one grassland region
Liu et al. (2020)	6 ML algorithms (downscaling)	ESA-CCI 25 km→1 km; 4 networks	Daily	Random forest best of six	Downscaling, not lead-time forecast
Carranza et al. (2021)	Random forest	In-situ RZSM; NL catchment	Daily	RF vs process+DA: slightly better interpolating, similar extrapolating	Poor at wet/dry extremes
Ren et al. (2023)	XGBoost + SHAP	70 stations; Jiangsu, China	Predict.	R=0.69, RMSE 11.11, NSE 0.50, ACC 88%	RH proxy; moderate R
P.-Y. Chen et al. (2025)	Random forest + rainfall feats	Hourly SWC; Taiwan	1–2 days	20 cm R^2=0.9, RMSE 2.0%; 10 cm R^2=0.5	Weaker in shallow layer / rainy season
Fang et al. (2017)	LSTM	SMAP L3 (36 km) + NLDAS-2; CONUS	SMAP recon.	Test RMSE 0.027 (median pixel), 0.036 (75th percentile); $R > 0.80$ at 75% of pixels	Extends SMAP, not multi-day-ahead
Y. Wang et al. (2024)	CNN, LSTM, Transf., hybrids vs. RF/SVR/ELM	30 ISMN sites; 5 depths	1,3,7 days	LSTM best basic network, beats RF/SVR/ELM; two LSTM hybrids better still	Point-scale only
L. Li et al. (2022)	AttConvLSTM (attention+enc–dec)	SMAP L4 18 km; US	Multistep (24 h)	24 h R^2=0.82, RMSE 0.02; +8.4% R^2 vs LSTM	Drops where autocorrelation low
Q. Li et al. (2022)	EDT-LSTM (enc–dec + residual)	FLUXNET sites	1–10 days	R^2 gain vs LSTM 7.95→19.71% (1→10 d)	Point-scale; no spatial context
Wu et al. (2025)	Causal-ConvLSTM	SMAP L4 RZSM + ERA5; Yangtze	1–14 days	R^2 gain vs ConvLSTM 10.2→25.7% (1→14 d)	Single basin; added complexity
Hosseinizadeh et al. (2025)	ConvLSTM + Sentinel-1	SAR SSM + terrain; Maryland	Mapping (2-day weather input)	R^2=0.72, r=0.85, RMSE 0.154	Local area; SAR-retrieval dependent
Zheng et al. (2024)	GRU + Transformer hybrid	8 stations; Hebei	1–2 days	1 d MSE 5.22%; mean R^2 89.9%	Degrades at longer horizon; station-based

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Table 2.1 (continued)

Study	Method	Data / area	Horizon	Reported performance	Key limitation
Y. Wang and Zha (2024)	Transformer vs LSTM + interpret.	SMAP L3 + well groundwater + GLDAS; Hetao, Inner Mongolia	1–10 days	1-day lag: Transformer R^2 0.523 vs LSTM 0.485	Modest R^2, below zero at 10 days
Pan et al. (2025)	GNN + ConvLSTM (GCCL)	SMAP L4; Hubei, China	1–7 days	RMSE 0.018–0.038 over 1–7 d; -14.3% vs. ConvLSTM at 1 d	Data on request; SMAP-dependent
W. Wang et al. (2025)	Spatiotemporal GCN	Wireless sensor network	48–192 h	+49.5/34.9/29.7% vs baselines	WSN-topology specific
Xi et al. (2025)	PGDL (physics + LSTM)	Climate + in-situ SSM	Daily	RMSE 0.04, R^2 0.66 (vs physics 0.13/-2.32)	Surface only; site-based
L. Li et al. (2024)	Hybrid attention; GFS-forced ConvLSTM	SMAP L4 target + GFS forecasts; China	1–16 days	Beats pure-DL at 79.5% stations (16 d)	Coarse GFS input; win-rate metric
L. Wang et al. (2025)	Self-supervised + physics Transformer	Synthetic SM	multi-depth	Consistently beats LSTM on extreme dryness, unseen weather and unseen depths	Validated on synthetic data only

Three patterns emerge. First, tree ensembles (random forest, XGBoost) are the consistently strong *conventional* baseline and the natural reference for this study (P.-Y. Chen et al., 2025; Lamichhane et al., 2025; Ren et al., 2023). Second, the deep-learning, Transformer, graph, and hybrid generations report accuracy gains over simpler learned baselines—gains that often widen with lead time (Q. Li et al., 2022; Wu et al., 2025), although neither of these two studies measures skill against persistence—but at the cost of data, compute, and transparency, and almost always benchmarked on large, data-rich, or in-situ-dense domains. Third, most studies target either a single point/station or a large continental grid; small, heterogeneous, data-scarce tropical provinces are under-represented, and few pair forecasting with a systematic explainability audit. The most recent regional synthesis, a roadmap for satellite soil moisture in Mainland Southeast Asia (Ganepola et al., 2026), confirms this from the other side: its four pillars are gap filling with region-specific validation, finer spatial and temporal resolution, multi-depth estimation, and capacity building. It treats soil moisture as an input to agriculture, weather and climate, and hazard applications such as flood and landslide forecasting, but forecasting soil moisture itself is not among its priorities.

2.6.1 Comparison of Prior Work

Table 2.2 compares the 21 studies of Table 2.1 on seven dimensions that bear directly on the gaps below. Every cell was assigned from the full text of the paper, not from its abstract, and the evidence for each is recorded in `gap_matrix.csv`, kept with the report. Four dimensions are fully met by none of them: no study reports skill against a persistence baseline, none tests the model comparison for significance, none compares a retrieval product with an assimilated one as forecast targets, and none is set in the tropics, although three include a few tropical sites among many. No prior study fully meets more than two of the seven. The marks for this study record what was done, not how well it turned out: explanation is by SHAP on the L4 random forest and by an information-gain ladder on L3. The L3/L4 mark is partial because the two products were forecast, but not under one protocol (gap G6). The L4 series is a $\pm 0.15^{\circ}$ box mean of daily-averaged 3-hourly values, while the L3 series is the value at one cell of the 9 km grid at one instant, 38.1 km from the centre of that box, so the comparison mixes processing level with footprint, time sampling and location. The two benchmarks also differ in horizons (1, 3 and 7 days on L4; 2, 3, 5 and 8 days on L3) and in how skill is aggregated: the L4 skill is a mean of per-fold skills, whereas the L3 skill is computed once over the pooled test samples.

Table 2.2: Seven-dimension comparison of prior work against this study. **Pers**: skill measured against persistence. **Sig**: significance test or interval on the model comparison. **Time**: validation respects time order. **XAI**: explains which inputs drive the predictions (○ = for one product or model only). **Fail**: analyses where or when the model fails. **L3/L4**: compares retrieval and assimilated products as targets (○ = uses one and states its level; for this study, forecasts both, but not under one protocol). **Trop**: tropical or Southeast Asian setting (○ = subtropical monsoon, or a few tropical sites among many). ✓ = yes, ○ = partial, – = no, ? = not stated.

Work	Pers	Sig	Time	XAI	Fail	L3/L4	Trop
Yao et al. (2013)	–	○	–	–	–	–	–
Orth and Seneviratne (2013)	○	–	–	–	○	–	–
Ahmad et al. (2010)	–	–	✓	–	○	–	–
Fu et al. (2023)	–	○	○	–	–	–	–
Liu et al. (2020)	–	–	–	✓	✓	–	–
Carranza et al. (2021)	–	–	○	✓	✓	–	–
Ren et al. (2023)	–	–	–	✓	○	–	○
P.-Y. Chen et al. (2025)	–	–	✓	–	✓	–	○
Fang et al. (2017)	–	–	✓	–	○	○	–
Y. Wang et al. (2024)	–	–	✓	✓	–	–	○
L. Li et al. (2022)	–	–	✓	○	✓	○	–
Q. Li et al. (2022)	–	–	✓	–	○	–	○
Wu et al. (2025)	–	–	✓	✓	–	○	○
Hosseinizadeh et al. (2025)	–	–	–	–	✓	–	–
Zheng et al. (2024)	–	–	?	–	–	–	–

Table 2.2 (continued)

Work	Pers	Sig	Time	XAI	Fail	L3/L4	Trop
Y. Wang and Zha (2024)	–	–	?	✓	○	○	–
Pan et al. (2025)	–	–	✓	○	○	○	○
W. Wang et al. (2025)	–	–	?	–	○	–	–
Xi et al. (2025)	–	–	✓	○	–	–	–
L. Li et al. (2024)	–	–	✓	–	✓	○	○
L. Wang et al. (2025)	–	–	✓	○	✓	–	?
This study	✓	✓	✓	○	✓	○	✓

2.7 Research Gaps

The literature reviewed above leaves five gaps that this study was designed to close. Executing it surfaced four further gaps that the forecasting literature does not discuss but that any follow-on work over this region must confront, and confirmed a fifth that had already been documented. Table 2.3 states each gap, how this study addressed it, and what was found. Gaps G1–G5 come from the literature; G6–G9 were identified during the work. Of these, G9 is a finding of this report, G6 a recommendation for how results should be reported, and G7 and G8 practical cautions rather than closed gaps. G10 was also met during the work, but the regional roadmap of Ganepola et al. (2026) had already documented it, and it is credited to them.

Table 2.3: Research gaps, how this study addressed each, and the outcome. G1–G5 are drawn from the literature; G6–G9 were surfaced by this study (G6 as a reporting recommendation, G7 and G8 as cautions); G10 was surfaced here and documented earlier by Ganepola et al. (2026).

#	Gap	How addressed here	Outcome
G1	No persistence-referenced benchmark exists for a tropical peri-urban province such as Pathum Thani (Table 2.2)	Persistence, climatology, random forest and XGBoost over 2016–2024 on two SMAP products, with the same walk-forward design but not the same folds (eight test years, 2017–2024, on L4; six or seven, to 2024, on L3) and with skill averaged over folds on L4 and pooled over folds on L3; on L3, ridge regression and two smoothing references added	**Partly closed.** Benchmark delivered for SMAP L4 over a $\pm 0.15^{\circ}$ box around the original point and for SMAP L3 at one non-urban cell, (3,6), with two further cells as a robustness check, and with significance tests and confidence intervals. The urban-flagged cells, which cover 76.2% of the province's area, returned no recommended-quality retrieval, and a further 1.9% lies in a cell with no retrieval, so 78.1% of the province cannot be benchmarked on recommended L3 retrievals

Table 2.3 (continued)

#	Gap	How addressed here	Outcome
G2	Persistence is rarely used as the reference (Table 2.2). Unless the seasonal cycle is removed, as Orth and Seneviratne (2013) do, reported "skill" can largely measure the annual cycle	Skill score against persistence adopted as the headline metric at every horizon and in every subgroup	**Closed at the sites studied.** Climatology loses to persistence on L4 at every horizon (-2.34 at 1 day to -0.67 at 7 days). On L3 it scores below persistence at 2, 5 and 8 days (-0.164, -0.068, -0.084) and above it at 3 days ($+0.025$), and no interval excludes zero. On L4 the best learned model scores above climatology (not tested) while failing to beat persistence, so the concern is real
G3	Explainability is reported in a minority of studies (e.g., Ren et al., 2023; Y. Wang & Zha, 2024), as Table 2.2 shows; accuracy is shown, physical plausibility usually is not	SHAP attribution for the L4 random forest; on L3, an information-gain ladder that adds Sentinel-1 and weather to the retrieval history	**Partly closed.** For the L4 random forest at 7 days, the current value has the largest mean absolute SHAP value, 0.054, about 4.5 times that of the next feature, the previous day's value (0.012); neither L4 model beat persistence. SHAP was computed only for this model and horizon, not for XGBoost or for any L3 model. On L3 the inputs were assessed by the ladder instead: neither Sentinel-1 nor weather added measurable additional information under the present data, quality filtering and evaluation protocol
G4	Advanced models are deployed without a local baseline to justify their cost and opacity	Benchmark-then-advance structure; no deep learning admitted to this study	**Closed at the sites studied.** The thesis now has a measured number to beat, and evidence about which direction is worth taking
G5	Failure characterisation is scarce, yet it should decide which advanced capability a follow-on model needs	On L4, errors of an untuned random forest conditioned on antecedent rainfall, current wetness and the magnitude of change; on L3, skill split by wet and dry season	**Partly closed.** On L4 the untuned random forest helps in none of the 21 forecast-time conditions tested (four rainfall bands and three wetness terciles at each of three horizons): it loses to persistence in every one, with skill from -0.017 to -0.567; only this one model was conditioned. On L3 the three-retrieval mean, ridge regression and the random forest gain over persistence in both seasons at every horizon. The largest seasonal differences are at 2 days, where the three-retrieval mean and ridge gain more in the dry season ($+0.174$ and $+0.121$, against $+0.100$ and $+0.078$ in the wet season), and at 8 days, where the random forest gains more in the wet season ($+0.203$ against $+0.106$); none of these differences was tested

Table 2.3 (continued)

#	Gap	How addressed here	Outcome
G6	Studies forecast either the retrieval, L3 (Fang et al., 2017; Y. Wang & Zha, 2024), or the assimilated product, L4 (L. Li et al., 2022, 2024; Pan et al., 2025; Wu et al., 2025), but none compares the two as targets under one protocol	L3 and L4 forecast over 2016–2024 with the same walk-forward design, in-fold tuning and significance tests, but not the same folds: eight test years (2017–2024) on L4, six or seven (to 2024) on L3. L4 skill is the mean of per-fold skills, with its interval and test on the pooled test samples; L3 skill is computed once over the pooled samples. L3 is the value at one cell of the 9 km grid, (3,6), at one instant, 38.1 km from the original point; L4 is a $\pm 0.15^{\circ}$ box mean of daily-averaged 3-hourly values around that point	**Reporting recommendation.** No learned model beats persistence on L4 (best skill -0.131 to -0.041); on recommended-quality L3 at cell (3,6) the best forecast at each horizon scores $+0.144$ to $+0.241$, much of it from smoothing recent retrievals, and cells (2,6) and (4,6) show the same broad pattern. At 3 days, the one shared horizon, the best learned skill is -0.066 on L4 (mean of per-fold skills) and $+0.241$ on L3 (the best of 12 learned configurations, pooled over folds). The contrast mixes processing level with footprint, time sampling and location, and the two benchmarks also differ in horizons and in skill aggregation, so this study cannot separate these effects. What it supports is a recommendation: state the product, footprint, time sampling and location of any target
G7	Forecast error is not compared with the reference product's stated accuracy; L. Li et al. (2024) compare forecast skill with the L4 product's own measured skill, but not with its accuracy requirement	Model error compared with SMAP's 0.04 m^3/m^3 mission accuracy requirement (Entekhabi et al., 2010)	**A caution, not a closed gap.** At cell (3,6) the best forecasts reach an ubRMSE of 0.031–0.035 m^3/m^3, below the requirement; persistence RMSE on the test samples is 0.040–0.042. The requirement is a design target for the retrieval's error against the ground, not the measured retrieval error at this cell, and the forecast error is measured against the retrieval, so the two are not comparable. The target's retrieval error, of unknown size, remains part of every measured forecast error
G8	An apparent saturation ceiling in retrieval records is not examined in forecasting studies	Every retrieval in the province scan (124,570 in 53 cells) checked against SMAP's quality flag; only recommended quality (flags 0 and 8) retained	**Practical caution.** The ceiling is not a measurement of saturated soil but a value at the upper limit of a valid retrieval that SMAP marks as not successful (flag 13): in every cell, 98.1–100% of flag-13 retrievals sit at that cell's maximum value, as SMAP's documented flag 13 implies; the point is that a forecasting study must apply the flag. The quality rule removes the flag-13 retrievals, but 77% of the retrievals it rejects at cell (3,6) fall in May–October and follow more rain than the accepted ones, so the filtered record under-represents the wettest conditions

Table 2.3 (continued)

#	Gap	How addressed here	Outcome
G9	Forecast horizons are chosen by convention (1, 3, 7 days) rather than from the satellite's sampling	Retrieval pairs counted at every gap over nine years, persistence error measured at each, and the horizons of 2, 3, 5 and 8 days chosen from the separations with enough retrieval pairs and approved by the advisor	**New finding.** At cell (3,6) exact pairs number 41 at 1 day, 333 at 2, 601 at 3, none at 4, 598 at 5, 326 at 6, 41 at 7 and 872 at 8 days. The 1- and 7-day pairs are too few for walk-forward evaluation, which rules those horizons out at this cell; 6 days was not a benchmark horizon, to keep the horizon set identical to the analysis presented on 23 September, and was run afterwards as a check. Its persistence RMSE over all pairs, 0.0430, is higher than at 5 or 8 days, so 6-day results cannot be read off the 5- and 8-day ones. Persistence error does not grow monotonically: over all exact-gap pairs, the 8-day pairs give an RMSE of 0.0394, below the 0.0398–0.0405 of the 2-, 3- and 5-day pairs; the differences are small and were not tested. There is no consistent evidence that orbit geometry affects it: the pooled 8-day figure is pulled down by pairs that join the morning and evening passes, which are not same-track, and within each pass the ordering is mixed or reversed
G10	In-situ networks are sparse outside the mid-latitudes (Dorigo et al., 2017, 2011), so a study in this region cannot assume ground truth	ISMN holdings checked directly; product intercomparison substituted and documented	**Confirmed.** No ISMN station exists in Thailand, as the regional roadmap also reports (Ganepola et al., 2026). Ten stations exist in the country (Jotisankasa et al., 2023), but their data are not publicly accessible; improvement cannot currently be verified against ground truth

Two of these deserve emphasis because they change what a follow-on study should attempt. G6 shows that a result reported for "SMAP" is not well defined without stating which product, footprint, time sampling and location were used. With walk-forward evaluation of the same design, though not on the same folds, no learned model beat persistence on L4, whereas on L3 at cell (3,6) every learned model on the retrieval history alone, and both smoothing references, had positive skill at 3, 5 and 8 days, with intervals above zero. Because the L4 series is a $\pm 0.15^{\circ}$ box mean of daily-averaged 3-hourly values and the L3 series the value at one cell of the 9 km grid at one instant, 38.1 km from the centre of that box, the contrast mixes processing level with footprint, time sampling and location, and this study cannot separate them. The two benchmarks also differ in horizons (1, 3 and 7 days on L4; 2, 3, 5 and 8 days on L3) and in skill aggregation, a mean of per-fold skills on L4 and a single score over the pooled samples on L3. A follow-on study should state the processing level, footprint, time sampling and location of any product it forecasts. G7 shows that an ubRMSE below the product's accuracy requirement, measured against the retrieval, does not establish accuracy against the true soil moisture. The requirement is a design target, not a limit on the error at any one cell, and the target

carries its own retrieval error of unknown size, which is part of every measured forecast error. A claim that some additional observation improves soil-moisture estimation therefore cannot be substantiated against L3 alone—it requires an independent reference: in-situ data, which is not currently obtainable for this region, or an indirect check such as triple collocation (Gruber et al., 2020).

Chapter 3

Methodology

This chapter describes the method as executed, not as planned. Where the proposal's design could not survive contact with the data, the departure is stated together with the measurement that forced it.

3.1 Research Framework

The study follows a fixed order of questions: the research question, then the availability of data to answer it, then baselines, then the information gain over those baselines, and only then the choice of model (Figure 3.1). Each stage must be settled before the next is attempted. This report covers the first four stages; the fifth is deferred to the thesis, where a more elaborate model or data assimilation is to be introduced only where stage four has shown that the additional data carries information.

Each stage answers a question on which the next depends. A skill score means little until the quality and coverage of the record are known, because they decide where a forecast can be scored at all and at which horizons. Skill is always measured against a reference, so the reference forecasts must be fixed before a learned model can be credited with anything. An input earns a place in a model only if it adds information beyond those references, so information gain is measured before a model is chosen. If a more elaborate model were chosen first, a gain could not be traced to the model, to the data or to a weak reference.

The work did not follow this order from the start, and the departures bear on how the results should be read. The first benchmark, on SMAP L4, was run before the quality of SMAP L3 had been examined; its result, that no learned model beat persistence, sent the study back to the research question and led to the comparison of the two products that frames Chapter 4. An earlier L3 run at the original, urban cell used retrievals of every quality, and it was there that the mean of the last three retrievals was found to score about as well as the tuned random forest (Section 1.4). The province-wide quality scan, the quality rule (25 September 2026), the choice of cell and the final protocol followed. The L3 benchmark reported here then ran in the stated order: quality scan, cell, horizons, reference forecasts, learned models, input ladder. Three choices were nevertheless made with knowledge of results on the same test years: the three-retrieval mean as a second reference, the damped anomaly of that mean as a diagnostic benchmark (Section 3.8.6), and the set of four horizons, carried over from the analysis presented on 23 September 2026 (Section 3.5.1). The first two raise the bar for the learned models and so work against them. A fourth choice was made later still: the family of twelve comparisons over which the Holm adjustment is taken (Section 3.8) was fixed after their p-values were known. It works in the learned models' favour, because over all 48 comparisons with the three-retrieval mean ridge regression's 8-day gain would not survive. Table 5.2 also lists the best of twelve learned configurations at each horizon, a choice that works in their favour and is not adjusted for (Section 3.8); it is given as a description only, and the bar a later model must pass is ridge regression on the L3 history (Section 5.2). No year after 2024 was held back untouched, so no test period is entirely free of these choices.

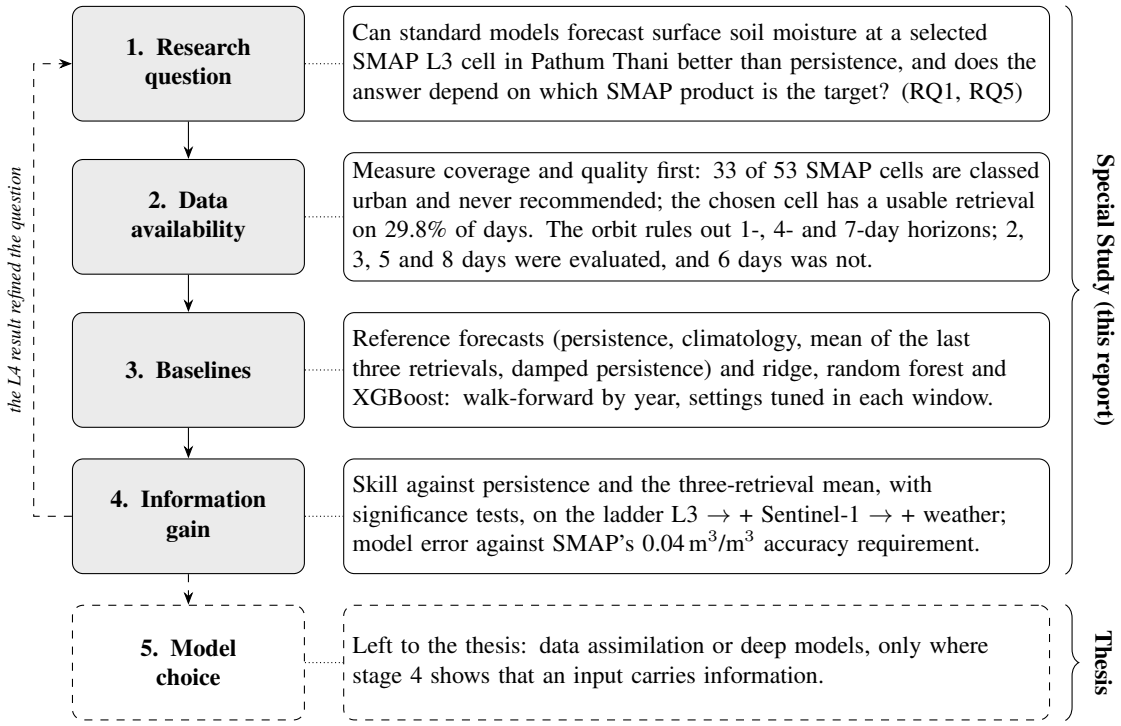


Figure 3.1: Research framework. Stages one to four are carried out in this report; model choice is left to the thesis and depends on the information gain measured in stage four.

3.2 System Overview

Figure 3.2 shows the system as built, from the sources to the output. Three open sources supply the SMAP L3 analysis: SMAP L3 itself, which is the target, and Sentinel-1 backscatter and ECMWF reanalysis weather, which are predictors. SMAP L4, the earlier target, and the comparison products pass through the same acquisition layer.

Acquisition streams one file at a time, reduces it to what the study needs, discards it, and can resume after an interruption (Section 3.10). For SMAP L3 it keeps every 9 km cell in a box around the province, together with SMAP's quality and surface flags, rather than a single pixel. A quality step then keeps only the retrievals SMAP recommends, sets aside the cells SMAP classes as urban, and chooses one cell at least 90% inside Pathum Thani (Section 3.3).

Three feature groups are built for the chosen cell: the L3 retrieval history, the latest Sentinel-1 scene, and trailing weather windows (Section 3.6). They feed two families of forecasters: reference forecasts, and three learned models (ridge regression, random forest and XGBoost), each fitted on four feature sets that add Sentinel-1 and weather to the L3 history in turn (Section 3.7). All L3 forecasters are scored under one evaluation protocol (Section 3.8). The output is the benchmark at cell (3,6), and the information gain of each input group, at 2, 3, 5 and 8 days. The earlier L4 target was scored with the same expanding-window, walk-forward design by calendar year (eight folds testing 2017–2024, against six or seven on L3), in-fold tuning and significance tests, but with calendar-space features, at 1, 3 and 7 days, and with a skill score averaged over folds rather than pooled over them (Section 3.8).

L4 also differs from L3 in product level, footprint, time sampling and location (Section 3.3).

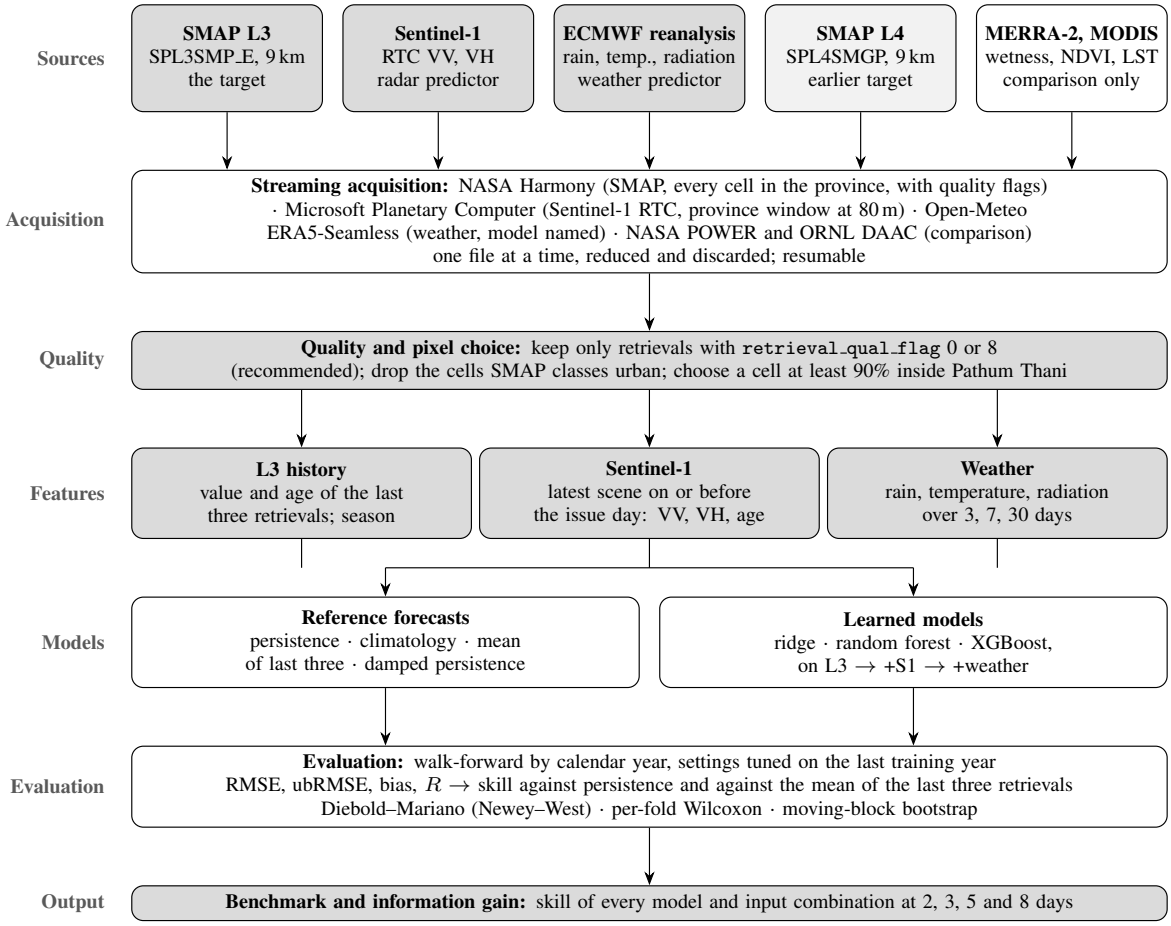


Figure 3.2: System overview. Among the sources, dark shading marks data used for the SMAP L3 target, light shading the earlier SMAP L4 target, and white comparison-only data. In the lower layers, dark shading marks the steps specific to the L3 target: the quality filter and pixel choice, the three feature groups, and the output.

3.3 Study Area and Period

The data cover Pathum Thani province over 2016-01-01 to 2024-12-31, a span of 3,288 days. The period was fixed in the study proposal and kept. It starts with 2016 because SMAP's record begins on 31 March 2015 (NSIDC, 2023), which makes 2016 its first complete calendar year, and it is made of whole years because the evaluation tests on one calendar year at a time (Section 3.8) and a seasonal normal needs complete annual cycles. Data after 2024 were not retrieved, which leaves the later years as a test period that this study did not use (Section 5.2). The province outline is the geoBoundaries ADM1 polygon (`https://www.geoboundaries.org`; derived from OpenStreetMap, ODbL licence), which encloses 1,514 km^2. The province has seven districts: Mueang Pathum Thani, Khlong Luang, Thanyaburi, Nong Suea, Lat Lum Kaeo, Lam Luk Ka and Sam Khok. Their boundaries

come from the geoBoundaries ADM2 layer (Royal Thai Survey Department and OCHA ROAP, 2019, CC BY 3.0 IGO); together the seven polygons cover 99.8% of the province outline. Two scopes are kept apart in what follows: the quality scan covers the whole province, and the forecasting benchmark one selected cell.

3.3.1 The province scan

SMAP L3 was acquired for the whole province rather than for a single point. The product is the enhanced SMAP Level 3 radiometer soil moisture, SPL3SMP_E version 006 (O'Neill et al., 2023) (NSIDC collection C2938664763-NSIDC_CPRD), posted on the 9 km EASE-2 grid. Its values are not measurements of 9 km squares: the radiometer's footprint is about 36 km across, and the measurements are interpolated onto the 9 km grid (Section 3.3.5). Level 3 is the target because the study concerns the satellite retrieval itself: Level 4 is the output of a land-surface model into which the retrievals are assimilated (Reichle et al., 2019), and on it no learned model beat persistence (Section 4.2). The enhanced product was used in place of the standard 36 km product for two reasons. Its 9 km posting lets each cell be related to the province outline and to SMAP's flags one by one, where a single 36 km cell would cover an area close to that of the whole province; and it was found to be as accurate as the standard product at validation sites (Chan et al., 2018). The province was scanned, and not one point, because the quality of the retrievals turned out to differ from cell to cell, which a single point cannot show. Every cell in a box from $13.856°$ to $14.336°$ N and from $100.272°$ to $101.012°$ E was requested through NASA Harmony, together with each retrieval's quality flag, surface flag, vegetation water content and static water-body fraction. The box holds 56 grid cells, of which 53 returned at least one retrieval; the counts below refer to those 53. Of the three that never did, only cell (5,3) overlaps the province, and the part of the province it covers, 1.9% of the area, therefore has no SMAP L3 data. All 108 months of the study period were downloaded without error, and granules exist on 3,207 of the 3,288 days. Cells are named (row, column) in the scan grid, counted from zero at its north-west corner.

3.3.2 The quality rule

The field `retrieval_qual_flag` is a set of bits. According to the product's user guide (NSIDC, 2023) and the file's own attributes, the bits with masks 1, 2, 4 and 8 stand for *recommended quality*, *retrieval attempted*, *retrieval successful* and *freeze/thaw retrieval successful*, and a set bit means that the condition was *not* met. Recommended quality is therefore the bit with mask 1 clear, which is the values 0 and 8; the user guide states that either value "indicates high-quality retrievals" (NSIDC, 2023, p. 13), a value of 8 only recording a failed freeze/thaw retrieval, which does not affect the soil-moisture retrieval. Chinthaka Ganepola, a PhD researcher at AIT who advised the study on the SMAP data, gave the same advice on 25 September 2026, and the rule was adopted. SMAP's own validation uses the same flag: the statistics at its core validation sites are computed only on retrievals whose quality flag shows recommended status (Colliander et al., 2017).

The reason for the rule is physical. The retrieval inverts a model of microwave emission from soil under vegetation, and it is reliable only where the conditions inside the sensed area resemble that

model. The product therefore withholds its recommendation when a surface condition crosses a threshold: more than 25% urban area, more than 5% open water, precipitation above 1 mm/h, steep terrain, vegetation water content above 5 kg/m^2, frozen ground or snow (NSIDC, 2023; O'Neill et al., 2021). These fractions refer to an area of about 33 to 36 km around the grid cell, not to the 9 km cell itself (NSIDC, 2023). A retrieval that is not recommended may still be close to the truth, but the product gives no assurance that it is, and a benchmark built on it would score forecasts against values of unknown quality.

The rule removes most of the record. Of 124,570 retrievals in the 53 cells, 33,822 carry flag 8 (recommended), 60,546 flag 9 (not recommended) and 30,202 flag 13 (not recommended, retrieval not successful). Overall, 27.2% are recommended; in cells at least half inside the province, 16.5% of 42,006. One bit of the surface flag, bit 7 (frozen ground, from the radiometer), is set on every cell and date. The user guide notes that this flag is validated only north of 45° N and gives false flags at low latitudes, and that the retrieval itself uses modelled soil temperatures instead (NSIDC, 2023); it was ignored.

3.3.3 Urban cells

The surface flag also carries SMAP's urban-area class, bit 3. It is set when more than 25% of the area around the cell, about 33 to 36 km across, is urban (NSIDC, 2023). The user guide gives the reason: the emission of built surfaces cannot be modelled, so "even a small amount of urban area in the radiometer footprint is likely to adversely bias the retrieved soil moisture" (NSIDC, 2023, p. 11). Urban areas are likewise excluded from SMAP's accuracy requirement (Entekhabi et al., 2010). The flag is set on 33 of the 53 cells, and over 2016–2024 none of their retrievals was recommended. That absence follows from how the recommendation is defined, since an urban fraction above 25% withholds it; what the scan adds is that the definition removes three quarters of this province. Of the province's 1,514 km^2, 76.2% (1,153 km^2) lies in these urban-flagged cells, a further 1.9% lies in cell (5,3), which never returned a retrieval, and the remaining 21.9% lies in non-urban cells. These are shares of the province's area by the flag of the grid cell it falls in, not measured fractions of built-up land: a flagged cell may hold much farmland, provided its surroundings are more than a quarter urban. Of the 12 cells at least 90% inside the province, 10 are urban and 2 are not.

The original study point, 14.02° N, 100.53° E in Mueang Pathum Thani district, lies in cell (4,2), centred at 14.0106° N, 100.5031° E. The cell is flagged urban, and every retrieval in it carries flag 9 or 13, so none was recommended. The earlier single-pixel L3 series is otherwise identical to this cell: the old downloader's fallback to the nearest valid pixel replaced only 2 of its 2,302 values. Every L3 result obtained at the original point therefore rests on retrievals that SMAP does not recommend.

3.3.4 Choice of cell

The study cell is (3,6), centred at 14.0833° N, 100.8766° E in the eastern part of Thanyaburi district, 38.1 km from the original point (Figure 3.3). Of its area, 90.6% lies inside Pathum Thani (43.1% in Nong Suea, 38.0% in Thanyaburi and 9.5% in Lam Luk Ka), and it is never flagged urban. It is one of only two non-urban cells among the 12 at least 90% inside the province, and of those two it has

the more recommended-quality days: 981, against 946 for cell (4,6). The three criteria have different reasons. A non-urban cell is required because urban-flagged cells carry no recommended-quality retrieval (Section 3.3). The number of recommended days decides how many forecasts can be scored. The 90% threshold is a practical rule that keeps the grid cell almost wholly inside the province; it was not tuned, and cell (2,6), at 89.8%, falls just short of it. The rule describes the grid cell only: the area that the radiometer senses is larger (Section 3.3.5). The cell's vegetation water content never exceeds 2.73 kg/m^2 (median 2.46), below the 5 kg/m^2 above which vegetated land is excluded from SMAP's accuracy requirement (Entekhabi et al., 2010), and its static water-body fraction is 0.002. Non-urban means here only that the urban flag is not set: the urban share of the cell's surroundings is at most 25%, not zero, so its retrievals may still carry an urban influence of unknown size. The water-body fraction is a static value and does not describe the seasonal flooding of fields. At this cell 70.4% of 2,376 retrievals are recommended. Land cover was not joined (Section 3.4), so the crops grown in the cell and whether they are irrigated are not known from the data used here; the cell is described as non-urban, not as farmland.

Figure 3.3 maps the scan. Each square is one cell of the 9 km grid: grey cells carry SMAP's urban flag, green cells do not, and the number in a green cell is its count of recommended-quality days. The map is included because three things on which the rest of the report depends can be seen in it. The non-urban cells lie only along the eastern and western edges of the province. The study cell and its two neighbours sit in one column in the east. And the area that the radiometer senses around cell (3,6), drawn as the dashed circle, is about as large as the province and reaches beyond its border (Section 3.3.5).

Two neighbouring non-urban cells serve as robustness checks. Cell (2,6), at 14.1561° N, 100.8766° E in Nong Suea district, has more recommended days (1,031) but lies 89.8% inside the province, just short of the 90% criterion. Cell (4,6), at 14.0106° N, 100.8766° E in Lam Luk Ka district, lies 90.6% inside and has 946. The results of this report therefore describe cell (3,6), checked at these two neighbours; they are not a result for the province as a whole, most of which lies in cells that returned no recommended retrieval.

3.3.5 Footprints and time sampling

The 9 km of the enhanced L3 product is the spacing of its grid, not the size of the area it senses. SMAP's radiometer observes a footprint that is roughly an ellipse of 36 by 47 km, and its native resolution is about 36 km (Chan et al., 2018; NSIDC, 2023). Because successive footprints overlap, the Backus–Gilbert interpolation can estimate the brightness temperature at the centre of every 9 km cell, and the soil moisture is retrieved there, but the estimate keeps the native resolution of the radiometer (Chan et al., 2018). The enhanced product was found to be as accurate as the standard 36 km product at validation sites, with an unbiased RMSE below 0.040 m^3/m^3 (Chan et al., 2018), none of them in Southeast Asia. Three consequences follow for this study. The value posted at cell (3,6) describes an area roughly the size of the province (an ellipse of 36 by 47 km covers about 1,330 km^2, against the province's 1,514 km^2), centred on the cell and reaching beyond the province's eastern edge. If the sensed area is taken as an ellipse of 36 by 47 km centred on the cell, between 53% and 59% of it lies inside the province, depending on the ellipse's orientation, which changes with the scan (57% for a circle of the same area). About 41% of it lies on grid cells that SMAP flags as urban, 47–55% on non-urban cells of the scan and the rest outside the 53 scanned cells (`l3_footprint_overlap.py`).

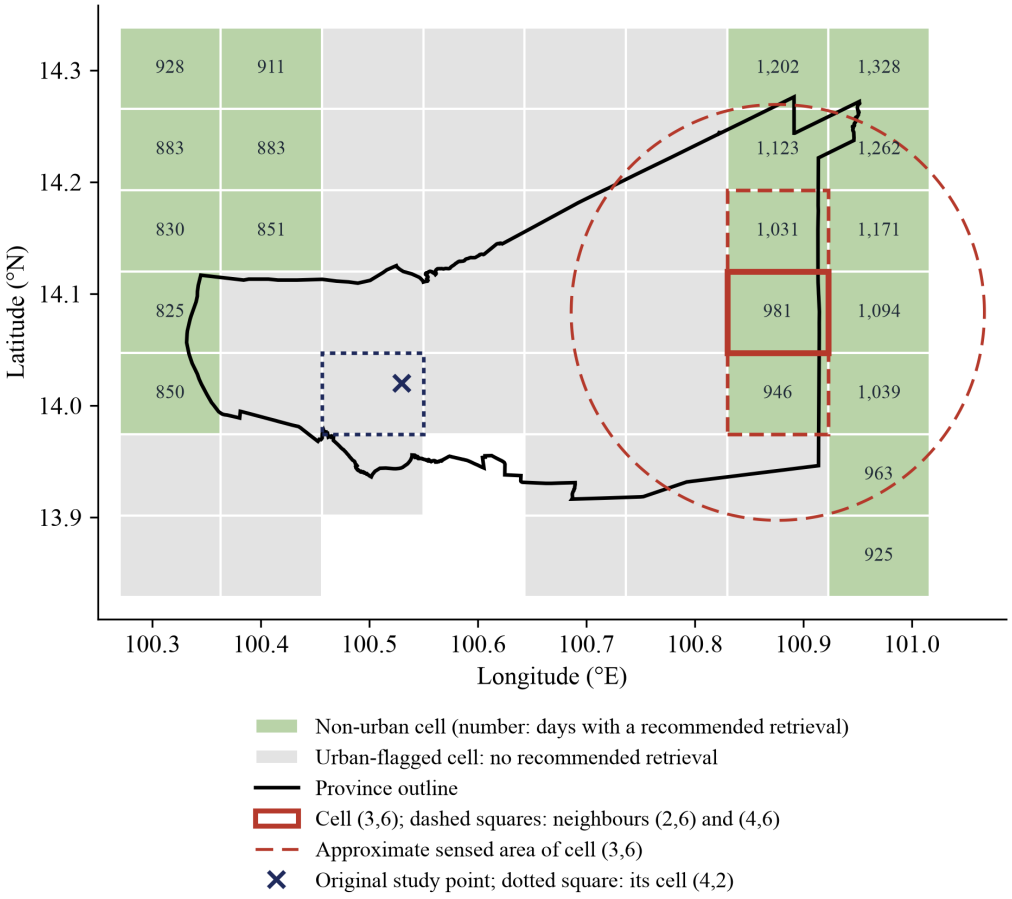


Figure 3.3: The province scan and the cells used. The 53 cells of the 9 km grid in the scan are shaded by SMAP's urban flag; in a non-urban cell the number is its count of days with a recommended-quality retrieval, 2016–2024. The dashed circle has the area of the 36 by 47 km footprint ellipse centred on cell (3,6); it outlines the sensed area and is not an antenna pattern.

These are shares of a geometric outline, not of what the radiometer weights, and a flagged cell is not built-up land (Section 3.3); they show that the value called non-urban here describes ground of which roughly two fifths lies under the urban flag. The neighbouring cells (2,6) and (4,6), whose centres lie 8 km to the north and south, are built largely from the same measurements, so they are not independent observations of other places (Section 4.3.2). And the 90% criterion of Section 3.3, the district shares and the urban flag describe the grid cell or its surroundings, not the footprint exactly.

The products differ in footprint and in time sampling, and this matters for how results may be compared. The L3 target is the value posted at one cell of the 9 km grid, observed at one instant: the morning overpass at about 06:00, or the evening overpass at about 18:00 where the morning one is missing. The SMAP L4 series of the earlier benchmark is a box of $\pm 0.15^{\circ}$ around the original point, 57% inside the province, and each daily value is the mean of the eight 3-hourly values of that day. L4 therefore shares neither the footprint nor the time sampling of L3, and its box is centred on the original point, 38.1 km from cell (3,6). Every comparison between the two in this report is a comparison of product level, footprint, time sampling and location together, and the study cannot separate their effects. The two benchmarks also differ in their horizons (Section 3.5.1) and in how the skill score is aggregated over folds (Section 3.8). The comparison products were sampled at the original point: a MODIS pixel lies wholly inside the province, while a MERRA-2 cell is larger than the province and extends well beyond it.

3.4 Data

The datasets were acquired from open services, none of which requires Google Earth Engine, whose registration was the original blocker. Table 3.1 gives the role, route and measured completeness of each. The grouping follows the shading of Figure 3.2. The last column carries the point of the table: the weather and the comparison products are complete, Sentinel-1 gives a usable scene on about one day in six, and the target itself has a recommended-quality value on 29.8% of days. The target, not the predictors, therefore limits how many forecasts can be scored.

Sentinel-1. Backscatter from the Sentinel-1 mission (Torres et al., 2012) was taken from the radiometrically terrain-corrected (RTC) collection of Microsoft Planetary Computer (`https://planetarycomputer.microsoft.com/dataset/sentinel-1-rtc`), which Catalyst produced from ESA's GRD scenes and which is licensed CC-BY-4.0. The collection is served as cloud-optimised GeoTIFFs, so only the window over the scan box was read, at about 80 m, a reduced-resolution overview of the 10 m product. A scene took about 5 s to read, and no whole scene of about 1 GB was downloaded. Linear VV and VH backscatter (γ^0) was averaged over each SMAP 9 km cell, and the valid fraction of the cell was recorded. The cell was chosen as the averaging area because it is the grid on which the target is posted, so radar and target refer to the same named place. It is, however, much smaller than the area the radiometer senses (Section 3.3.5), so the radar and the target do not describe the same ground. Over 2016–2024, 846 scenes intersect the box (742 from Sentinel-1A and 104 from Sentinel-1B; 443 ascending and 403 descending), and none failed to read. The ascending pass is at about 11:28 UTC (18:28 Thai time) and the descending pass at about 23:08 UTC (06:08), close to SMAP's evening and morning overpasses. Sentinel-1B lost power on 23 December 2021, so scene counts fall from 2022. At cell (3,6), 559 dual-polarisation scene-days

Table 3.1: Datasets, acquisition routes, and measured completeness over 2016–2024 (3,288 days). Completeness was counted from the retrieved files, not taken from product documentation. For SMAP L3 and Sentinel-1 it is given at cell (3,6) unless stated.

Dataset	Variable	Resolution	Route	Days with data
Used for the SMAP L3 target				
SMAP L3 (SPL3SMP_E v006)	surface soil moisture, AM and PM, with quality and surface flags	9 km grid (native ∼36 km), daily; province scan	NASA Harmony	981 (29.8%) at recommended quality; 1,252 at any quality
Sentinel-1 RTC	VV and VH backscatter	10 m, read at ∼80 m; mean per 9 km cell	Microsoft Planetary Computer	846 scenes in the box; 559 dual-polarisation scene-days
ECMWF reanalysis (ERA5/ERA5-Land, Open-Meteo ERA5-Seamless)	rainfall, temperature, shortwave radiation, ET_0	daily (UTC), one grid point	Open-Meteo	3,288 (100%)
Earlier SMAP L4 target				
SMAP L4 (SPL4SMGP v008)	surface and root-zone soil moisture	9 km, 3-hourly; $\pm 0.15^\circ$ box	NASA Harmony	3,288 (100%)
ECMWF blend (Open-Meteo best match)	rainfall, temperature, ET_0	daily (UTC), one grid point	Open-Meteo	3,288 (100%)
Comparison only				
ECMWF blend (Open-Meteo best match)	soil moisture, 0–7 cm; tested on L4	hourly, averaged to daily	Open-Meteo	3,288 (100%)
MERRA-2	surface wetness (0–1)	$0.5^\circ \times 0.625^\circ$	NASA POWER	3,288 (100%)
MODIS NDVI (MOD13Q1)	vegetation index; tested on L4	250 m, 16-day	ORNL DAAC	16-day composites
MODIS LST (MOD11A2)	land-surface temperature; tested on L4	1 km, 8-day	ORNL DAAC	composite gaps to 80 d
Copernicus DEM	elevation	90 m	Open-Meteo	static

have at least half of the cell valid. All of them come from two relative orbits, 172 on the ascending pass (216 scene-days) and 62 on the descending pass (343), derived from the orbit numbers in the scene identifiers and checked against the catalogue, so within a pass direction the cell is always viewed from the same position.

Weather. Daily weather for the L3 work was taken from the Open-Meteo historical archive (`https://open-meteo.com/en/docs/historical-weather-api`) with the ERA5-Seamless model, which Open-Meteo describes as merging temperature and humidity from ERA5-Land (Muñoz-Sabater et al., 2021) with wind and solar radiation from ERA5 (Hersbach et al., 2020). The merged product was needed because Open-Meteo's ERA5-Land alone carries no rainfall or radiation. It is labelled in this report as ECMWF reanalysis (ERA5/ERA5-Land, Open-Meteo ERA5-Seamless). The variables are rainfall, 2 m mean temperature, shortwave radiation (MJ/m^2/day) and reference evapotranspiration ET_0, on UTC days; all 3,288 days are present. The grid point for cell (3,6) is at 14.10° N, 100.90° E. That point describes the temperature, on ERA5-Land's 0.1° grid; rainfall and radiation come from ERA5's 0.25° grid, on which cells (3,6) and (4,6) share one cell, so their rainfall and radiation series are identical. Of the robustness cells, only (2,6) tests the weather features with different rainfall. Reanalysis rainfall is a model estimate for an area of about 27 km and was not compared with a rain gauge; local storms can differ from it substantially. The earlier L4 pipeline requested Open-Meteo's default, which Open-Meteo calls Best Match and describes as combining ECMWF IFS HRES (9 km, from 2017), ERA5 (0.25°) and ERA5-Land (0.1°). The IFS is ECMWF's operational forecasting and analysis system, not a reanalysis, so this series, weather and soil moisture, is called the ECMWF blend (Open-Meteo best match) in this report. Which model supplied a given day was not recorded, and the IFS enters only from 2017, so the composition of the blend may change within the record.

Comparison products. The ECMWF blend soil moisture and MERRA-2 surface wetness, the latter from NASA POWER (`https://power.larc.nasa.gov`), are used for product intercomparison with SMAP, not for validation. The ECMWF blend soil moisture, MODIS NDVI and MODIS land-surface temperature were also tested as inputs on L4, and on L4 only. The two MODIS products (MOD13Q1 and MOD11A2) were obtained through the subsetting service of the ORNL DAAC (ORNL DAAC, 2017); the collection it served at the time of download was not recorded.

Absent and added datasets. Two datasets named in the proposal are absent. GLDAS has no working access route—it is not carried by AppEEARS, its GES DISC OPeNDAP service is retired, and its native files are global and 3-hourly—so MERRA-2 occupies its role as the land-surface-model comparator. ESA WorldCover was reachable but not joined: at a single cell it is a constant, and the land-cover information that mattered for the choice of cell, the urban class, came from SMAP's own surface flag. Two gaps of the earlier analysis are closed. Downward shortwave radiation is now included in the weather features, and Sentinel-1 is now used as a predictor.

3.4.1 Three absences that the assimilated product conceals

Three features of the observational record are invisible in SMAP L4, which has a value on all 3,288 days, and were found only by working with L3.

First, the gaps are set by the orbit. Cell (3,6) is overpassed on three of every eight days of SMAP's eight-day repeat: slots 0, 3 and 5 of the cycle before 28 September 2022, and slots 0, 3 and 6 after it. The retrieval days are therefore a fixed sampling pattern, not a random subset of the calendar. L4 fills the remaining days from the land-surface model.

Second, the satellite was in safe mode from 20 June to 22 July 2019 and from 7 August to 19 September 2022. L4 shows no discontinuity across either period, because the land-surface model continued to integrate.

Third, many of the retrievals that exist are ones SMAP itself does not recommend. Flag 13 marks a retrieval that was not successful, and in every cell of the scan 98.1–100% of flag-13 retrievals (99.86% on average) sit exactly at that cell's maximum value. The product documentation gives the upper limit of a valid retrieval as the soil's porosity (NSIDC, 2023); it does not say whether these values sit at the limit because the retrieval failed, or are marked unsuccessful because they reached it. Either way they are not measurements of the soil. The value this study earlier read as a saturation ceiling, 0.4634 m^3/m^3 on 28.6% of AM retrievals at the original point, is this value and not a measurement of saturated soil. At cell (3,6) the rejected retrievals are flag 13 (597) and flag 9 (107). Before filtering, the cell's maximum, 0.4692, is a flag-13 value; after filtering, the maximum is 0.4689, reached by a single retrieval. The quality rule removes these retrievals, at a cost: 77% of the rejected retrievals fall in May–October; the monthly counts peak in May (114) and from September to November (101, 166 and 106). They are also wetter occasions within the same months. In each of the nine months with at least ten accepted and ten rejected retrievals, the reanalysis rainfall of the preceding 3, 7 and 30 days was higher before rejected retrievals than before accepted ones, by 7, 13 and 28 mm on average over those months; and 178 of the 704 rejected retrievals, against none of the accepted, carry the surface flag for precipitation above 1 mm/h. The filtered record therefore under-represents the wettest conditions. Whether the soil was saturated on the rejected days cannot be known from the data used here, because their values are not measurements.

The three absences combine as follows at cell (3,6). Of 1,234 expected overpass days, 29 were lost to safe mode, 1 had no granule, 9 had a granule but no retrieval in the cell, and 258 carried only retrievals that are not recommended. That leaves 937 retrieval days on the expected pattern; a further 44 retrievals fall off the pattern where adjacent swaths overlap, for 981 retrieval days, 29.8% of the period.

3.5 Forecast Formulation

Let $\theta(t)$ denote surface soil moisture on day t. On L3, $\theta(t)$ is the recommended-quality morning retrieval on UTC day t, or the evening retrieval where the morning one is missing; on L4 it is the daily mean. At this longitude the evening pass (about 18:00 local time) falls near 11 UTC and the morning pass (about 06:00 local time on the following local day) near 23 UTC, so within one UTC

file date the morning value is the later observation of the two. For a horizon h, the task is to predict $\theta(t+h)$ from information available at the issue date t. Each horizon is trained as a separate model—direct multi-horizon forecasting—rather than by iterating a one-step model, which compounds error. All dates are UTC days, matching the dates of the SMAP files, and a forecast is taken to be issued at the end of UTC day t, after every input dated t exists. Neither SMAP nor the reanalysis is available that soon in practice: both are published some time after the day they describe, so an operational system would issue later, and the reanalysis stands in for weather observations that a forecaster would hold. Chan et al. (2018) list a latency of 50 hours for the enhanced Level 3 product; the delay was not measured in this study. With that latency a forecast from the retrieval history alone could be issued about two days after the retrieval, which would leave about 0, 1, 3 and 6 days of the 2-, 3-, 5- and 8-day horizons. The lead times are therefore counted from the date of the retrieval, not from the arrival of the data.

The morning retrieval is preferred because SMAP's retrieval assumes that the air, the vegetation and the near-surface soil are close to thermal equilibrium, which is most nearly the case near the 6 a.m. overpass (Entekhabi et al., 2010). That reasoning is the mission's and was not tested here. At this cell the evening retrievals are no noisier by two simple measures: taken alone, they have the smaller spread (standard deviation 0.046 against $0.048\,\mathrm{m^3/m^3}$) and the higher correlation between consecutive retrievals (0.735 against 0.654; `l3_extra_checks.py`). The evening retrieval fills the days without a morning one, so that the record is not thinned further. The cost of this choice is a series that mixes the two passes: the morning-first series is the least autocorrelated of the three target constructions in Table 4.3, and some pairs join a morning value to an evening one (Section 3.5.1). One robustness run uses the mean of the two passes instead (Section 3.8).

3.5.1 Choice of horizons

The proposal specified horizons of 1, 3 and 7 days. That design presumes a daily record, and on L3 it does not survive. In neither slot pattern of the eight-day repeat are two overpasses one or seven days apart, so exact 1- and 7-day pairs of retrievals do not arise from the regular pattern. At cell (3,6) they exist, 41 of each, because the cell lies where adjacent swaths overlap and 44 of its retrievals fall off the three-in-eight pattern; at the original cell there were none. Table 3.2 counts the pairs.

Forty-one pairs spread over nine years cannot support the evaluation protocol, which needs at least 120 training and 20 test samples in every walk-forward fold (Section 3.8). The orbit therefore rules out 1 and 7 days at this cell for lack of pairs, and it provides no 4-day pairs at all. The L3 benchmark runs at $h \in \{2, 3, 5, 8\}$, taken from the separations with enough retrieval pairs. Six-day pairs also exist, 326 of them, enough for the protocol; 6 days was not made a benchmark horizon, to keep the horizon set identical to the analysis presented to the advisor on 23 September 2026. This is a reason of consistency, not of data. The benchmark and its robustness runs score no 6-day forecast; a 6-day run made afterwards as a check is reported separately (Section 4.7.1). The 5- and 8-day results could not have stood in for a 6-day horizon: persistence error over all 6-day pairs ($0.0430\ \mathrm{m^3/m^3}$) is higher than over all 5- or 8-day pairs (0.0405 and 0.0394; Table 4.21). The horizons were chosen by the study on these grounds; the advisor approved SMAP L3 as the target on 23 September 2026 and the four horizons on 6 October 2026. A tolerance window of ± 1 day would make every nominal horizon available, but the realised lead time would then vary from sample to sample; exact separations were preferred. The separations are exact in file dates only: because the target takes the morning value

Table 3.2: Pairs of recommended-quality L3 retrievals separated by exactly h days at cell (3,6), 2016–2024, from 981 retrieval days. One- and seven-day pairs are too few for walk-forward evaluation, and no two retrievals are exactly four days apart. Six-day pairs are numerous enough but were not a benchmark horizon, to keep the horizon set identical to the analysis presented on 23 September 2026; 6 days was run afterwards as a check (Section 4.7.1).

Horizon h (days)	1	2	3	4	5	6	7	8
Retrieval pairs	41	333	601	0	598	326	41	872
Evaluated	no	yes	yes	no	yes	no	no	yes

where it exists and the evening value otherwise, 18–22% of test samples join a morning value to an evening one and are $h \pm 0.5$ days apart. The L4 benchmark, being gap-free, retains the proposal's 1, 3 and 7 days.

The horizons also differ in geometry. Pairs 8 days apart come from the same slot of the repeat cycle and so largely from the same track (four pairs in five; the rest join a morning and an evening value), whereas 2-, 3- and 5-day pairs join different tracks, which view the cell from different positions in the swath. Section 4.7 examines persistence error by separation and by pass and finds no consistent evidence that this difference affects it; its effect on the learned models' gains was not examined.

3.6 Feature Construction

Every input is dated on or before the issue date t. Because the target lies at least one day ahead on L4 and at least two days ahead on L3, no input overlaps the target day.

3.6.1 Calendar-space features for SMAP L4

For the gap-free L4 series the features are conventional: soil moisture at lags $0, 1, 2, 3, 7, 14, 30$ days; rainfall sums and mean temperature over 3, 7 and 30 day windows, from the ECMWF blend (Open-Meteo best match); and day-of-year encoded as $(\sin, \cos)$ so that 31 December and 1 January are adjacent rather than maximally distant. A fuller set adds reference evapotranspiration, MODIS NDVI and land-surface temperature taken only from composites that closed before the issue date, and the previous day's ECMWF blend soil moisture and L4 root-zone soil moisture.

Including the lag-zero value is essential rather than optional. Persistence predicts $\hat{\theta}(t+h) = \theta(t)$; a learned model denied $\theta(t)$ is solving an $(h{+}1)$-step problem while being scored against an h-step baseline. An earlier version of the pipeline contained precisely this error, and every skill score

computed before it was corrected is invalid.

3.6.2 Observation-space features for SMAP L3

Fixed calendar lags cannot be computed on a record with a value on 29.8% of days. Requiring six earlier lags simultaneously would discard the great majority of rows. Interpolating across the gaps is not an acceptable alternative: a linear fill depends on both endpoints, so a filled value at t embeds an observation made after t. This is not hypothetical—the MODIS LST feature was built that way in an earlier version and was found to embed observations up to 71 days ahead of their own date in 36% of rows, because monsoon cloud produced composite gaps as wide as 80 days—a known limitation of MODIS land-surface temperature under the region's persistent cloud cover (Ganepola et al., 2026).

The features are therefore defined over the sequence of retrievals rather than over the calendar:

- $\theta(t)$, the retrieval at issue time;
- $\theta_{-1}, \theta_{-2}, \theta_{-3}$, the three preceding retrievals, whenever they occurred;
- $\tau_{-1}, \tau_{-2}, \tau_{-3}$, the age in days of each of those retrievals, counted back from t;
- the season, as the day of year of t encoded as $(\sin, \cos)$.

The age variables are what make the construction work: they let the model distinguish a two-day-old antecedent from an eight-day-old one instead of treating them as equivalent. A sample exists only where a retrieval is present both at t and at $t + h$; persistence accordingly means carrying the last real observation forward, which is what a forecaster holding this product could do.

Three preceding retrievals were used. The number was fixed in advance and not tuned. With a median gap of three days between retrievals, the four values span about nine days, a little more than the longest horizon, and they include the three that the mean of the last three retrievals averages, so a learned model can reproduce that reference when the data offer nothing better. The season enters as the sine and cosine of the day of year, as on L4, so that the last day of December and the first day of January are neighbours.

3.6.3 Sentinel-1 features, paired backward only

For each issue date t, the Sentinel-1 features come from the latest dual-polarisation scene acquired on or before t (UTC) that covers at least half of the cell. The features are VV and VH in decibels, converted from the linear cell means, and the age of the scene in days. Decibels are the conventional scale for backscatter. The age is given so that a model can discount an old scene, as the ages of the retrievals let it discount an old retrieval. The half-cell threshold keeps out a mean taken from a strip at the edge of a scene; it was not tuned. Of the 981 retrieval days at cell (3,6), 978 have such a scene. Its median age is 2 days, 86% of days have one no more than 6 days old, and the oldest is 23 days.

The pairing is backward only by design. A two-sided window of $\pm N$ days around t would pair more retrieval days with a scene, but part of those scenes would be acquired after t. Such a scene does not exist when the forecast is issued, and a scene acquired between the issue date and the target date partly observes the state being forecast. Using such scenes would leak future information into the predictors and would tend to inflate the measured skill.

3.6.4 Weather features

The weather features are rainfall sums, and means of temperature and shortwave radiation, over 3, 7 and 30 days, trailing and including the issue day. They come from the ECMWF reanalysis (ERA5/ERA5-Land, Open-Meteo ERA5-Seamless) at the grid point of the chosen cell. The weather is gap-free, so ordinary rolling windows are used. The three windows describe the last few days, the last week and the last month, so that both a recent shower and a wet or dry month can enter; they were not tuned. ET_0 was retrieved but is not an L3 feature: it is computed largely from temperature and radiation, which are already features. Irrigation and field flooding, which change surface wetness independently of rain, are not represented by any input.

3.7 Models

3.7.1 Reference forecasts

Four reference forecasts are compared on L3. None uses any input other than the target series:

$$\hat{\theta}_{\text{pers}}(t+h) = \theta(t), \tag{Equation 3.1}$$

$$\hat{\theta}_{\text{mean3}}(t+h) = \tfrac{1}{3}\left[\theta(t) + \theta_{-1} + \theta_{-2}\right], \tag{Equation 3.2}$$

$$\hat{\theta}_{\text{damp}}(t+h) = b_0 + b_1\,\theta(t), \tag{Equation 3.3}$$

where $\hat{\theta}_{\text{pers}}$, $\hat{\theta}_{\text{mean3}}$ and $\hat{\theta}_{\text{damp}}$ are the persistence, three-retrieval-mean and damped-persistence forecasts for day $t+h$; $\theta(t)$ is the retrieval on the issue date t; θ_{-1} and θ_{-2} are the two retrievals preceding it, whenever they occurred; h is the horizon in days; and b_0 and b_1 are an intercept and a slope fitted by least squares on the training years for each horizon.

Persistence (Equation 3.1) carries the last retrieval forward and has no parameters. The **mean of the last three retrievals** (Equation 3.2) is also parameter-free; it smooths out short-lived variation between retrievals, whether retrieval noise or genuine change. The number three was carried over from the earlier run in which this smoother was first found to be competitive; means of two or four retrievals were not tried. A longer mean smooths more but reaches further into the past, where the soil may genuinely have changed. **Damped persistence** (Equation 3.3) relaxes the last retrieval towards the training-period mean when $0 < b_1 < 1$. **Seasonal climatology** predicts the mean of the training-year targets whose day of year lies within ± 7 days of the target date's day of year, and so encodes the annual cycle and nothing else. The ± 7-day window is this study's own choice, made in the study plan

to smooth a sparse record; it is not taken from a published convention. The validation guidance of Gruber et al. (2020) computes a soil-moisture climatology by averaging all years within a 4–8-week window around each day of year, and only when at least 5–10 years of data are available. The training record here grows from 2 to 8 years across the folds, and the window is narrower than that guidance. Section 4.6.5 repeats the comparisons under windows of that length and under other definitions. The L4 benchmark uses persistence and seasonal climatology as its reference forecasts.

3.7.2 Learned models

Three learned models are fitted on L3. **Ridge regression** (Hoerl & Kennard, 1970) is a linear model fitted by least squares with a penalty on the size of its weights:

$$\hat{\theta}_{\text{ridge}}(t+h) = w_0 + \sum_{j=1}^{J} w_j \, z_j(t), \tag{Equation 3.4}$$

$$(w_0, \ldots, w_J) = \arg\min_{w} \sum_{i=1}^{n_{\text{tr}}} \Big(\theta_i - w_0 - \sum_{j=1}^{J} w_j \, z_{ij}\Big)^2 + \lambda \sum_{j=1}^{J} w_j^2, \tag{Equation 3.5}$$

where $\hat{\theta}_{\text{ridge}}(t+h)$ is the ridge forecast for day $t+h$; $z_j(t)$ is the j-th input on the issue date t, standardised to zero mean and unit variance over the training samples; J is the number of inputs (9, 12, 18 or 21 in the four feature sets below); w_0 is the intercept and $w_1, \ldots, w_J$ are the weights; n_{tr} is the number of training pairs; θ_i and z_{ij} are the target and the standardised inputs of training pair i; and $\lambda \geq 0$ is the penalty strength (the parameter `alpha` of scikit-learn), which is tuned. With $\lambda = 0$ the model is ordinary least squares. A larger λ shrinks the weights towards zero and so pulls the forecast towards the mean of the training targets. **Random forest** (Breiman, 2001) averages 300 regression trees, each grown on a bootstrap resample of the training samples. Every feature is considered at every split, which is the scikit-learn default for regression (`max_features` = 1.0), so the trees differ through the resampling alone; the model is therefore a set of bagged regression trees, without the random feature subset at each split of Breiman's original method. **XGBoost** (T. Chen & Guestrin, 2016) grows 500 trees sequentially, each fitted to the residual of the ensemble so far and each built on a random subset of the features; the library's default penalty on the leaf weights is kept, and no penalty on the number of leaves is set. The numbers of trees, 300 and 500, were fixed and not tuned. In a random forest more trees only reduce the variation between one fit and another, which the seed check shows to be small here (Section 5.4); in XGBoost the number of trees trades off against the learning rate, and the learning rate is the one that is tuned (Section 3.8.3). The L4 benchmark uses the random forest and XGBoost only; ridge regression was not run on L4.

On L3 each learned model is fitted on four feature sets, which form an information-gain ladder:

- **L3**: the retrieval history and season (Section 3.6.2);
- **L3+S1**: with the Sentinel-1 features added;
- **L3+met**: with the weather features added;
- **L3+S1+met**: with both added.

The three models were chosen to answer three different questions on a few hundred training pairs. Ridge regression asks whether a tuned linear weighting of the recent retrievals does better than the fixed weights of the three-retrieval mean; with nine inputs and a penalty it has few effective parameters and is stable on small samples. The penalty matters because the inputs are correlated with one another: successive retrievals have a lag-1 autocorrelation of 0.676 (Table 4.3), and the 3-, 7- and 30-day weather windows overlap by construction. Weights fitted by least squares alone can then be unreliable (Hoerl & Kennard, 1970; Kang et al., 2018). An unpenalised linear fit is kept among the references as damped persistence (Equation 3.3). The random forest asks whether non-linear responses and interactions, for example a different behaviour when the soil is already wet, add anything; averaging many trees keeps its variance low with little tuning. XGBoost (T. Chen & Guestrin, 2016), the most flexible of the three, asks whether more aggressive fitting adds anything further; with the random forest it represents the tree ensembles that are the consistently strong conventional baseline in the studies of Table 2.1. Together they span a range of flexibility, so that a gain or its absence can be related to model capacity.

Their limits at this sample size are known in advance. A fold trains on 126 to 761 pairs (Table 3.3), few for the tree ensembles, which can fit the noise of a small sample, and XGBoost most of all. Trees cannot forecast beyond the range of the training targets, so the wettest states, which the quality rule removes, cannot be predicted by them. The ensembles depend on a random seed, and each was fitted with one (Section 5.4). Deep learning is deliberately excluded. The purpose of this study is a clean reference, and admitting a model whose behaviour cannot be audited at this sample size would compromise it.

3.8 Evaluation Protocol

3.8.1 Walk-forward evaluation

Random k-fold cross-validation is inadmissible here. On L4, adjacent days correlate at 0.987, so hiding a day while training on its neighbours allows the model to interpolate rather than forecast, and reports a skill that does not exist. On L3 the reason is more direct: every retrieval appears in several samples, as the target of one and as the issue value or a lagged value of others, so random folds would put the same retrieval on both sides of the split. Evaluation is instead expanding-window and walk-forward by calendar year: each fold trains on all years before a test year and is evaluated on that year alone, with the training window growing by one year per fold. Testing on whole later years also matches the use of a forecast, which is always issued from the past. The window expands rather than rolls because the record is short: a rolling window of a few years would leave the early folds with too few pairs to fit, and would discard data that the expanding window keeps. A rolling window would test whether the relation changes over time, for example after the change of orbit slots in September 2022 (Section 3.4.1); that was not examined. On L4 the expanding window gives eight folds, testing 2017 to 2024. On L3 a fold is used only if it has at least 120 training and at least 20 test samples. These are practical minimums, not tuned values: 20 test samples keep a year's RMSE from resting on a handful of pairs, and at cell (3,6) the threshold of 120 removes exactly the folds trained on 2016 alone and, at 2 days, also the fold trained on 2016–2017 (93 pairs), so every fold kept has at least two years of training data. This admits six folds at 2 days, testing 2019 to 2024, and seven at 3, 5 and 8

days, testing 2018 to 2024 (Table 3.3). The two products are therefore not scored on the same folds.

Table 3.3: Training and test samples in each walk-forward fold at cell (3,6), main run. A fold is named by its test year and trains on all earlier years; "–" marks a fold that falls below the minimum of 120 training or 20 test samples and is not used.

	2 days		**3 days**		**5 days**		**8 days**	
Test year	Train	Test	Train	Test	Train	Test	Train	Test
2017	–	–	–	–	–	–	–	–
2018	–	–	160	59	156	59	222	89
2019	126	35	219	68	215	69	311	103
2020	161	44	287	75	284	75	414	103
2021	205	31	362	64	359	64	517	95
2022	236	25	426	44	423	45	612	65
2023	261	30	470	55	468	54	677	84
2024	291	38	525	69	522	68	761	100
Total test		203		434		434		639

Within a horizon, every model is scored on identical samples. A sample is kept only if its target exists and every feature of the largest feature set is available; on L3 this includes a Sentinel-1 scene on or before the issue date. The same samples then serve every reference forecast and every learned model on every feature set, so differences in skill are not differences in the test data.

3.8.2 Main and robustness runs

The main L3 run uses cell (3,6), recommended-quality retrievals and the morning-first target of the forecast formulation above. The whole protocol, with the same horizons, feature sets and tests, was repeated at the two neighbouring non-urban cells (2,6) and (4,6), each with weather requested at its own centre, and at the original cell (4,2) with retrievals of every quality. The last run places the setting of the earlier, unfiltered analysis under the current protocol, so that the old setting and the new are scored in the same way; that comparison changes the cell and the retrieval quality together, and so cannot show which of the two matters. A further run at cell (3,6) replaced the target with the mean of the morning and evening retrievals (the mean of both where both exist, otherwise the one available), to test whether averaging the two passes alters the pattern of results. No run used an evening-only target.

Two further checks stand behind the comparison with the three-retrieval mean and are reported with it (Table 4.10). The *purged run* repeats the main run without the training pairs whose target falls in the test year. Training pairs are selected by the year of their issue date, so a pair issued in the last days of December has its target in early January of the test year; there are 0 to 3 such pairs among the 126 to 761 of a fold, and the purged run removes them (option `--purge` of `soil_moisture_l3_v3.py`). The *seed check* refits the random forest and XGBoost with 20 further random seeds, on the same folds and with the settings chosen in the main run (`l3_seed_sensitivity.py`); ridge regression and the reference forecasts have no random component. A third check, on how Sentinel-1 scenes that split a

cell are combined, is described in Section 5.4.

Four further checks were made after the benchmark was complete. They use the same data and protocol and do not enter the family of twelve comparisons. The two parameter-free forecasts and ridge regression were scored at each of the 20 non-urban cells of the scan, on the retrieval history alone (Section 4.3.3). The 6-day horizon was run at cell (3,6) (Section 4.7.1). Three variants of the main run were made: a wider penalty range for ridge regression, the training pairs and tuning choices of the main and purged runs exchanged for the random forest, and bootstrap blocks of half and double the length (Section 4.3). Last, a test built for nested models was computed on the saved forecasts of ridge regression (Section 3.8.5).

3.8.3 Tuning inside the fold

Hyperparameters are selected within each training window, and the last training year is held out for the selection: the configuration with the lowest RMSE on that year is chosen. On L3, if that year has fewer than 20 samples, or the earlier training years together have fewer than 40, the last 20% of the training window is held out instead. Neither condition arose at cells (3,6), (2,6) or (4,6); at the original cell it arose once, in the first 8-day fold. The chosen configuration is then refitted on the whole training window before the test year is touched. The random forest search covers maximum tree depth (unlimited, 10 or 20) and minimum leaf size (1, 5 or 10); the XGBoost search covers learning rate (0.03, 0.06 or 0.1), maximum depth (3, 5 or 7) and row subsampling (0.7 or 1.0); and the ridge search covers the penalty strength λ of Equation 3.5 (0.1, 1, 10 or 100). The random-forest and XGBoost grids are the same on L4 and L3, and the configuration chosen in every fold is recorded with the results. On L3 the depth limits of 10 and 20 rarely bind at these sample sizes, so several random-forest settings tie in validation error and the depth recorded among them is arbitrary; the effect on any skill score is below 0.001.

The ranges are small on purpose. Settings are selected on one held-out year of 25 to 113 pairs, and a large search on so few pairs would select noise. Each range therefore spans the values that matter most for overfitting on a small sample: for ridge regression the strength of the penalty, from light to strong on standardised inputs; for the random forest the minimum leaf size, which sets how finely the trees can follow single pairs; and for XGBoost the learning rate, the depth of each tree and the share of pairs each tree sees. The standardisation of ridge regression's inputs is fitted on the training data only, in selection on the part of the window used for fitting, and in the final fit on the whole window.

The record of choices shows where the ranges were too narrow. Across the 108 fits of each model (four horizons, six or seven folds, four feature sets), ridge regression chose the strongest penalty, 100, in 69; XGBoost chose its lowest learning rate in 54 and its greatest depth in 50; and the random forest chose its largest minimum leaf size in 43. On the L3 history alone, the configuration behind the comparison with the three-retrieval mean, ridge regression chose the strongest penalty in 4 of 6 folds at 2 days, 5 of 7 at 3 days, 3 of 7 at 5 days and 5 of 7 at 8 days. A choice at the edge of a range suggests that a wider range might have been chosen, so the scores are those of the ranges stated. For ridge regression a range widened to 10,000 was tried afterwards: a penalty of 1,000 was chosen in one fold at each of 2, 5 and 8 days, and no score improved (Section 4.3). For XGBoost, which was also run without early stopping and without tuning of its penalties, the scores are best read as a lower bound for that kind of model at this sample size.

3.8.4 Metrics

Four metrics are computed for every forecast, because a single number conceals distinct failures, and all four are kept in the result files. Three of them are

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2}, \tag{Equation 3.6}$$

$$\mathrm{bias} = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{\theta}_i - \theta_i\right), \tag{Equation 3.7}$$

$$\mathrm{ubRMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(e_i - \bar{e}\right)^2}, \tag{Equation 3.8}$$

where n is the number of test samples at one horizon; θ_i is the retrieval that arrived on the target date of test sample i, and $\hat{\theta}_i$ the forecast made for it; $e_i = \theta_i - \hat{\theta}_i$ is the error of that forecast; and $\bar{e}$ is the mean of the errors. All three are in $\mathrm{m^3/m^3}$, and $\mathrm{RMSE}^2 = \mathrm{bias}^2 + \mathrm{ubRMSE}^2$. The fourth metric is the Pearson correlation R between the forecasts and the retrievals. In this report R always denotes that correlation, and r the correlation between any other two series.

RMSE (Equation 3.6) gives the typical size of the error. Bias (Equation 3.7) reveals a systematic offset, positive when the forecasts are too wet. The unbiased RMSE (Equation 3.8) removes that offset; it is the conventional soil-moisture metric and the one in which SMAP's products are assessed at validation sites (Colliander et al., 2017). And R measures whether the forecasts move in phase with the retrievals, irrespective of scale. Chapter 4 reports the skill score with RMSE and ubRMSE, and R where it bears on the argument; bias is not tabulated.

The headline is not an error but a skill score, because an error means little until it is set against the error of a forecast that needs no model (Murphy, 1988):

$$S = 1 - \frac{\mathrm{RMSE}_{\mathrm{model}}}{\mathrm{RMSE}_{\mathrm{ref}}}, \tag{Equation 3.9}$$

where S is the skill score; $\mathrm{RMSE}_{\mathrm{model}}$ is the root-mean-square error of the model's forecasts over a set of test samples at one horizon; and $\mathrm{RMSE}_{\mathrm{ref}}$ is the root-mean-square error of the reference forecast over the same samples. $S > 0$ indicates an improvement over the reference, $S = 0$ a tie, and $S < 0$ a loss.

The two benchmarks form S over different sets of samples. On L3, S is computed once, over the test samples of all folds pooled together. The L4 benchmark (`soil_moisture_v5.py`) instead computes S within each of its eight folds and reports the mean of the eight per-fold values, and it averages RMSE, ubRMSE, bias and R over folds in the same way. Its bootstrap interval and Diebold–Mariano test, however, use the pooled test samples, so on L4 the point estimate and its interval describe different statistics. The L4 and L3 skill scores are therefore not computed identically, and this difference adds to the differences of product level, footprint, time sampling and location (Section 3.3) and of horizons (Section 3.5.1).

On both products the reference is persistence. On L3, skill is also computed against the mean of the

last three retrievals. In an earlier, unfiltered run at the urban cell that parameter-free smoother scored higher than the tuned random forest at three of the four horizons; scoring against it ensures that an improvement which smoothing alone would deliver is not credited to a learned model. Damped persistence and climatology are scored against persistence like any other forecast, but neither serves as the reference of a skill score in the main benchmark; the later re-scoring of Section 3.8.6 also uses climatology as a reference.

RMSE is additionally computed separately for issue dates in the wet season (May–October) and the dry season (November–April), and skill within each season is taken against persistence in the same season. No significance test is run within season, so the season split describes a pattern rather than a tested difference.

3.8.5 Significance

Because the error series are paired and autocorrelated, a smaller RMSE is not by itself evidence of superiority. Three tests are applied, in the same way in the L4 benchmark (`soil_moisture_v5.py`) and the L3 benchmark (`soil_moisture_l3_v3.py`); a fourth, for nested models, was added afterwards on L3 and is described at the end of this subsection.

The first is the Diebold–Mariano test (Diebold & Mariano, 1995), which asks whether two forecasts differ in squared error on the same samples:

$$d_i = e_{\text{ref},i}^2 - e_{\text{model},i}^2, \tag{Equation 3.10}$$

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{V}/n}} \sqrt{\frac{n+1-2h+h(h-1)/n}{n}}, \tag{Equation 3.11}$$

$$\hat{V} = \hat{\gamma}_0 + 2\sum_{k=1}^{q}\Big(1-\frac{k}{q+1}\Big)\hat{\gamma}_k, \tag{Equation 3.12}$$

where d_i is the squared-error differential on test sample i; $e_{\text{ref},i}$ and $e_{\text{model},i}$ are the errors of the reference forecast and of the model on that sample; $\bar{d}$ is the mean of d_i over the n test samples of the horizon h; $\hat{V}$ is the Newey–West estimate of the variance of d_i, which allows for its autocorrelation (Newey & West, 1987); $\hat{\gamma}_k$ is the autocovariance of d_i at a lag of k test samples; q is the bandwidth, the larger of $h-1$ and $\lfloor 4(n/100)^{2/9} \rfloor$; and the second square root in Equation 3.11 is the small-sample correction of Harvey et al. (1997). DM is positive when the model has the smaller squared error, and its p-value is two-sided, from a t distribution with $n-1$ degrees of freedom. The bandwidth q of Equation 3.12 grows with the sample size and is counted in test samples, not days: 4, 5, 5 and 7 samples at 2, 3, 5 and 8 days. The samples of a horizon are pooled in date order over the test years, with gaps between them, and the small-sample correction takes the horizon in days as if it were a number of steps. The code averages each autocovariance over the $n-k$ available products instead of dividing by n as the Newey–West estimator does; using n changes no p-value by more than 0.0012 and moves none across 0.05.

The second is the Wilcoxon signed-rank test (Wilcoxon, 1945). It is applied to the mean absolute error (MAE, the mean of $|e_i|$) of the model and of persistence in each fold, rather than to per-day errors, because the folds are far closer to independent than the days. With only six or seven folds on

L3, however, the test cannot return a p-value below 0.031 or 0.016 respectively, so it serves as a check that a gain is consistent across folds, not as strong evidence on its own. The need for the per-fold form is not theoretical. An earlier version of the L4 benchmark applied Wilcoxon to roughly 1,490 daily errors as though independent, and every p-value returned as 0.0000—an apparent certainty that was an artefact of the assumption.

The third is a moving-block bootstrap (Künsch, 1989), which gives a confidence interval on the skill score itself by resampling contiguous blocks, so that serial correlation is preserved. It uses 2,000 resamples, above the minimum of 1,000 that validation guidance asks for (Gruber et al., 2020). The blocks are 30 samples long at 2 and 3 days, 50 at 5 days and 80 at 8 days (ten times the horizon, at least 30), and the interval is the 2.5th to 97.5th percentile of the resampled skills. The lengths follow that fixed rule and were not tuned. A resample therefore draws only 7, 15, 9 and 8 blocks at 2, 3, 5 and 8 days, so the intervals are approximate. Their sensitivity to the block length was examined afterwards by halving and doubling the blocks (Section 4.3). The blocks do not wrap around the end of the series, so the first and last test years are drawn less often than the others, and a bound that lies very close to zero can depend on the bootstrap seed. On L3 the Diebold–Mariano test and the bootstrap interval are computed against the three-retrieval mean as well as against persistence.

The tests establish different things, and none establishes accuracy against the true soil moisture, because every error is measured against the retrieval. A skill score measures how much smaller a forecast's error is than that of a reference on the same samples. A Diebold–Mariano p-value says how often a difference at least that large would arise on these samples if the two forecasts were equally accurate; a small value does not show that the difference would recur at another cell or in other years. A bootstrap interval gives the range of skill consistent with resampling the test record. The per-fold Wilcoxon test asks whether a gain appears year after year. Because it works on MAE while the skill score works on squared error, the two can disagree: a forecast that removes a few large errors lowers RMSE in every year without always lowering MAE. A result that fails to reach significance shows that no difference could be measured on these samples, not that none exists; the width of its interval shows how large a difference could have gone unseen.

Every test compares one forecast with a reference forecast. The learned models nest the references: ridge regression on the L3 history could reproduce persistence or the three-retrieval mean by its choice of weights. The Diebold–Mariano test is read here as a comparison of the forecasts as they were produced, with their estimated parameters, not of the underlying models. Between nested models such a comparison tends to favour the smaller one, because the larger carries the noise of estimating parameters it may not need (Clark & West, 2007). A gain that passes the test is therefore, if anything, understated, and a finding of no measurable gain is weaker evidence of no gain than it would be between forecasts that are not nested. No test compares two learned models, so differences between them, including those between ridge regression and the random forest, are untested. Many forecasts are compared, so some small p-values are expected by chance. For the question on which the benchmark rests, whether learning adds to the three-retrieval mean, the twelve comparisons of Table 4.9 (three models at four horizons, on the L3 history) are treated as one family, and their p-values are also given after the Holm adjustment (Holm, 1979), which controls the chance of any false positive in the family:

$$\tilde{p}_{(j)} = \max_{k \le j} \min\Big\{1,\ (m-k+1)\, p_{(k)}\Big\}, \qquad \text{(Equation 3.13)}$$

where $p_{(1)} \le \cdots \le p_{(m)}$ are the unadjusted p-values of the family in ascending order; m is the number of comparisons in the family, twelve here; and $\tilde{p}_{(j)}$ is the adjusted value of the j-th smallest.

The smallest p-value is thus multiplied by twelve, the next by eleven and so on, and no adjusted value may be smaller than the one before it. A result is called significant only if it survives that adjustment, that is, if its adjusted value is below 0.05; one that does not is called suggestive, and a p-value below 0.05 that was not adjusted is reported as unadjusted, not as significant. The rule covers these twelve comparisons. Outside them, in the robustness runs, the comparisons with persistence and climatology and the L4 benchmark, a p-value is quoted as it stands, unadjusted, and the word significant is not used for it, unless an adjustment is stated. The twelve-member family is this study's primary question, but it was fixed after the p-values were known (Section 3.1). The verdict depends on that choice: had all 48 comparisons with the three-retrieval mean on the four feature sets of Section 3.7.2 been treated as one family, ridge regression's 8-day gain would have an adjusted p of about 0.10 (0.095 in the purged run), and only the random forest's 8-day gain in the main run (adjusted 0.033) would survive. With the three 6-day comparisons of the later check (Section 4.7.1) the family would hold fifteen: ridge regression's 8-day gain would still survive (adjusted 0.029) and its 6-day gain would not (adjusted 0.056). With the wider penalty range tried afterwards, ridge regression's 8-day gain would have an adjusted p of 0.051 (Section 4.3). The verdict for that gain therefore holds for the benchmark as specified and sits close to the threshold. Where the report picks the best of the 12 learned configurations (three models on four feature sets) at a horizon, its p-value is not adjusted for that selection.

A test built for nested models (Clark & West, 2007) was computed afterwards as a check, for ridge regression on the L3 history against the three-retrieval mean at the four horizons, on the forecasts the benchmark saved (`l3_nested_test.py`). It removes from the squared error of the larger model the part that estimating unneeded weights is expected to add,

$$f_i = e^2_{\text{mean3},i} - \left[e^2_{\text{ridge},i} - \left(\hat{\theta}_{\text{mean3},i} - \hat{\theta}_{\text{ridge},i} \right)^2 \right], \qquad \text{(Equation 3.14)}$$

where f_i is the adjusted squared-error differential on test sample i; $e_{\text{mean3},i}$ and $e_{\text{ridge},i}$ are the errors of the three-retrieval mean and of ridge regression on that sample, each the retrieval that arrived minus the forecast; and $\hat{\theta}_{\text{mean3},i}$ and $\hat{\theta}_{\text{ridge},i}$ are the two forecasts. The statistic is the mean of f_i over its standard error, computed as the Diebold–Mariano statistic of Equation 3.11 with f_i in place of the d_i of Equation 3.10. It is read one-sided, as the test's authors prescribe: a value above 1.645 rejects at the 5% level. Because $f_i = 2\, e_{\text{mean3},i}\, (\hat{\theta}_{\text{ridge},i} - \hat{\theta}_{\text{mean3},i})$, the test asks whether ridge regression departs from the three-retrieval mean in the direction of that forecast's error. Its null hypothesis is that the three-retrieval mean is the correct model, so that its error cannot be predicted from the retrieval history. It therefore asks whether the history holds information that the mean does not use, not whether the ridge forecast as produced is the more accurate; that question stays with the Diebold–Mariano test. The test's authors argue for normal critical values for models fitted by least squares; ridge regression with a penalty chosen in each fold lies outside those conditions, so the p-values are a guide. The test was added after the benchmark's results were known. Which comparisons to make with it was fixed before its values were seen; they stand outside the family of twelve and are given with a Holm adjustment over the four.

The change in skill when an input group is added (Section 3.9) is given a 95% interval by the same moving-block bootstrap, applied to both feature sets with the same resampled blocks, so that the interval is for the paired difference (`l3_real_data_evaluation.py`). No Diebold–Mariano test is applied to it, and none to XGBoost's changes.

The earlier, unfiltered L3 run did not apply this protocol: its Diebold–Mariano variance used too few

lags, and it ran no Wilcoxon test. Its results are superseded.

3.8.6 Additional yardsticks

After the main benchmark, the saved day-by-day forecasts of the main run were re-scored with yardsticks common in weather forecasting and hydrology, to check whether the conclusions depend on the choice of persistence as the reference and of RMSE as the metric (`l3_climatology_evaluation.py`). No model was refitted. Every climatology, fitted fraction and category threshold below is computed from the training years of each fold only.

Departures from normal are measured against a training-year climatology,

$$a(t) = \theta(t) - c(t), \qquad \text{(Equation 3.15)}$$

where $a(t)$ is the anomaly, the departure from normal, of the retrieval on date t; $\theta(t)$ is that retrieval; and $c(t)$ is the mean of all recommended-quality retrievals in the training years whose day of year lies within ± 7 days of the day of year of t.

Two reference forecasts in the tradition of soil-moisture memory studies (Delworth & Manabe, 1988) forecast the normal for the target date plus a fraction of the present departure:

$$\hat{\theta}_{\mathrm{DAP}}(t+h) = c(t+h) + \rho_h \, a(t), \qquad \text{(Equation 3.16)}$$

$$\hat{\theta}_{\mathrm{DAM3}}(t+h) = c(t+h) + \rho_h^{(3)} \, \tfrac{1}{3} \left[a(t) + a_{-1} + a_{-2} \right], \qquad \text{(Equation 3.17)}$$

where $\hat{\theta}_{\mathrm{DAP}}$ is damped anomaly persistence and $\hat{\theta}_{\mathrm{DAM3}}$ the damped anomaly of the three-retrieval mean, both for day $t+h$; $c(t+h)$ is the climatology of Equation 3.15 on the target date; a_{-1} and a_{-2} are the anomalies of the two retrievals preceding the issue date, each against the climatology of its own date; and ρ_h and $\rho_h^{(3)}$ are fractions fitted by least squares through the origin on the training-year pairs of retrievals exactly h days apart. With $\rho_h = 1$, Equation 3.16 becomes anomaly persistence, which carries the present departure over unchanged. A linear combination of climatology and persistence has long been proposed as a standard of reference for skill scores (Murphy, 1992), and Equation 3.16 is such a combination written in anomalies. Equation 3.17 applies it to the three-retrieval mean, combining the two simple ideas of Section 3.7, smoothing and damping. It was defined after the main results were known, so it is a post hoc diagnostic benchmark and not a planned reference, and it is not proposed as a forecasting model. Its purpose is to show how much a very simple seasonal adjustment of recent retrievals already achieves, and so how hard it is for a learned model to demonstrate information beyond it. Its climatology c, built from all training-year retrievals, differs from the reference climatology above, which uses only the training targets; both are compared in Section 4.6.5.

Skill against climatology is Equation 3.9 with the seasonal climatology of Section 3.7 as the reference, the usual reference of weather and seasonal forecasting (Murphy, 1988). The anomaly correlation asks whether a forecast catches departures from normal:

$$\mathrm{ACC} = \mathrm{corr}\Big(\hat{\theta}(t+h) - c(t+h),\ \theta(t+h) - c(t+h) \Big), \qquad \text{(Equation 3.18)}$$

where ACC is the anomaly correlation; corr is the Pearson correlation over the pooled test samples of

one horizon; $\hat{\theta}(t+h)$ is the forecast; $\theta(t+h)$ is the verifying retrieval; and $c(t+h)$ is as in Equation 3.15. Climatology itself scores close to zero.

The Kling–Gupta efficiency, widely used in hydrology (Gupta et al., 2009), is

$$\mathrm{KGE} = 1 - \sqrt{(R-1)^2 + (\alpha-1)^2 + (\beta-1)^2}, \qquad \text{(Equation 3.19)}$$

where R is the correlation between forecasts and verifying retrievals, as in Section 3.8.4; $\alpha = \sigma_f/\sigma_o$ is the amplitude ratio, the standard deviation of the forecasts divided by that of the retrievals; and $\beta = \mu_f/\mu_o$ is the ratio of their means. Unlike Equation 3.9, KGE penalises a forecast that varies less than the observations.

Finally, each anomaly is placed in a dry, normal or wet category by the 33rd and 67th percentiles of the training-year anomalies, and each forecast anomaly by the same thresholds. The Heidke skill score is

$$\mathrm{HSS} = \frac{N_c - E}{N - E}, \qquad E = \frac{1}{N}\sum_{k=1}^{3} n_k^o \, n_k^f, \qquad \text{(Equation 3.20)}$$

where HSS is the Heidke skill score; N is the number of forecasts; N_c the number in the correct category; E the number expected to be correct by chance; and n_k^o and n_k^f the numbers of observations and of forecasts in category k. $\mathrm{HSS} = 0$ for chance and 1 for a perfect forecast. For the wet category, the hit rate is the share of observed wet cases that were forecast wet, and the false-alarm ratio the share of wet forecasts that were not wet.

The Diebold–Mariano test and the moving-block bootstrap of Section 3.8 are applied to skill against climatology and against Equation 3.17, and the bootstrap also to the anomaly correlation. In the sensitivity analysis of Section 4.6.5 only the Diebold–Mariano test is used. KGE and the category scores are not tested.

3.9 Interpretation

The question of which inputs matter is asked differently for the two targets.

On L3 it is answered by the information-gain ladder of Section 3.7. Each learned model is fitted on the L3 history alone, then with Sentinel-1, with weather, and with both, on identical samples and folds. Every rung is scored against persistence and against the three-retrieval mean with the tests of Section 3.8, and the information gain of an input group is the change in skill that adding it produces, given with the paired interval of Section 3.8. The upper end of that interval is the largest gain that the data still allow, and is reported with each null result. The ladder replaces the group permutation ablation of the earlier, unfiltered run, which ranked the datasets on a single train–test split at the urban cell. A ladder was preferred to an attribution method on L3 because the question is whether an input improves forecasts of dates the model has not seen. An attribution describes what a fitted model uses, whether or not that use makes its forecasts better; a change in skill on the test years measures the improvement directly.

On L4, SHAP (Lundberg & Lee, 2017) attributes each individual prediction among the features via

Shapley values, giving both a global ranking and local attribution. It was computed only at 7 days, for the random forest on the full feature set, on the 359 held-out rows of the last fold (test year 2024). It was run once, with the L4 benchmark, to describe what that model relied on. XGBoost, the best learned model on L4, was not analysed, and SHAP was not computed for L3, whose inputs are assessed by the ladder instead.

3.10 Data Pipeline and Reproducibility

The pipeline is scripted end to end (Figure 3.4), and `code/README.md` serves as its runbook, giving the command and output of every step. Three acquisition lanes serve the L3 analysis. `download_smap_l3_province.py` submits one Harmony request per month for the scan box, streams each returned granule, extracts every cell with its flags, and deletes the file. `extract_sentinel1_rtc.py` reads only the box window of each Sentinel-1 RTC scene and averages it over each 9 km cell. `download_weather_point.py` fetches the ERA5-Seamless weather for a chosen cell, naming the model in the request. A fourth lane is the earlier L4 route (`download_smap_harmony.py`, `download_datasets.py` and `merge_smap.py`), which feeds `soil_moisture_v5.py` and `l4_reassessment.py`.

Three steps follow. `l3_province_pixels.py` applies the quality rule, marks the urban cells and tabulates, for every cell, its share inside the province and its number of recommended-quality days; the cell of Section 3.3 was chosen from that table by the criteria stated there. The script's own candidate filter, which also requires 90% of a cell's retrievals to be recommended, admits no cell and was not used. `l3_pixel_profile.py` profiles the chosen cell: its coverage, gaps, exact-gap pairs, orbit breakdown and Sentinel-1 pairing. `soil_moisture_l3_v3.py` runs the reference forecasts, the learned models and the information-gain ladder at 2, 3, 5 and 8 days, for the main and the robustness runs.

Streaming keeps disk use small: the nine-year L4 record—some 26,000 files—never occupies disk, and no whole Sentinel-1 scene is downloaded. Every acquisition step is resumable. A SMAP month counts as complete only when nothing in it is missing: every day present for L4, and every returned granule read for L3. An earlier rule that accepted any data for a month silently left a 30-day hole in December 2019 while reporting success. Sentinel-1 scenes already read are skipped on a re-run. All code, intermediate tables and result files are kept with the project repository, which is private; they are available from the author on request. The benchmark was run in Python with scikit-learn, XGBoost, NumPy, pandas and SciPy (versions 3.14, 1.9, 3.4, 2.5, 3.0 and 1.18 at the time of writing). Every random number generator was seeded with 0, the settings chosen in every fold are saved beside the results (`results_l3_v3_*_hyperparams.csv`), and refitting ridge regression from them reproduces the saved forecasts exactly; refitting the random forest with seed 0 reproduces them to within 10^{-15} and XGBoost to within 2×10^{-8} (`l3_seed_sensitivity.py`). The Diebold–Mariano statistics are not stored with the results; they can be recomputed from the saved day-by-day forecasts of the main and purged runs (`results_l3_v3_main_px3_6_pred.csv` and its `_purge` counterpart). The later checks are produced by `l3_extra_checks.py`, `l3_nested_test.py`, `l3_spatial_references.py`, `l3_footprint_overlap.py` and `l3_scan_map.py`, and by the benchmark script run with `--horizons6`. The check that combined two Sentinel-1 frames (Section 5.4) was run outside this pipeline.

Use of generative AI. Generative AI was used to assist with coding and writing in this study. The author is responsible for its content.

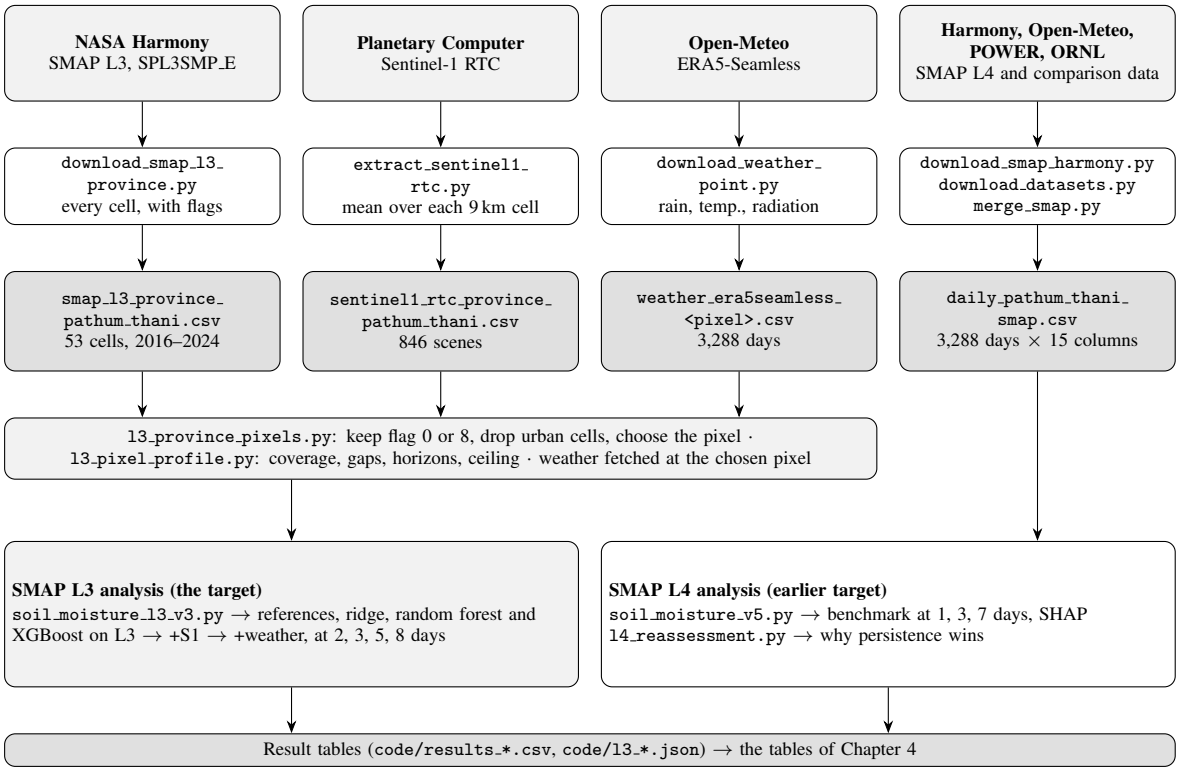


Figure 3.4: Data pipeline. Each lane runs from an online service through a script to a raw table. The three SMAP L3 lanes meet at the quality filter and pixel choice, which also fixes where the weather is fetched, and feed the L3 benchmark. The fourth lane builds the merged daily table for the earlier SMAP L4 benchmark.

3.11 Summary of Design Decisions and Settings

Table 3.4 gathers the design decisions of this chapter in one place: what was chosen, with the setting used, why it was chosen, and the section that gives the detail. It is meant for a reader who wishes to check a decision or to repeat the experiment, and it adds nothing that the sections above do not state. The decisions fall into four groups: what is forecast (the first seven rows), what the forecasts are made from (the next three), which forecasters are compared (the next two), and how they are judged (the last six).

Table 3.4: Design decisions of the SMAP L3 benchmark, the setting used for each, and the reason. λ is the penalty strength of Equation 3.5; S is the skill score of Equation 3.9.

Decision	Choice and setting	Reason	Section
Target product	SMAP enhanced Level 3 radiometer soil moisture, SPL3SMP_E version 006, on the 9 km grid	It is the satellite retrieval itself; Level 4 is a model output on which persistence could not be beaten. The 9 km posting lets cells be matched to the province and to SMAP's flags	3.3.1
Quality rule	Retrievals with quality flag 0 or 8 only	The product gives no assurance for the others, and SMAP's own validation uses the same flag	3.3.2
Cell	(3,6), with cells (2,6) and (4,6) as checks	Urban-flagged cells carry no recommended retrieval. Of the two non-urban cells at least 90% inside the province it has the more recommended days, 981 against 946	3.3.4
Period	2016–2024, nine whole years	2016 is SMAP's first complete year; the folds and the seasonal normal need whole years	3.3
Target value	The morning retrieval; the evening one where the morning one is missing	The retrieval's assumptions hold best near 6 a.m.; the evening value keeps the record from thinning further	3.5
Horizons	2, 3, 5 and 8 days, exact separations in file dates	The separations with enough retrieval pairs: 1 and 7 days have 41 pairs each and 4 days none. Six days was left out to match the earlier analysis and run later as a check	3.5.1
Forecast form	A separate model for each horizon	Iterating a one-step model compounds its error	3.5
Inputs from the target	The issue-day retrieval, the three before it, their ages, and the sine and cosine of the day of year (9 inputs)	Calendar lags cannot be formed on a record with a value on 29.8% of days, and filling the gaps would use later observations. The ages let a model discount an old value	3.6.2
Sentinel-1 inputs	VV and VH in decibels from the latest scene on or before the issue date that covers at least half of the cell, and its age (3 inputs)	A scene acquired after the issue date does not exist when the forecast is made	3.6.3
Weather inputs	Rainfall sums, and means of temperature and shortwave radiation, over 3, 7 and 30 days ending on the issue date (9 inputs)	The last few days, the last week and the last month; the windows were not tuned	3.6.4
Reference forecasts	Persistence; seasonal climatology with a $\pm$7-day window; the mean of the last three retrievals; damped persistence	Skill has meaning only against a reference. The mean of three is a second reference so that a learned model is not credited with what averaging alone supplies	3.7.1

continued on next page

Table 3.4 (continued)

Decision	Choice and setting	Reason	Section
Learned models	Ridge regression; random forest, 300 trees; XGBoost, 500 trees; each on four feature sets	A tuned linear weighting, then non-linear responses, then more aggressive fitting: a range of flexibility on a few hundred pairs. Deep learning is left to the thesis	3.7.2
Evaluation design	Expanding window, walk-forward by calendar year. A fold needs at least 120 training and 20 test samples: 6 folds at 2 days, 7 at 3, 5 and 8 days	Random folds would put the same retrieval on both sides of the split, and a forecast is always issued from the past. The record is too short for a rolling window	3.8.1
Tuning	Grid search scored on the last training year, then a refit on all training years. Ridge: λ of 0.1, 1, 10 or 100. Random forest: depth unlimited, 10 or 20; minimum leaf 1, 5 or 10. XGBoost: learning rate 0.03, 0.06 or 0.1; depth 3, 5 or 7; row subsample 0.7 or 1.0	The test year is never used for a choice. A large search on 25 to 113 held-out pairs would select noise	3.8.3
Metrics	RMSE, bias, ubRMSE and R. Headline: skill S against persistence and against the mean of three, over the pooled test samples	An error means little without a reference. Four metrics, because one number hides distinct failures	3.8.4
Significance	Diebold–Mariano test with a Newey–West variance and a small-sample correction; moving-block bootstrap, 2,000 resamples; Wilcoxon test on per-fold MAE; Holm adjustment over twelve comparisons	The errors of two forecasts are paired and autocorrelated, and many comparisons are made	3.8.5
Information gain	The change in S when Sentinel-1, weather or both are added, on identical samples and folds, with a paired bootstrap interval	A change in skill on the test years measures directly whether an input helps	3.9
Reproducibility	Every random seed 0; the chosen settings and the day-by-day forecasts saved; Python 3.14, scikit-learn 1.9, XGBoost 3.4	So that every table can be regenerated from the scripts	3.10

Chapter 4

Results

This chapter reports, in order, the quality and coverage of the data (Section 4.1), the benchmark on SMAP L4 (Section 4.2), the benchmark on SMAP L3 at cell (3,6), with its split by season, its repetition at three other cells and with a second target definition, and a check at all 20 non-urban cells (Section 4.3), the information that Sentinel-1 and weather add to the L3 history (Section 4.4), the behaviour of the forecasts on the real record (Section 4.5), the same forecasts under other yardsticks (Section 4.6), the dependence of persistence error on horizon and orbit (Section 4.7), the effect of the quality rule on the former saturation ceiling (Section 4.8), the forecast errors set against SMAP's accuracy requirement (Section 4.9), and the product intercomparison (Section 4.10). Every count was taken from the retrieved files. Unless stated otherwise, L3 results use recommended-quality retrievals only (`retrieval_qual_flag` 0 or 8).

4.1 Data and Retrieval Quality

4.1.1 The province scan

The scan box holds 56 grid cells, of which 53 returned at least one retrieval over 2016–2024. Table 4.1 summarises the quality of the 124,570 SMAP L3 retrievals in those 53 cells, under the quality rule of Section 3.3. Area shares refer to the province outline of 1,514 km^2.

Recommended retrievals are a minority of the record: 27.2% overall, and 16.5% in cells at least half inside the province. What decides where they exist is SMAP's own urban class. The 33 urban-flagged cells carry no recommended retrieval in nine years, and 76.2% of the province's area lies in them; a further 1.9% lies in cell (5,3), which never returned a retrieval, so the non-urban cells cover only 21.9% of the province. Of the 12 cells at least 90% inside the province, only 2 are not urban. This absence recurs in the same places every time, unlike the gaps from orbital sampling and outages discussed below. It is the first answer to the fourth research question: where the baselines can be evaluated at all is decided by the quality of the retrievals, before any model is fitted.

Pathum Thani has seven districts, and the scan box covers all of them in full. Table 4.2 shows that the usable record does not. Recommended-quality retrievals exist over parts of four districts, all on the province's eastern and western edges: 55% of Nong Suea, 28% of Thanyaburi, 26% of Lam Luk Ka and 17% of Lat Lum Kaeo by area. Khlong Luang, Sam Khok and Mueang Pathum Thani lie wholly in urban-flagged cells or in the one cell that returned no retrieval, and have none. The quality-filtered benchmark and its two robustness cells lie in three adjoining eastern districts, Thanyaburi, Nong Suea and Lam Luk Ka. In the other four districts the only run of the full protocol is the contrast at the original urban cell (4,2), which uses retrievals that are not of recommended quality; the later check at the 20 non-urban cells (Section 4.3.3) also scores two western cells that reach into Lat Lum Kaeo.

The cell of the original study point, (4,2), is one of the urban cells, and every retrieval in it carries flag

Table 4.1: Quality of SMAP L3 retrievals in the province scan, 53 cells, 2016–2024. Recommended quality is `retrieval_qual_flag` 0 or 8.

Item	Value
Retrievals by quality flag	
Flag 8: recommended (only the freeze/thaw retrieval failed)	33,822
Flag 9: not recommended	60,546
Flag 13: not recommended, retrieval not successful	30,202
All retrievals	124,570
Recommended, all cells	27.2%
Recommended, cells at least 50% inside the province (42,006 retrievals)	16.5%
Cells	
Grid cells in the box; cells with at least one retrieval	56; 53
Cells flagged urban (surface flag bit 3)	33 of 53
Recommended share in the urban cells	0%
Share of the province's area (1,514 km^2) in urban cells	76.2%
Share in cell (5,3), which never returned a retrieval	1.9%
Share in non-urban cells	21.9%
Cells at least 90% inside the province	12 (10 urban, 2 not)
Cell (4,2), containing the original study point	urban; every retrieval flag 9 or 13

Table 4.2: The seven districts of Pathum Thani and the SMAP L3 scan, 2016–2024: district area, and the share of that area in cells flagged urban (no recommended-quality retrieval), in non-urban cells (recommended-quality retrievals exist) and in the cell that returned no retrieval. District boundaries: geoBoundaries ADM2.

District	Area (km^2)	Urban	Non-urban	No retrieval	Use in this study
Nong Suea	340	45%	55%	0%	robustness cell (2,6); part of main cell
Thanyaburi	114	70%	28%	2%	centre of main cell (3,6)
Lam Luk Ka	304	70%	26%	4%	robustness cell (4,6)
Lat Lum Kaeo	192	83%	17%	0%	—
Mueang Pathum Thani	142	89%	0%	11%	original point, cell (4,2)
Khlong Luang	305	100%	0%	0%	—
Sam Khok	120	100%	0%	0%	—

9 or 13; its recommended share is zero. The earlier L3 results at that point, from an earlier unfiltered run at the urban cell, therefore rest entirely on retrievals that SMAP does not recommend, and are superseded by the results of Section 4.3. The same protocol, applied to all retrievals at that cell, is reported in Section 4.3.2.

4.1.2 The chosen cell

Table 4.3 profiles the recommended-quality record at cell (3,6), and Table 4.4 traces how its 981 retrieval days arise from the orbit.

The record at cell (3,6) holds 981 retrieval days, 29.8% of the period. Gaps between consecutive retrieval days have a median of 3 days and a mean of 3.35, with a maximum of 99; most are of 3 days (522) or 2 days (333), as the eight-day repeat dictates. These gap counts refer to consecutive retrievals only. The pair counts of Table 3.2 and Table 4.21 count every pair of retrievals exactly h days apart, consecutive or not, and are larger. In the terms of Ganepola et al. (2026), the gaps from orbital sampling and instrument outages are *random* gaps, in the sense that they arise because the satellite did not pass or did not operate, not from the state of the soil or the land surface; the orbital gaps nonetheless follow a fixed calendar pattern (Section 3.4.1). The gaps added by the quality rule are not random in this sense: 77% of the rejected retrievals fall in May–October (Section 4.8).

The morning and evening retrievals of the same file date, on the 691 days that have both, correlate at 0.739 (the morning pass of a UTC file date follows its evening pass by about 12 hours); the morning value is higher by 0.0067 on average, with a standard deviation of $0.0332\,\mathrm{m}^3/\mathrm{m}^3$. The target series is the least autocorrelated of the three constructions examined: 0.676 between consecutive retrievals, against 0.735 for the evening retrievals alone and 0.745 for the mean of the two passes.

The quality rule is by far the largest loss. It removes 258 of the 1,234 expected overpass days, against 29 lost to safe mode, 1 with no granule and 9 with no retrieval in the cell. The 44 off-pattern retrievals are what make exact 1- and 7-day pairs possible at this cell (Section 4.7).

Table 4.3: Data profile of cell (3,6), 2016–2024, recommended quality. The target takes the morning retrieval where it exists and the evening retrieval otherwise.

Item	Value
Retrieval days	981 of 3,288 (29.8%)
Days with a morning / evening / both retrievals	806 / 866 / 691
Days with any retrieval, any quality	1,252
Retrievals recommended	70.4% of 2,376
Retrievals rejected	flag 13: 597; flag 9: 107
Gap between consecutive retrieval days	median 3, mean 3.35, maximum 99 days
Number of gaps of 1 / 2 / 3 / 5 / 6 / 8 days	41 / 333 / 522 / 27 / 21 / 13
Lag-1 autocorrelation, consecutive retrievals	0.676 (target); 0.735 (evening only); 0.745 (mean of the two passes)
Morning against evening, 691 days with both	$r = 0.739$; morning minus evening: mean $+0.0067$, s.d. $0.0332\,\mathrm{m^3/m^3}$
Maximum value	0.4689, reached by one retrieval (0.1%); before filtering 0.4692, a flag-13 value
Median vegetation water content	$2.46\,\mathrm{kg/m^2}$
Static water-body fraction	0.002

Table 4.4: How the 981 retrieval days at cell (3,6) arise. The cell is overpassed on slots 0, 3 and 5 of the eight-day repeat before 28 September 2022, and on slots 0, 3 and 6 after it.

Step	**Days**
Expected overpass days (3 of every 8)	1,234
lost to safe mode (2019-06-20 to 07-22; 2022-08-07 to 09-19)	-29
no granule at all	-1
granule, but no retrieval in the cell	-9
retrieval not recommended (flag 9 or 13)	-258
Retrieved on expected days	937
retrieved off the pattern, where adjacent swaths overlap	$+44$
Retrieval days	981

4.1.3 Sentinel-1 coverage

All 846 Sentinel-1 RTC scenes that intersect the scan box over 2016–2024 were read, and none failed: 742 from Sentinel-1A and 104 from Sentinel-1B, 443 on ascending and 403 on descending passes. The ascending pass is at about 11:28 UTC (18:28 Thai time) and the descending pass at about 23:08 UTC (06:08), close to SMAP's evening and morning overpasses. At cell (3,6), 559 dual-polarisation scene-days have at least half of the cell valid. Table 4.5 gives the counts by year.

Table 4.5: Sentinel-1 RTC coverage by year. Scenes in the scan box, and dual-polarisation scene-days covering at least half of cell (3,6).

Year	Scenes in the box	Scene-days at cell (3,6)
2016	70	29
2017	103	53
2018	113	77
2019	106	79
2020	110	81
2021	112	83
2022	86	57
2023	79	54
2024	67	46
Total	846	559

Coverage is not stationary. Sentinel-1B lost power on 23 December 2021, so the counts fall from 2022: at the cell, from 83 scene-days in 2021 to 57, 54 and 46 in 2022–2024. The radar input is therefore sparser in the last three test years than in the years before them.

Under backward-only pairing (Section 3.6), 978 of the 981 retrieval days have a scene on or before the issue date. The scene is from the issue day itself on 213 days, at most 1 day old on 394, at most 3 days old on 584, at most 6 days old on 840 (86%) and at most 12 days old on 969. The median age is 2 days and the maximum 23 days.

4.1.4 Weather

The weather for cell (3,6), from the ECMWF reanalysis (ERA5/ERA5-Land, Open-Meteo ERA5-Seamless; temperature at the grid point 14.10° N, 100.90° E, rainfall and radiation from ERA5's 0.25° cell), is complete: all 3,288 days are present, with no missing value. The weather of the earlier L4 benchmark, from the ECMWF blend (Open-Meteo best match), is likewise complete (Table 3.1).

4.2 Benchmark on SMAP L4

Table 4.6 gives the L4 benchmark: eight expanding-window folds testing 2017–2024, with hyperparameters tuned in-fold. The learned models on L4 are the random forest and XGBoost; ridge regression was fitted on L3 only. The skill in the table is the mean of the eight per-fold skill scores, whereas its 95% moving-block bootstrap interval and the Diebold–Mariano test are computed on the test samples of all folds pooled. The L3 benchmark (Section 4.3) instead computes skill once over the pooled samples, so the two benchmarks do not summarise skill in an identical way. The L4 series is the box of $\pm 0.15^\circ$ around the original study point, 57% inside the province, and each daily value is the mean of the eight 3-hourly values of that day (Section 3.3).

Table 4.6: Skill against persistence with SMAP L4 as the target. Best learned model at each horizon. Skill is the mean of the eight per-fold skill scores; the 95% moving-block bootstrap interval and the Diebold–Mariano test are computed on the test samples of all folds pooled. Negative skill indicates a loss to persistence.

Horizon	Best model	Skill	95% interval	DM p
1 day	XGBoost	-0.131	$[-0.235, -0.065]$	5×10^{-6}
3 days	XGBoost	-0.066	$[-0.150, -0.012]$	0.0045
7 days	XGBoost	-0.041	$[-0.190, +0.034]$	0.048

No configuration beats persistence. At 1 and 3 days the interval lies entirely below zero and the Diebold–Mariano p is below 0.05, *against* the learned models. At 7 days the interval straddles zero while the Diebold–Mariano test gives $p = 0.048$, unadjusted for the three horizons: a loss by the point estimate that is not established. Seasonal climatology is worse still, from -2.34 at 1 day to -0.67 at 7 days.

For comparison, the same L4 protocol was run with the ECMWF blend soil moisture (0–7 cm, Open-Meteo best match) as the target, giving $+0.016$, $+0.041$ and $+0.048$ at 1, 3 and 7 days. Only the 3-day value excludes zero, and it is not corroborated by the Wilcoxon test ($p = 0.15$); among the many comparisons performed across models, feature sets, horizons and the two targets, one such result is close to what chance alone produces. It is reported as suggestive, not as a finding.

4.2.1 Why persistence is difficult to beat

A likely explanation lies in the target rather than in the models. Table 4.7 gives the autocorrelation of each candidate target series.

Table 4.7: Autocorrelation of candidate target series, 2016–2024. SMAP L4 (the $\pm 0.15^\circ$ box), the ECMWF blend soil moisture (0–7 cm, Open-Meteo best match) and MERRA-2 surface wetness are daily series for the original study point, evaluated at daily lags. SMAP L3 is evaluated between consecutive recommended-quality retrievals at cell (3,6), which are a median of 3 days apart.

Series	**lag 1**	**lag 3**	**lag 7**
SMAP L4	0.987	0.969	0.951
ECMWF blend	0.956	0.862	0.772
MERRA-2 surface wetness	0.989	0.966	0.933
SMAP L3, cell (3,6)	0.676	–	–

At a lag-one autocorrelation of 0.987, today's value already accounts for almost all of tomorrow's variance, and little remains for a model to supply. SMAP L4 is the output of a land-surface model into which observations have been assimilated, and the series used here is also a $\pm 0.15^\circ$ box mean of daily-averaged 3-hourly values; both plausibly contribute to its smoothness. Without in-situ data the smoothness of the soil itself cannot be measured, so the share due to processing and the share due to averaging cannot be separated. MERRA-2 surface wetness, also the output of a land-surface model, is about as smooth (0.989 at lag 1, 0.933 at lag 7), while the ECMWF blend loses its memory faster (0.772 at lag 7).

The L3 target at cell (3,6) is far less persistent: 0.676 between consecutive retrievals. This contrast is not a clean measure of processing level. L4 is a box mean of $\pm 0.15^\circ$ of daily-averaged 3-hourly values, whereas L3 is the value at one cell of the 9 km grid, observed at one instant; the L3 lag is taken between retrievals that are mostly two or three days apart rather than one; and cell (3,6) lies 38.1 km from the original point on which the L4 box is centred. The contrast therefore mixes product level with footprint, time sampling and location.

SHAP attribution shows which input the L4 random forest weighted most. It was computed for the random forest on the full feature set at 7 days, on 359 held-out rows (`code/results_v5_shap_top_drivers_h7.txt`); XGBoost was not analysed. The current soil-moisture value had the largest mean absolute SHAP value, 0.054, about four and a half times that of the next feature, the previous day's value (0.012). Even with the current value weighted this heavily, no L4 model beat persistence at any horizon (Table 4.6).

4.2.2 Conditional analysis

The advisor asked whether the models add skill when soil moisture changes substantially. Both analyses in this subsection use a single untuned random forest (300 trees, other hyperparameters at their defaults) on the base feature set (`code/l4_reassessment.py`), not the tuned models of Table 4.6. Partitioning test days by the magnitude of the change that actually occurred suggests that this random forest does add skill: $+0.118$ at 3 days on the upper 5% of changes (146 test days). That partition is not admissible: the bands are defined by the realised change, which is persistence's own error, so persistence is exact by construction in the quiet band and maximally wrong in the extreme band. Any competing model appears favourable at the extremes for arithmetical rather than physical reasons.

Repeating the analysis on conditions known at forecast time—antecedent three-day rainfall in four bands, and current wetness in terciles, seven bands at each of the three horizons—the random forest loses to persistence in all 21 combinations, with skill from -0.017 to -0.567. None of the 21 tested conditions identifiable in advance favours this random forest over persistence.

4.3 Benchmark on SMAP L3

Tables 4.8 and 4.9 give the L3 benchmark at cell (3,6): recommended-quality retrievals, the morning-first target, and horizons of 2, 3, 5 and 8 days, chosen by the study from the separations with enough retrieval pairs (Section 3.5.1), under the protocol of Section 3.8. The learned models are ridge regression, the random forest and XGBoost. Skill is computed once over the test samples of all walk-forward folds pooled, whose test years are 2019–2024 at 2 days and 2018–2024 at 3, 5 and 8 days; this differs from the L4 table, whose skill is a mean of per-fold scores over eight folds testing 2017–2024 (Section 4.2). The test sets hold 203 samples at 2 days (6 folds), 434 at 3 and at 5 days, and 639 at 8 days (7 folds each). Within a horizon, every forecast is scored on the same samples. Because a benchmark sample also needs every feature, including a Sentinel-1 scene, and must fall in a test year, the persistence RMSE on these samples (0.0398 to 0.0420) differs slightly from the all-pairs values of Table 4.21. The learned models in these two tables use the L3 history alone; the added inputs are the subject of Section 4.4. RMSE, ubRMSE, bias and R were computed for every forecast and are kept in the result files; the tables of this chapter report RMSE and skill, with R for persistence, and the ubRMSE of the best forecasts is reported in Section 4.9. In both tables a positive skill means a smaller error than the reference of that table, which is persistence in Table 4.8 and the mean of the last three retrievals in Table 4.9. The first table shows whether a forecast is better than assuming no change; the second shows whether a learned model adds anything to a plain average, which is the comparison on which the benchmark rests.

With recommended-quality retrievals at cell (3,6), a non-urban cell, every forecast in Table 4.8 other than climatology has positive skill against persistence at 3, 5 and 8 days, and in every such case the 95% interval lies above zero. One of these bounds, $+0.006$ for the mean of three at 8 days, is close enough to zero to depend on the bootstrap seed; there the Diebold–Mariano test ($p = 0.0056$) carries the claim. At 2 days only the mean of three and ridge regression beat persistence with an interval above zero; the random forest's interval ($[-0.020, +0.187]$) includes zero, and damped persistence and XGBoost could not be distinguished from persistence at this sample size (intervals

Table 4.8: Skill against persistence with SMAP L3 as the target, cell (3,6), recommended quality. Learned models on the L3 history alone. Intervals are moving-block bootstrap 95% intervals; DM is Diebold–Mariano against persistence; Wilcoxon is on per-fold MAE; *Folds* counts the test years with positive skill. RMSE in m^3/m^3. Persistence: Equation 3.1; mean of three: Equation 3.2; damped persistence: Equation 3.3.

Forecast	RMSE	Skill	95% interval	DM p	Wilcoxon p	Folds
$h = 2$ *days:* $n = 203$*, 6 folds; persistence RMSE 0.0407,* $R = 0.688$						
Climatology	0.0474	-0.164	$[-0.313, +0.014]$	0.0874	0.2188	2/6
Mean of three	0.0347	$+0.147$	$[+0.103, +0.202]$	< 0.0001	0.0312	6/6
Damped persistence	0.0404	$+0.008$	$[-0.074, +0.099]$	0.8391	0.4375	5/6
Ridge	0.0364	$+0.106$	$[+0.044, +0.183]$	0.0006	0.0312	6/6
Random forest	0.0379	$+0.070$	$[-0.020, +0.187]$	0.0872	0.1562	4/6
XGBoost	0.0429	-0.053	$[-0.199, +0.125]$	0.5105	1.0000	3/6
$h = 3$ *days:* $n = 434$*, 7 folds; persistence RMSE 0.0414,* $R = 0.587$						
Climatology	0.0403	$+0.025$	$[-0.139, +0.181]$	0.7253	0.6875	3/7
Mean of three	0.0329	$+0.204$	$[+0.131, +0.277]$	< 0.0001	0.0781	7/7
Damped persistence	0.0359	$+0.133$	$[+0.089, +0.175]$	< 0.0001	0.0312	7/7
Ridge	0.0315	$+0.238$	$[+0.170, +0.309]$	< 0.0001	0.0312	7/7
Random forest	0.0316	$+0.237$	$[+0.173, +0.301]$	< 0.0001	0.0312	7/7
XGBoost	0.0335	$+0.192$	$[+0.113, +0.275]$	0.0002	0.1094	7/7
$h = 5$ *days:* $n = 434$*, 7 folds; persistence RMSE 0.0420,* $R = 0.574$						
Climatology	0.0449	-0.068	$[-0.160, +0.058]$	0.2041	0.4688	4/7
Mean of three	0.0360	$+0.144$	$[+0.106, +0.191]$	< 0.0001	0.0156	7/7
Damped persistence	0.0397	$+0.054$	$[+0.009, +0.099]$	0.0045	0.0156	6/7
Ridge	0.0363	$+0.136$	$[+0.076, +0.187]$	< 0.0001	0.0469	6/7
Random forest	0.0348	$+0.172$	$[+0.133, +0.230]$	< 0.0001	0.0156	7/7
XGBoost	0.0362	$+0.139$	$[+0.085, +0.219]$	< 0.0001	0.0156	7/7
$h = 8$ *days:* $n = 639$*, 7 folds; persistence RMSE 0.0398,* $R = 0.638$						
Climatology	0.0431	-0.084	$[-0.187, +0.075]$	0.1539	0.4688	3/7
Mean of three	0.0371	$+0.069$	$[+0.006, +0.136]$	0.0056	0.2969	6/7
Damped persistence	0.0363	$+0.089$	$[+0.046, +0.118]$	< 0.0001	0.0312	6/7
Ridge	0.0348	$+0.125$	$[+0.079, +0.173]$	< 0.0001	0.0312	7/7
Random forest	0.0341	$+0.144$	$[+0.060, +0.219]$	< 0.0001	0.0469	6/7
XGBoost	0.0350	$+0.120$	$[+0.058, +0.179]$	< 0.0001	0.1094	6/7

Table 4.9: Skill against the mean of the last three retrievals (Equation 3.2), with its moving-block 95% interval and the Diebold–Mariano p-value against that reference. Same samples and models as Table 4.8; the learned models use the L3 history. Positive skill means the forecast improves on smoothing alone. Holm: p-value after the Holm adjustment over the twelve learned-model comparisons of the table (Section 3.8); it is not applied to the two reference forecasts.

h	**Forecast**	**Skill**	**95% interval**	**DM** p	**Holm** p
2 days	Climatology	-0.366	$[-0.536, -0.162]$	0.0053	–
	Damped persistence	-0.164	$[-0.270, -0.034]$	0.0244	–
	Ridge	-0.049	$[-0.110, +0.018]$	0.1768	0.86
	Random forest	-0.090	$[-0.185, +0.010]$	0.1035	0.62
	XGBoost	-0.235	$[-0.399, -0.062]$	0.0320	0.26
3 days	Climatology	-0.225	$[-0.396, -0.069]$	0.0031	–
	Damped persistence	-0.090	$[-0.161, -0.028]$	0.0051	–
	Ridge	$+0.042$	$[+0.018, +0.066]$	0.0092	0.092
	Random forest	$+0.040$	$[-0.001, +0.080]$	0.0764	0.53
	XGBoost	-0.016	$[-0.062, +0.033]$	0.5753	1
5 days	Climatology	-0.248	$[-0.334, -0.117]$	0.0002	–
	Damped persistence	-0.105	$[-0.193, -0.038]$	0.0151	–
	Ridge	-0.010	$[-0.059, +0.029]$	0.6971	1
	Random forest	$+0.032$	$[-0.014, +0.085]$	0.1725	0.86
	XGBoost	-0.007	$[-0.086, +0.074]$	0.8586	1
8 days	Climatology	-0.164	$[-0.230, -0.042]$	0.0027	–
	Damped persistence	$+0.021$	$[-0.034, +0.054]$	0.3780	–
	Ridge	$+0.060$	$[+0.021, +0.086]$	0.0020	0.022
	Random forest	$+0.081$	$[+0.041, +0.116]$	0.0007	0.008
	XGBoost	$+0.054$	$[+0.020, +0.076]$	0.0170	0.15

$[-0.074, +0.099]$ and $[-0.199, +0.125]$), which is not evidence that they are equally accurate.

Most of the apparent gain of the learned models over persistence comes from smoothing, so the meaningful question is whether a learned model adds information beyond a simple smoothing baseline. The mean of the last three retrievals, which has no parameters, scores $+0.147$, $+0.204$, $+0.144$ and $+0.069$ at 2, 3, 5 and 8 days. Table 4.9 measures what the learned models add to it, with the Holm adjustment over its twelve learned-model comparisons (Equation 3.13). At 8 days the random forest adds $+0.081$ ($p = 0.0007$, adjusted 0.008) and ridge $+0.060$ ($p = 0.0020$, adjusted 0.022); both gains survive the adjustment. XGBoost's $+0.054$ ($p = 0.017$) does not (adjusted 0.15). At 3 days ridge adds $+0.042$, with an interval above zero ($[+0.018, +0.066]$) and $p = 0.0092$, but it does not survive the adjustment (adjusted 0.092), and it is not repeated at cell (4,6) or with the two-pass target (Section 4.3.2); it is therefore a suggestive gain, not an established one. The random forest's $+0.040$ has $p = 0.076$, a value that depends on its random seed (0.037 to 0.076 over 21 seeds; Section 5.4), and XGBoost falls slightly below the mean (-0.016, $p = 0.58$). At 5 days nothing significant is added; the best is the random forest at $+0.032$ ($p = 0.17$), and ridge and XGBoost fall slightly below the mean. At 2 days nothing is added at all. The mean of three is the best forecast at that horizon, and every learned model is below it: ridge by 0.049 ($p = 0.18$), the random forest by 0.090 ($p = 0.10$) and XGBoost by 0.235 ($p = 0.032$, adjusted 0.26). The 2-day result rests on the fewest samples, 203 over 6 folds.

Table 4.10: Two checks on the gains over the mean of the last three retrievals at cell (3,6): the purged run, without the training pairs whose target falls in the test year, and the refits of the random forest and XGBoost with 21 random seeds (Section 3.8). Skill and Diebold–Mariano p are against the mean of three; Holm is the adjustment over the twelve comparisons of each run. The seed refits use the training pairs and settings of the main run; ridge regression has no random component.

		Main run			**Purged run**			**21 seeds (range)**	
h	**Model**	**Skill**	p	**Holm**	**Skill**	p	**Holm**	**Skill**	p
3	Ridge	$+0.042$	0.0092	0.092	$+0.042$	0.0094	0.10	–	–
	Random forest	$+0.040$	0.0764	0.53	$+0.040$	0.0952	0.67	$+0.040$ to $+0.047$	0.037 to 0.076
8	Ridge	$+0.060$	0.0020	0.022	$+0.060$	0.0020	0.024	–	–
	Random forest	$+0.081$	0.0007	0.008	$+0.071$	0.0139	0.12	$+0.078$ to $+0.083$	0.0004 to 0.0013
	XGBoost	$+0.054$	0.0170	0.15	$+0.058$	0.0104	0.10	$+0.049$ to $+0.065$	0.003 to 0.043

Table 4.10 sets the main run beside the two checks defined in Section 3.8. In the purged run ridge regression's values are unchanged ($p = 0.0094$ at 3 days, 0.0020 at 8 days), and its 8-day gain still survives the adjustment (adjusted 0.024). The random forest's 8-day gain becomes $+0.071$ ($p = 0.014$). Three comparisons of that run have smaller p-values (ridge regression at 8 and 3 days and XGBoost at 8 days), so the random forest's is multiplied by nine and no longer survives (adjusted 0.12). Over the 21 seeds of the main run its p stayed between 0.0004 and 0.0013, so the change is larger than seed noise. The purged run changed some tuning choices as well as the training pairs (Section 5.4); a check below separates the two. Ridge regression's 8-day gain is therefore the most stable result of the table. Its Diebold–Mariano statistic is 3.10 on 638 degrees of freedom (the random forest's is 3.42 and XGBoost's 2.39; ridge regression's at 3 days is 2.62 on 433).

By test year, ridge regression's 8-day gain over the mean of three was -0.092 in 2018, the fold with only two training years, and $+0.092$, $+0.075$, $+0.067$, $+0.039$, $+0.073$ and $+0.108$ in 2019 to 2024: positive in six of the seven years. Its size depends on which years are pooled. Without 2024, the year of the largest gain, the pooled gain is $+0.046$ (unadjusted $p = 0.023$, which by itself would not survive an adjustment for twelve comparisons); without 2018 it is $+0.081$.

Three further checks were made afterwards with the same data and protocol (`l3_extra_checks.py`). The first two show how close to the threshold both 8-day gains sit. The first takes the purged run apart for the random forest. With the purged training pairs but the tuning choices of the main run, its 8-day gain is $+0.079$ ($p = 0.0009$); with the main run's training pairs but the tuning choices of the purged run, it is $+0.070$ ($p = 0.016$). The fall in the purged run therefore comes from the tuning choices, which differ in four of the seven folds and in substance in one (a minimum leaf size of 1 in place of 5 in the fold testing 2024), and not from the removed pairs themselves. The random forest's 8-day gain is thus sensitive to tuning choices that are selected on a single held-out year. The second check widens ridge regression's penalty range from 0.1–100 to 0.1–10,000. A penalty of 1,000 was then chosen in one fold at each of 2, 5 and 8 days, and it improved nothing: the gain over the mean of three became -0.168 at 2 days, -0.045 at 5 days and $+0.057$ at 8 days ($p = 0.0047$), and was unchanged at 3 days. With the wider range, ridge regression's 8-day gain would have an adjusted p of 0.051 in the family of twelve, just above the threshold. The cap of 100 therefore did not hold ridge regression back, but the adjusted verdict at 8 days depends on it. Both 8-day gains are positive in every variant, and neither passes the adjusted test in every variant. The third recomputed the bootstrap intervals with blocks of half and of double the length. Of the 28 intervals for the mean of three and the three learned models on the L3 history, two change sides, the random forest's against the mean of three at 3 and at 5 days, whose lower bounds move from just below zero to just above it with the longer blocks; no interval on which the text relies changes.

A test built for nested models (Section 3.8.5, Equation 3.14; `l3_nested_test.py`) was then computed on the saved forecasts of ridge regression. Its statistic against the mean of three is -0.30 at 2 days (one-sided $p = 0.62$), 6.28 at 3 days, 3.49 at 5 days ($p = 0.0003$) and 6.36 at 8 days; at 3 and 8 days p is below 0.0001, and after adjustment over the four horizons the largest of the three is 0.0005. The purged run gives the same statistics to within 0.03, and the adjusted differential is positive in each of the seven test years at 3, 5 and 8 days and in four of six at 2 days. At 3, 5 and 8 days the hypothesis that the mean of three is the correct model is therefore rejected: the retrieval history holds information that the mean does not use, which agrees with DAM3 scoring above it at 3 and 8 days (Section 4.6). The test does not enlarge any gain. At 5 days ridge regression departs from the mean of three in the right direction on average, but its departures cost slightly more squared error than they save (0.00019 against 0.00016 in $(\mathrm{m}^3/\mathrm{m}^3)^2$), so the forecast as produced is no more accurate than the mean (-0.010, $p = 0.70$). The absence of a gain at 5 days is thus a matter of how well the weights could be estimated from this record, not of the history holding nothing. At 2 days neither test finds anything, on 203 samples.

Damped persistence is weaker than the mean of three at 2, 3 and 5 days, each difference with an unadjusted p below 0.05, and level with it at 8 days ($+0.021$, $p = 0.38$). At 8 days the mean of three has its lowest skill against persistence ($+0.069$), and the learned models' advantage over it is at its largest. A reading based on orbit geometry, that an 8-day target usually lies on the same track as the issue-day retrieval whereas the two earlier retrievals averaged into the mean of three generally do not, finds no consistent support in Section 4.7. Another reading, also untested, is that the mean of three keeps the whole of a recent departure from normal, which costs most at the longest horizon; DAM3,

which damps that departure towards the season and has no track preference, also scores above the mean of three at 8 days ($+0.046$, $p = 0.016$, unadjusted; Section 4.6).

What a learned model does with the retrieval history can be read directly from ridge regression, which is linear in its inputs (Equation 3.4). Refitted with the penalty chosen in each fold, which reproduces the saved forecasts exactly, it gives the issue-day retrieval and the three before it average weights of 0.23, 0.22, 0.23 and 0.15 at 2 days, and at the longer horizons weights that favour the issue day: 0.27, 0.19, 0.16 and 0.11 at 8 days. The four weights sum, on average, to 0.83, 0.74, 0.80 and 0.72 at 2, 3, 5 and 8 days, not to one. The rest of each forecast comes from the intercept, the season and the ages of the retrievals, whose coefficients are not reported here. Ridge regression thus keeps between about three quarters and four fifths of the recent level; that the remainder is a pull towards a seasonal level is a reading of those other terms. The three-retrieval mean weights three retrievals equally and keeps their whole level. For ridge regression, then, its weights show the increment over that mean to be a re-weighting of the same retrievals, read here as a damping towards the season; it uses no input other than the retrieval history, the ages of the retrievals and the season. Part of the damping may reflect the penalty, which was the strongest of its range in 17 of its 27 fits on the L3 history (Section 3.8.3). The damped anomaly of the three-retrieval mean keeps a similar fraction, 0.70 to 0.82 (Section 4.6), fitted without a penalty. The same weights help to explain the pattern across horizons. At 2 days, with the fewest training pairs, the weights spread almost equally over four retrievals, and ridge regression scores below the mean of three. At 8 days, where the mean of three is at its weakest against persistence, a weighting that favours the issue-day retrieval and damps towards the season gains most over it. The weights do not explain why 5 days gives no gain while 3 and 8 days do. The random forest and XGBoost were not examined in this way.

Among the learned models, XGBoost is the weakest on the L3 history at 2, 3 and 8 days; at 5 days it and ridge are within 0.003 of each other ($+0.139$ and $+0.136$). These rankings among the learned models rest on point estimates: the differences between learned models, including those between ridge regression and the random forest, were not tested, only each model's difference from persistence and from the mean of three. Seasonal climatology has negative skill at 2, 5 and 8 days (-0.164, -0.068 and -0.084), although none of these intervals excludes zero, and it ties with persistence at 3 days ($+0.025$). Against the mean of three, climatology scores lower at every horizon, each with an unadjusted $p \leq 0.0053$.

The per-fold Wilcoxon test is reported as a check on consistency across folds, not as strong evidence: with six folds it cannot return a p-value below 0.031, and with seven folds not below 0.016. Across folds the learned models are consistent at 3 days, where all three have positive skill in all seven folds, and at 5 and 8 days, where each has positive skill in at least six of seven; their Wilcoxon p-values at these horizons range from 0.016 to 0.109.

4.3.1 Season

Table 4.11 splits the scores of Table 4.8 into the wet season (May–October) and the dry season (November–April), by the month of the issue date; each skill is computed against persistence in the same season. Persistence itself is less accurate in the wet season at 3, 5 and 8 days, and more accurate at 2 days. The mean of three, ridge regression and the random forest gain on persistence in both seasons at every horizon. The largest seasonal differences are at 2 and 8 days. At 2 days, the

horizon with the fewest samples, the mean of three and ridge gain more in the dry season ($+0.174$ and $+0.121$, against $+0.100$ and $+0.078$ in the wet season), while the random forest is level ($+0.070$ and $+0.071$). At 8 days all three gain more in the wet season, the random forest most ($+0.203$ against $+0.106$). At 3 days the mean of three also gains more in the dry season ($+0.225$ against $+0.178$), while ridge and the random forest are nearly level; at 5 days the differences run in both directions, ridge gaining more in the wet season ($+0.156$ against $+0.122$) and the random forest in the dry season ($+0.180$ against $+0.159$). None of these differences was tested.

Table 4.11: Skill against persistence by season at cell (3,6), on the samples of Table 4.8, split by the month of the issue date (wet: May–October; dry: November–April). Learned models on the L3 history alone. Persistence RMSE in m^3/m^3; each skill is computed against persistence in the same season. No significance test was run within season.

h **(days)**	**Persistence RMSE** wet / dry	**Mean of three** wet / dry	**Ridge** wet / dry	**Random forest** wet / dry
2	0.0389 / 0.0418	$+0.100$ / $+0.174$	$+0.078$ / $+0.121$	$+0.070$ / $+0.071$
3	0.0445 / 0.0394	$+0.178$ / $+0.225$	$+0.234$ / $+0.241$	$+0.234$ / $+0.239$
5	0.0443 / 0.0406	$+0.147$ / $+0.143$	$+0.156$ / $+0.122$	$+0.159$ / $+0.180$
8	0.0420 / 0.0385	$+0.100$ / $+0.048$	$+0.148$ / $+0.110$	$+0.203$ / $+0.106$

These differences are a pattern, not a finding. No significance test was run within season, and each season holds only part of the samples of Table 4.8. The wet-season scores also describe the wet season only as the recommended-quality record samples it: the rejected retrievals fall mostly in May–October and follow more rain than the accepted ones, so the filtered record under-represents the wettest wet-season states (Section 4.8), and the table says little about skill in those states.

4.3.2 Robustness

Table 4.12 repeats the benchmark, under the same protocol, in three further settings, and in a fourth, at cell (3,6) with the mean of the two passes as the target, discussed at the end of this subsection. Two of the three are the cells immediately north and south of cell (3,6), with recommended-quality retrievals: cell (2,6) in Nong Suea district, 89.8% inside the province, with 1,031 recommended-quality retrieval days, and cell (4,6) in Lam Luk Ka district, 90.6% inside, with 946. The third is the old urban cell (4,2) with all its retrievals, every one of which carries flag 9 or 13 (Section 4.1). Table 4.13 gives the gains of ridge regression and the random forest over the mean of three in all four settings. The p-values of these runs are not adjusted for multiple comparisons, except where stated for the two-pass target, so a value below 0.05 is reported as unadjusted, not as significant.

The two neighbouring cells are not independent of cell (3,6). They are adjacent cells of a grid whose values are interpolated from footprints about 36 km across (Section 3.3.5), and on the 915 days on which all three have a target value, the series of cell (3,6) correlates at 0.934 with that of cell (2,6) and at 0.943 with that of cell (4,6); the changes between consecutive common days correlate at 0.86. The

Table 4.12: Robustness of the L3 benchmark, same protocol as Table 4.8. Skill against persistence of the mean of three, ridge regression and the random forest on the L3 history alone, in three further settings and, in the last block, at cell (3,6) with the mean of the two passes as the target, and of the random forest with Sentinel-1 (RF+S1) or weather (RF+met) added. *RF vs mean3* is the skill of the random forest on the L3 history relative to the mean of three, with its Diebold–Mariano p-value in brackets (unadjusted). Persistence RMSE in m^3/m^3. Bootstrap intervals against persistence are not shown; they are kept in the result files, and not all of them exclude zero (see text). Table 4.13 gives both learned models against the mean of three, with intervals.

h	n	Pers. RMSE	Mean3	Ridge	RF	RF vs mean3 (p)	RF+S1	RF+met
Cell (2,6), Nong Suea, recommended quality								
2	220	0.0418	+0.165	+0.117	+0.081	−0.101 (0.087)	+0.064	+0.064
3	458	0.0425	+0.216	+0.252	+0.242	+0.033 (0.209)	+0.237	+0.233
5	458	0.0431	+0.145	+0.137	+0.129	−0.019 (0.473)	+0.124	+0.135
8	670	0.0404	+0.094	+0.136	+0.139	+0.050 (0.027)	+0.127	+0.129
Cell (4,6), Lam Luk Ka, recommended quality								
2	185	0.0411	+0.161	+0.100	+0.080	−0.096 (0.023)	+0.063	+0.086
3	414	0.0401	+0.220	+0.244	+0.248	+0.036 (0.080)	+0.258	+0.249
5	408	0.0420	+0.145	+0.135	+0.128	−0.020 (0.422)	+0.109	+0.124
8	621	0.0384	+0.044	+0.124	+0.104	+0.063 (0.010)	+0.083	+0.086
Old cell (4,2), urban, all retrievals (flag 9 or 13)								
2	262	0.0526	+0.216	+0.023	+0.216	+0.001 (0.988)	+0.161	+0.185
3	606	0.0520	+0.172	+0.185	+0.166	−0.007 (0.723)	+0.156	+0.149
5	604	0.0546	+0.205	+0.130	+0.219	+0.018 (0.450)	+0.204	+0.209
8	1,041	0.0489	+0.089	+0.100	+0.091	+0.003 (0.941)	+0.086	+0.037
Cell (3,6), mean of the two passes as target								
2	203	0.0343	+0.168	+0.131	+0.118	−0.060 (0.040)	+0.105	+0.095
3	434	0.0340	+0.179	+0.199	+0.230	+0.061 (0.008)	+0.225	+0.225
5	434	0.0354	+0.145	+0.102	+0.168	+0.027 (0.267)	+0.155	+0.158
8	639	0.0341	+0.050	+0.113	+0.146	+0.102 (< 0.001)	+0.143	+0.125

neighbour runs therefore test how sensitive the result is to the grid position, on largely shared data, rather than whether it holds at other places.

Cells (2,6) and (4,6) reproduce the broad pattern of cell (3,6). In both, every forecast in the table has a positive point estimate of skill against persistence at every horizon, but not every bootstrap interval excludes zero: at cell (2,6) the intervals of the random forest and of the random forest with Sentinel-1 or weather include zero at 2 days (the random forest: $+0.081$, $[-0.026, +0.225]$), and at cell (4,6) those of the mean of three and of the random forest with Sentinel-1 or weather include zero at 8 days (the mean of three: $+0.044$, $[-0.025, +0.111]$). The mean of three is the best forecast at 2 and 5 days in both cells. The random forest falls below the mean at those horizons, with p below 0.05 only at 2 days in cell (4,6) (-0.096, $p = 0.023$). At 5 days it lies slightly below the mean in both of these cells and slightly above it at cell (3,6), with $p \geq 0.17$ in all three. At 8 days its gain over the mean has p below 0.05 in both cells ($+0.050$, $p = 0.027$, and $+0.063$, $p = 0.010$), as at cell (3,6), and so has ridge regression's ($+0.046$, $p = 0.013$, and $+0.083$, $p < 0.001$; Table 4.13). At 3 days the result varies with the cell. The random forest's gain over the mean stays above $p = 0.05$ in both cells ($+0.033$, $p = 0.21$, and $+0.036$, $p = 0.080$), as at cell (3,6); ridge regression's falls below it at cell (2,6) ($+0.046$, $p = 0.003$), as at cell (3,6), but not at cell (4,6) ($+0.031$, $p = 0.090$), where no model on the L3 history alone adds a 3-day gain over the mean with p below 0.05. Adding Sentinel-1 or weather to the random forest changes its skill by at most 0.02 in either direction at 2, 3 and 5 days and lowers it at 8 days, so neither input shows a consistent gain.

At the old urban cell (4,2), every retrieval is flag 9 or 13, and 28% of the retrieval days of its series (338 of 1,194) carry a flag-13 value, which sits at the cell's upper limit. There persistence is less accurate (RMSE 0.049 to 0.055, against 0.040 to 0.042 at cell (3,6)), and the mean of three and the random forest gain substantially on it: the mean of three scores from $+0.089$ to $+0.216$. Ridge regression gains less at 2 and 5 days ($+0.023$ and $+0.130$), where its intervals include zero, as do those of the random forest with Sentinel-1 or weather at 8 days; at 3 and 8 days ridge regression is level with or above the other two ($+0.185$ and $+0.100$). The random forest, however, adds no measurable gain over the three-retrieval mean at any horizon ($+0.001$, -0.007, $+0.018$ and $+0.003$; Diebold–Mariano p from 0.45 to 0.99), and neither does ridge regression (-0.246 to $+0.015$, p from 0.22 to 0.67) or XGBoost (-0.055 to -0.021, p from 0.23 to 0.50). Their gain over persistence there is matched by smoothing alone.

Read together, a gain over smoothing with p below 0.05 appeared only where the retrievals were of recommended quality at a non-urban cell. There the increment is small and depends on the horizon: at 8 days the gains of ridge regression and the random forest over the mean of three have unadjusted p below 0.05 in all three non-urban cells; at 3 days ridge regression's gain does at two of them, and no model's on the L3 history alone at the third; at 5 days no gain reaches $p = 0.05$; and at 2 days every learned model on the L3 history falls below the mean of three. With not-recommended retrievals at the urban cell, no learned model's gain reaches $p = 0.05$ at any horizon. The two settings differ in both the cell and the quality of the retrievals, and the urban series includes values at the retrieval's upper limit, so these runs do not show which of these removes the increment.

With the mean of the morning and evening retrievals as the target at cell (3,6), persistence error fell to 0.034–$0.035\,\mathrm{m^3/m^3}$, against 0.040–$0.042\,\mathrm{m^3/m^3}$ for the morning-first target, and the broad pattern was unchanged (Tables 4.12 and 4.13): the three-retrieval mean was the best forecast at 2 days ($+0.168$), no learned model's gain over it reached $p = 0.05$ at 5 days (the random forest $+0.027$, $p = 0.27$), and at 8 days ridge regression added $+0.067$ over it (Diebold–Mariano $p = 0.0003$,

interval $[+0.027, +0.096]$) and the random forest $+0.102$ ($p < 0.001$). Adjusted by Holm over the twelve comparisons of this run, as in Table 4.9, both 8-day gains survive (adjusted 0.004 and < 0.001). At 3 days, however, the gain with p below 0.05 moved from ridge regression to the random forest: with this target ridge adds $+0.024$ ($p = 0.15$), while the random forest adds $+0.061$ ($p = 0.008$, adjusted 0.068); as with the morning-first target, no 3-day gain survives the adjustment. The four XGBoost comparisons that complete the twelve of this run are -0.138 ($p = 0.004$, adjusted 0.036), $+0.023$ ($p = 0.44$), -0.024 ($p = 0.46$) and $+0.045$ ($p = 0.14$) at 2, 3, 5 and 8 days, so with this target XGBoost's loss to the mean at 2 days also survives the adjustment. The broad pattern is therefore not specific to the morning-first target, but which learned model's 3-day gain over the mean falls below $p = 0.05$ depends on the target. Only the mean of the two passes was tested as an alternative target; a target of evening retrievals alone was not run.

Table 4.13: Gains over the mean of the last three retrievals in the robustness runs: skill of ridge regression and the random forest on the L3 history relative to the mean of three, with the moving-block 95% interval and the Diebold–Mariano p-value against the mean. Same samples as Table 4.12. These p-values are not adjusted for multiple comparisons; Holm-adjusted values are given in the text for the two–pass target only.

	Ridge regression vs mean3			**Random forest vs mean3**		
h	**Skill**	**95% interval**	**DM** p	**Skill**	**95% interval**	**DM** p
Cell (2,6), Nong Suea, recommended quality						
2	-0.058	$[-0.138, +0.030]$	0.153	-0.101	$[-0.203, +0.039]$	0.087
3	$+0.046$	$[+0.018, +0.069]$	0.003	$+0.033$	$[-0.019, +0.075]$	0.209
5	-0.010	$[-0.050, +0.027]$	0.664	-0.019	$[-0.072, +0.033]$	0.473
8	$+0.046$	$[+0.012, +0.072]$	0.013	$+0.050$	$[+0.019, +0.081]$	0.027
Cell (4,6), Lam Luk Ka, recommended quality						
2	-0.073	$[-0.142, +0.024]$	0.114	-0.096	$[-0.154, -0.006]$	0.023
3	$+0.031$	$[-0.0005, +0.055]$	0.090	$+0.036$	$[+0.002, +0.071]$	0.080
5	-0.012	$[-0.066, +0.038]$	0.617	-0.020	$[-0.082, +0.053]$	0.422
8	$+0.083$	$[+0.053, +0.101]$	< 0.001	$+0.063$	$[+0.027, +0.094]$	0.010
Cell (3,6), mean of the two passes as target						
2	-0.044	$[-0.105, +0.018]$	0.151	-0.060	$[-0.108, +0.009]$	0.040
3	$+0.024$	$[-0.009, +0.049]$	0.149	$+0.061$	$[+0.021, +0.102]$	0.008
5	-0.050	$[-0.158, +0.021]$	0.339	$+0.027$	$[-0.009, +0.094]$	0.267
8	$+0.067$	$[+0.027, +0.096]$	< 0.001	$+0.102$	$[+0.063, +0.137]$	< 0.001
Old cell (4,2), urban, all retrievals (flag 9 or 13)						
2	-0.246	$[-0.685, +0.041]$	0.218	$+0.001$	$[-0.098, +0.101]$	0.988
3	$+0.015$	$[-0.012, +0.035]$	0.260	-0.007	$[-0.046, +0.027]$	0.723
5	-0.094	$[-0.348, +0.034]$	0.358	$+0.018$	$[-0.038, +0.078]$	0.450
8	$+0.013$	$[-0.043, +0.060]$	0.666	$+0.003$	$[-0.092, +0.104]$	0.941

4.3.3 The other non-urban cells

The main benchmark concerns one cell. As a check made after it was complete, the two parameter-free forecasts and ridge regression on the L3 history were scored at each of the 20 non-urban cells of the scan (`l3_spatial_references.py`), under the same protocol with one difference: a sample needs only the retrieval history, not a Sentinel-1 scene or weather. Sentinel-1 had been extracted for every cell of the scan, but weather had been downloaded only for the four cells of the main and robustness runs. The random forest, XGBoost, Sentinel-1 and weather were not run at these cells. At cell (3,6) the scores agree with the main run to within 0.002. At cell (2,6), where requiring a Sentinel-1 scene removes more samples, they differ from the robustness run of Table 4.13: ridge regression's 8-day gain is $+0.035$ ($p = 0.074$) on these 787 samples against $+0.046$ ($p = 0.013$) on the 670 of that run. The cells fall into two groups (Figure 3.3): twelve in the east, within 26 km of cell (3,6), whose series correlate with its series at 0.71 to 0.94, and eight on the western edge of the province, 51 to 65 km away, which correlate with it at 0.45 to 0.55 and are closer to separate observations. The p-values are not adjusted, and with 20 cells and four horizons some small values are expected by chance.

Table 4.14 gives the result cell by cell. For each cell it lists the number of recommended-quality days, the distance from cell (3,6) and the correlation of its series with that cell's, the persistence RMSE at 3 and 8 days, the skill of the mean of three against persistence at the four horizons, and ridge regression's gain over the mean of three at 3 and 8 days. It is included to show how far the pattern found at one cell recurs at the others, and above all at the western cells, the only ones that do not share most of their footprint with cell (3,6).

Table 4.14: Reference forecasts and ridge regression at the 20 non-urban cells of the scan, on the retrieval history alone. Days: recommended-quality retrieval days. km and r: distance from cell (3,6) and correlation of the target series with its series. RMSE: persistence, m^3/m^3. Mean of three: skill against persistence. Ridge: skill of ridge regression on the L3 history against the mean of three, with its unadjusted Diebold–Mariano p.

				RMSE		**Mean of three, skill**				**Ridge vs mean of three (p)**	
Cell	**Days**	**km**	r	**3 d**	**8 d**	**2 d**	**3 d**	**5 d**	**8 d**	**3 days**	**8 days**
East											
(0,6)	1,202	24	0.72	0.049	0.047	+0.170	+0.155	+0.152	+0.084	+0.042 (<0.001)	+0.008 (0.82)
(0,7)	1,328	26	0.71	0.049	0.049	+0.166	+0.135	+0.166	+0.079	+0.049 (<0.001)	+0.050 (0.017)
(1,6)	1,123	16	0.82	0.046	0.044	+0.162	+0.171	+0.156	+0.107	+0.047 (0.002)	+0.045 (0.016)
(1,7)	1,262	19	0.80	0.044	0.044	+0.122	+0.141	+0.147	+0.070	+0.033 (0.056)	+0.072 (<0.001)
(2,6)	1,031	8	0.93	0.043	0.039	+0.165	+0.216	+0.145	+0.095	+0.046 (0.003)	+0.035 (0.074)
(2,7)	1,171	13	0.89	0.041	0.040	+0.114	+0.178	+0.155	+0.070	+0.023 (0.13)	+0.050 (0.036)
(3,6)	981	0	1.00	0.041	0.040	+0.147	+0.204	+0.144	+0.069	+0.043 (0.009)	+0.061 (0.002)
(3,7)	1,094	10	0.92	0.039	0.037	+0.138	+0.195	+0.156	+0.044	+0.022 (0.19)	+0.051 (0.006)
(4,6)	946	8	0.94	0.040	0.038	+0.161	+0.220	+0.145	+0.044	+0.031 (0.088)	+0.083 (<0.001)
(4,7)	1,039	13	0.88	0.039	0.036	+0.181	+0.217	+0.167	+0.020	+0.025 (0.12)	+0.060 (0.006)
(5,7)	963	19	0.81	0.041	0.037	+0.201	+0.251	+0.155	+0.026	+0.036 (0.015)	+0.080 (<0.001)
(6,7)	925	26	0.75	0.042	0.038	+0.194	+0.209	+0.164	+0.044	+0.045 (0.002)	+0.079 (<0.001)
West											
(0,0)	928	65	0.52	0.045	0.047	+0.124	+0.133	+0.033	−0.065	−0.061 (0.20)	+0.104 (<0.001)

Cell	Days	km	r	3 d	8 d	2 d	3 d	5 d	8 d	3 days	8 days
(0,1)	911	56	0.55	0.044	0.047	$+0.132$	$+0.154$	$+0.070$	-0.017	-0.059 (0.25)	$+0.085$ (<0.001)
(1,0)	883	63	0.50	0.042	0.045	$+0.124$	$+0.118$	$+0.080$	-0.065	-0.410 (0.19)	$+0.059$ (0.16)
(1,1)	883	53	0.53	0.044	0.047	$+0.155$	$+0.132$	$+0.110$	-0.010	-0.002 (0.91)	$+0.081$ (<0.001)
(2,0)	830	61	0.47	0.046	0.045	$+0.166$	$+0.170$	$+0.159$	-0.024	$+0.008$ (0.63)	$+0.042$ (0.28)
(2,1)	851	51	0.52	0.046	0.047	$+0.153$	$+0.151$	$+0.151$	$+0.023$	-0.088 (0.32)	$+0.038$ (0.17)
(3,0)	825	60	0.45	0.050	0.045	$+0.101$	$+0.237$	$+0.188$	$+0.051$	-0.021 (0.36)	$+0.054$ (0.009)
(4,0)	850	61	0.45	0.058	0.049	$+0.166$	$+0.237$	$+0.232$	$+0.061$	-0.001 (0.95)	$+0.053$ (0.010)

Forecast error does vary across these cells. Persistence is less accurate in the west (RMSE 0.043 to 0.060 at 2 days and 0.045 to 0.049 at 8 days, against 0.039 to 0.050 and 0.036 to 0.049 in the east; the table prints the 3- and 8-day values). The pattern of the benchmark is nonetheless not specific to cell (3,6). The mean of three beat persistence at every one of the 20 cells at 2, 3 and 5 days ($+0.03$ to $+0.25$; unadjusted p below 0.05 at 14, 20 and 18 cells). At 8 days it did so at 15 cells (-0.065 to $+0.107$); the five where it did not are all in the west. Ridge regression's gain over the mean of three at 8 days was positive at all 20 cells ($+0.008$ to $+0.104$), with unadjusted p below 0.05 at 15 of them, 10 of the 12 eastern and 5 of the 8 western. At 3 days it was positive at 13 cells, with p below 0.05 at 7, all in the east; at 2 days it reached $p = 0.05$ at no cell and at 5 days at one. At no cell and horizon did ridge regression lose to the mean of three with p below 0.05. The small 8-day gain of a learned model over smoothing thus recurs in sign across every non-urban cell, including the western cells that share little with cell (3,6). These cells cover 21.9% of the province; the urban-flagged cells, 76.2% of it, cannot be scored on recommended retrievals at all, so this is still not a result for the province.

4.3.4 Comparison with the L4 benchmark

Set beside Section 4.2, the outcome reverses: on L4 no learned model beat persistence, whereas at cell (3,6) every forecast in Table 4.8, which uses the L3 history alone, other than climatology beats it at 3, 5 and 8 days. This is not a clean comparison of processing levels. The L4 series is a $\pm 0.15^{\circ}$ box mean of daily-averaged 3-hourly values around the original point, while the L3 target is the value at one cell of the 9 km grid, 38.1 km from that point, observed at one instant. The two benchmarks also differ in their horizons (1, 3 and 7 days against 2, 3, 5 and 8), in their folds (on L4, eight testing 2017–2024; on L3, six testing 2019–2024 at 2 days and seven testing 2018–2024 at the other horizons) and in how skill is aggregated (on L4 a mean of per-fold skills, on L3 computed once over the pooled samples), and ridge regression was fitted on L3 only. The contrast therefore mixes product level with footprint, time sampling and location, and the differences in horizon and skill aggregation add to it; this study cannot separate them. The L4 and L3 experiments are therefore reported separately, and the pair should not be read as a direct product-against-product comparison of models. On L3, moreover, most of the gain over persistence is delivered by smoothing alone.

4.4 Information Gain from Sentinel-1 and Weather

Table 4.15 gives the information-gain ladder of Section 3.7: the skill of each learned model against persistence on the L3 history alone, and the change in skill when Sentinel-1 backscatter, weather, or

both are added. All rungs are scored on the same samples and folds as Table 4.8.

Table 4.15: Information-gain ladder at cell (3,6). Skill against persistence for each feature set, and in the Δ columns the change relative to the L3 history alone. S1: Sentinel-1 backscatter; met: rainfall, temperature and shortwave radiation from the ECMWF reanalysis (ERA5/ERA5-Land, Open-Meteo ERA5-Seamless). Each Δ is computed from the unrounded skills, so it can differ by 0.001 from the difference of the rounded skills shown.

		L3	**L3+S1**		**L3+met**		**L3+S1+met**	
h	**Model**	Skill	Skill	Δ	Skill	Δ	Skill	Δ
2	Ridge	$+0.106$	$+0.103$	-0.003	$+0.112$	$+0.006$	$+0.104$	-0.002
	RF	$+0.070$	$+0.055$	-0.015	$+0.068$	-0.002	$+0.052$	-0.019
	XGBoost	-0.053	-0.016	$+0.036$	$+0.025$	$+0.078$	$+0.009$	$+0.062$
3	Ridge	$+0.238$	$+0.233$	-0.004	$+0.235$	-0.003	$+0.233$	-0.005
	RF	$+0.237$	$+0.241$	$+0.005$	$+0.228$	-0.008	$+0.229$	-0.007
	XGBoost	$+0.192$	$+0.223$	$+0.032$	$+0.223$	$+0.031$	$+0.221$	$+0.029$
5	Ridge	$+0.136$	$+0.136$	$+0.001$	$+0.147$	$+0.012$	$+0.140$	$+0.004$
	RF	$+0.172$	$+0.161$	-0.011	$+0.161$	-0.011	$+0.161$	-0.011
	XGBoost	$+0.139$	$+0.132$	-0.007	$+0.112$	-0.027	$+0.110$	-0.028
8	Ridge	$+0.125$	$+0.125$	-0.001	$+0.130$	$+0.004$	$+0.128$	$+0.002$
	RF	$+0.144$	$+0.136$	-0.009	$+0.119$	-0.025	$+0.121$	-0.023
	XGBoost	$+0.120$	$+0.071$	-0.049	$+0.094$	-0.025	$+0.069$	-0.051

For ridge regression and the random forest, adding Sentinel-1, weather, or both changes skill by at most 0.02 in either direction at 2, 3 and 5 days (from -0.019 to $+0.012$), and by -0.025 to $+0.004$ at 8 days. Ridge regression's changes have no consistent sign (six negative, six positive); the random forest's are negative in 11 of 12 cases. The largest single entry, the random forest with Sentinel-1 at 3 days ($+0.241$), lies 0.005 above the same model on the L3 history alone ($+0.237$), a difference far smaller than the width of either bootstrap interval ($[+0.175, +0.308]$ and $[+0.173, +0.301]$).

XGBoost changes more, from -0.051 to $+0.078$, but not in a way that points to information in the added inputs. Its gains at 2 and 3 days bring it closer to, but not above, ridge and the random forest fitted on the L3 history alone (at 2 days its best rung is $+0.025$, against $+0.106$ for ridge on L3; at 3 days $+0.223$, against $+0.238$). At 5 and 8 days the added inputs lower its skill, and at 8 days none of its three rungs with added inputs clearly beats persistence: with Sentinel-1 and with both groups the interval includes zero ($+0.071$, $[-0.048, +0.154]$; $+0.069$, $[-0.032, +0.147]$), and with weather the lower bound, $+0.002$, is close enough to zero to depend on the bootstrap seed. These changes are more plausibly a matter of how XGBoost fits a small sample than of what the inputs contain.

The conclusion is that neither Sentinel-1 backscatter nor weather (rainfall, temperature and shortwave radiation) added measurable additional information beyond the retrieval history under the present data, quality filtering and evaluation protocol. The statement is bounded by the inputs as constructed here—backscatter averaged over the 9 km cell from the latest scene on or before the issue date, and

weather over trailing windows that end on the issue date—and by the single cell and the four horizons. It is not a finding that Sentinel-1 or weather is uninformative about soil moisture in general.

4.5 Behaviour on the Real Record

The skill scores above summarise many forecasts in a single number. Following the advisor's order of work—research question, data availability, baseline, information gain, model choice—this section examines what the forecasts at cell (3,6) actually do on the retrieval record: when a forecast can be made at all, what the gain over persistence consists of, how the forecasts behave when the retrieval changes substantially, and whether Sentinel-1 or weather help in the situations where they should. The numbers come from the day-by-day forecasts of the main run and are produced by `l3_real_data_evaluation.py`.

4.5.1 When a forecast can be made

A forecast can be issued only on a day with a recommended-quality retrieval. Such a day occurs on 36–40% of days from January to April, 22–33% from May to September, and 10% in October, when the quality rule rejects 82% of the retrievals made (166 of 203) and leaves 74% of the days that have a retrieval without a usable one. On an ordinary day the latest usable retrieval has a median age of 1 day in the dry season (November–April) and 2 days in the wet season (May–October). From September to November the median age is 4 days; the latest retrieval is at least 5 days old on 47% of days and at least 10 days old on 37%. The record contains 23 gaps of 10 days or more. In 2022 there was almost no usable retrieval from August to late November, the period of the year's heaviest rain (Figure 4.1). Two causes overlap there. From 7 August to 19 September the satellite was in safe mode and made no retrieval at all; outside that period, 24 overpass days from August to November carried only retrievals that are not recommended, and 4 carried a recommended one (2 and 5 August, 12 and 20 November). The quality-filtered record is therefore thinnest exactly when a soil-moisture forecast would matter most for flood preparedness.

Figure 4.1 shows that year, which holds the longest gap of the record, 99 days. The upper panel plots every recommended-quality retrieval as a dot, together with three 3-day forecasts drawn at their target dates: persistence, the mean of the last three retrievals and the random forest on the L3 history. The lower panel gives the daily reanalysis rainfall. Three things can be read from the figure. The dots thin out as the rain increases, and from August to late November there are almost none. The persistence line repeats each dip and spike of the dots three days later, while the other two lines stay closer to the level that neighbouring dots share, which is the smoothing measured in Section 4.3. And no line crosses the gap, because a forecast needs a retrieval to start from.

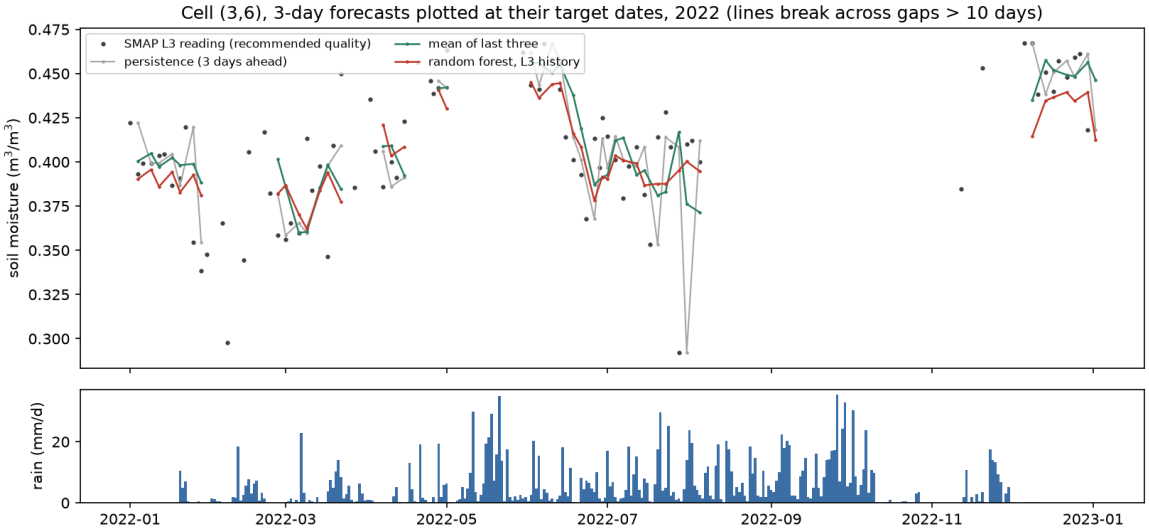


Figure 4.1: Cell (3,6), 2022: recommended-quality SMAP L3 retrievals (dots) and the 3-day forecasts of persistence, the mean of the last three retrievals and the random forest on L3 history, plotted at their target dates; daily rainfall from the ECMWF reanalysis (ERA5-Seamless) below. Lines break across gaps of more than 10 days.

4.5.2 What the gain over persistence consists of

Of the 981 retrievals, 51 (5%) are isolated: each differs from both of its neighbours by more than 0.05 m^3/m^3 in the same direction while the neighbours agree within 0.03 m^3/m^3 (32 dips and 19 spikes). The two thresholds are fixed choices of this study and were not varied. In 17 of the 34 cases where the same date also has the other pass, that pass lies closer to the isolated value than to its neighbours, so these are not all noise that is independent between the two passes; the agreement may reflect a brief real change or an error that both passes share. Either way, such a departure does not last. Persistence copies an isolated value into its forecast; a smoothing forecast does not. Table 4.16 shows that at 3 days more than half of the reduction in squared error relative to persistence comes from the forecasts issued from an isolated retrieval; at 8 days the share is smaller and differs between forecasts. The decomposition is descriptive. A retrieval is labelled isolated with the help of the retrieval after it, which at 3 days is the forecast's own target in 93% of cases, so persistence is bound to err badly on these forecasts and part of the share is built into the label.

At 3 days, then, the forecasts issued from an isolated retrieval, 7% of all forecasts, supply 54–62% of the reduction in squared error: much of the 3-day gain is concentrated where a departure does not last. In skill, however, more than half of the 3-day gain remains once the isolated retrievals are excluded ($+0.112$ to $+0.155$, against $+0.204$ to $+0.238$ with them), so filtering short-lived departures is a large part of the gain, not all of it. At 8 days the combined share is smaller (12–23%), but it is a net of two opposite parts. Forecasts issued from an isolated retrieval supply 50% of the gain of the mean of three, 30% of ridge regression's and 20% of the random forest's. Forecasts whose target alone is isolated lose to persistence, mostly in 2024, the year that holds 16 of the 51 isolated retrievals; without that fold the combined 8-day shares are 39%, 33% and 29%, and the 3-day shares 58%, 53% and 54%. Removing only the forecasts issued from an isolated retrieval lowers 8-day skill from $+0.069$ to $+0.039$ for the mean of three, from $+0.125$ to $+0.101$ for ridge regression and from $+0.144$ to

Table 4.16: Share of the reduction in squared error relative to persistence, by the position of an isolated retrieval: forecasts issued from one (6.7% of forecasts at 3 days, 5.8% at 8 days), forecasts whose target alone is one (3.7% and 5.5%), and both groups together. The shares are nets of signed terms and are rounded independently. Last column: skill against persistence once both groups are removed. Cell (3,6). Descriptive only: the label uses the retrieval after the issue date.

	3 days				8 days			
Forecast	**Issued**	**Target**	**Both**	**Skill w/o**	**Issued**	**Target**	**Both**	**Skill w/o**
Mean of last three	62%	+3%	65%	+0.112	50%	−27%	23%	+0.070
Ridge, L3	54%	+5%	59%	+0.155	30%	−7%	22%	+0.129
Random forest, L3	54%	+5%	59%	+0.151	20%	−8%	12%	+0.169

+0.133 for the random forest. The random forest's 8-day gain is therefore largely independent of isolated retrievals; that of the mean of three is not. The two horizons are also not strictly comparable: at 3 days the target is the retrieval next after the issue retrieval in 93% of forecasts, so an isolated issue value almost guarantees a large persistence error, whereas at 8 days it is so in 2%.

Year by year, the random forest on L3 history beat persistence at 3 days in all seven test years (+0.124 in 2021 to +0.351 in 2024), and at 8 days in six of seven (2024: −0.087; the mean of three −0.115).

4.5.3 When the retrieval changes substantially

The advisor asked whether the models add skill when soil moisture changes substantially. Table 4.17 groups the 3-day forecasts by the change that actually occurred. The grouping uses information available only afterwards, so it describes behaviour and cannot rank the models: persistence is nearly exact in the level class by construction.

Table 4.17: 3-day forecasts at cell (3,6) grouped by the change between the issue and target retrievals (m^3/m^3): number of cases, mean change, and mean forecast change for each model. Descriptive only: the cases are selected by the change that occurred.

Change	n	**Observed**	**Mean of three**	**Ridge**	**RF**	**RF + weather**
Rose by ≥ 0.04	55	+0.077	+0.031	+0.035	+0.039	+0.044
Within ± 0.04	323	−0.001	−0.002	−0.004	−0.003	−0.003
Fell by ≥ 0.04	56	−0.064	−0.021	−0.022	−0.021	−0.021

The forecasts move in the right direction but not as far as the change between retrievals: about half as far on the rises (41–57%) and about a third as far on the falls (32–35%). Part of this shortfall is expected of any forecast that minimises squared error against a target with short-lived variation: when cases are chosen because the target changed, a part of that change is variation that no forecast

could anticipate. The proportions describe behaviour; they do not by themselves measure a missed signal. The forecasts rose on 84–87% of the wetting cases, yet much of that rise reflects recovery from a low retrieval rather than an anticipation of rain: none of the models has information about rain still to come. When more than 20 mm of rain fell during the 3-day window (71 cases), soil moisture rose by $+0.017$ on average while the forecasts changed by between -0.002 (mean of three) and $+0.004$ (random forest with weather), and skill fell to $+0.077$ for the mean of three and $+0.170$ for the random forest. Wetting that followed heavy reanalysis rainfall is therefore not forecast from the retrieval history; whether the rain caused it was not tested (Section 4.5.4).

4.5.4 Information gain where it should appear

An average may hide a gain confined to particular situations, so the ladder comparison was repeated in subsets defined by what is known on the forecast day. At 3 days, adding Sentinel-1 to the random forest changed skill by -0.004 $[-0.021, +0.012]$ when the radar scene was from the same or the previous day (200 forecasts), and adding weather changed it by $+0.001$ $[-0.042, +0.047]$ after at least 20 mm of rain in the previous three days (70 forecasts). In every subset examined—fresh or older radar scene, recent heavy rain or none, wet or dry season—and for both ridge and the random forest at 3 and 8 days, the bootstrap interval of the change includes zero. These subset intervals resample forecasts independently and so ignore the dependence between successive forecasts; they describe the size of the changes and are not a formal test. Over all forecasts of a horizon the moving-block bootstrap of Section 3.8 can be applied, to all 24 changes for ridge regression and the random forest at the four horizons. Every interval for ridge regression includes zero. For the random forest three lie below zero, all for changes that add weather: at 8 days -0.025 $[-0.053, -0.003]$; and, by margins small enough to depend on the bootstrap seed, at 5 days -0.011 $[-0.024, -0.0003]$ and with both groups at 8 days -0.023 $[-0.055, -0.00004]$. Where the added inputs did anything, they made the random forest slightly worse. The intervals also show how large a gain could have gone unseen. The largest gain still consistent with them is $+0.023$ at 2 days (the random forest with weather), $+0.013$ at 3 days, $+0.040$ at 5 days (ridge regression with weather) and $+0.020$ at 8 days, a few hundredths of skill, or at most about 0.002 m^3/m^3 of RMSE. Gains of that size cannot be ruled out. No interval was computed for XGBoost's changes.

The links measured with these inputs are weak at this cell. On the 213 days with both products, Sentinel-1 backscatter correlates with the SMAP retrieval at 0.24 (VV) and 0.32 (VH), or 0.38–0.39 for the morning, descending passes; changes between consecutive descending scenes correlate at 0.51 over 28 pairs, which is consistent with the radar responding to wetting and drying, though on few pairs. Rainfall between retrievals up to 3 days apart correlates with the change in the retrieval at only 0.16 (896 pairs); after more than 30 mm the retrieval rose on average ($+0.021$) but in only 66% of cases. These correlations describe the inputs as built here, and several features of that construction may hide information. The backscatter is averaged over the 9 km grid cell, much smaller than the area the radiometer senses (Section 3.3.5); the latest scene of either pass is used without a pass indicator, although the correlation is higher within the descending pass; the scene is typically two days older than the issue retrieval; and the rainfall is a reanalysis estimate for an area of about 27 km that was never compared with a gauge. Irrigation and field flooding, which wet the surface without rain, would also weaken the link between rain and the retrieval, and the land use of the cell is not known from the data used here (Section 3.3). Finally, every score is taken against a retrieval with its own error, so an input that tracks the soil but not that error would earn no credit. The result therefore shows that,

with the inputs built as described, no information beyond the retrieval history could be measured. It does not show whether that information is absent, already contained in the history, or lost in the construction.

4.5.5 Two kinds of change

The analyses above separate two kinds of change in the retrieval record, and the skill scores treat them very differently. The first is the isolated, short-lived change: a retrieval that departs from both of its neighbours and returns. Whether it is retrieval noise or a brief real event, it does not last, and a forecast gains by not copying it. More than half of the 3-day reduction in squared error arises there, though much less at 8 days (Table 4.16). The second is wetting that follows heavy reanalysis rainfall and the wet season: the retrieval rose by $+0.017$ on average when more than 20 mm of reanalysis rain fell during the forecast window, and the forecasts did not follow. That the rain caused the rise is plausible but not established: the reanalysis rainfall was never compared with a gauge, it correlates with the change in the retrieval at only 0.16, and irrigation or field flooding can wet the surface without rain. The benchmark thus credits the handling of the first kind and gives no credit for the second, which none of the inputs can anticipate.

The second kind is also under-sampled. The quality rule rejects 82% of October retrievals, and 77% of all the retrievals it rejects fall in May–October (Section 4.8), so the wettest conditions, in which the response to rain is likely to be largest, are the ones the filtered record represents least. At 3 days the skill reported in this chapter is therefore to a large extent skill at not copying short-lived changes; at 8 days much less so. It is measured on a record that under-represents the wettest conditions, and it should not be read as skill at forecasting wetting after heavy rain.

4.6 The Same Forecasts under Other Yardsticks

The skill scores so far use persistence as the reference and RMSE as the metric. This section re-scores the same forecasts of the main run with the yardsticks of Section 3.8.6: skill against climatology, the anomaly correlation, two references from soil-moisture memory, the Kling–Gupta efficiency and a three-category score. The re-scoring reproduces the benchmark's skill against persistence exactly and its bootstrap intervals to within 0.0005.

Figure 4.2 plots the first two yardsticks against the horizon for seven forecasts. Panel (a) gives the skill against seasonal climatology, where zero means no better than the seasonal normal, and panel (b) the anomaly correlation, where a higher value means that departures from normal are caught more closely. What the figure shows is a separation into two groups. In panel (a) persistence and anomaly persistence lie lowest at every horizon and close to zero at 3 days, while the forecasts that average several retrievals or learn from them lie well above, between 0.14 and 0.27; damped anomaly persistence lies between the two groups at 2, 3 and 5 days. In panel (b) the three forecasts built on a single retrieval lie together at the bottom, and the others lie above them at 2, 3 and 5 days. At 8 days the mean of three and its damped anomaly fall back to their level (0.56), and only ridge regression and the random forest stay above it (0.58 and 0.60). The two subsections that follow give the numbers.

4.6.1 Against climatology

At this cell persistence and seasonal climatology could not be told apart. Persistence scored $+0.141$, -0.026, $+0.063$ and $+0.077$ against climatology at 2, 3, 5 and 8 days, and no difference had p below 0.05 (Diebold–Mariano p from 0.087 to 0.73). In the wet season persistence scored below climatology at 3, 5 and 8 days (-0.056, -0.026 and -0.032). A single retrieval could thus not be shown to be a better forecast than the seasonal normal. Against climatology, the three-retrieval mean scored $+0.141$ to $+0.268$, and ridge regression and the random forest on the L3 history $+0.190$ to $+0.232$ and $+0.202$ to $+0.224$ (Figure 4.2a). Every learned model examined—the three models on the L3 history and the random forest with Sentinel-1, weather or both—beat climatology at every horizon (Diebold–Mariano $p \leq 0.015$), and scored above it in both seasons, where it was not tested. The finding that the forecasts beat a naive reference therefore does not depend on which naive reference is used.

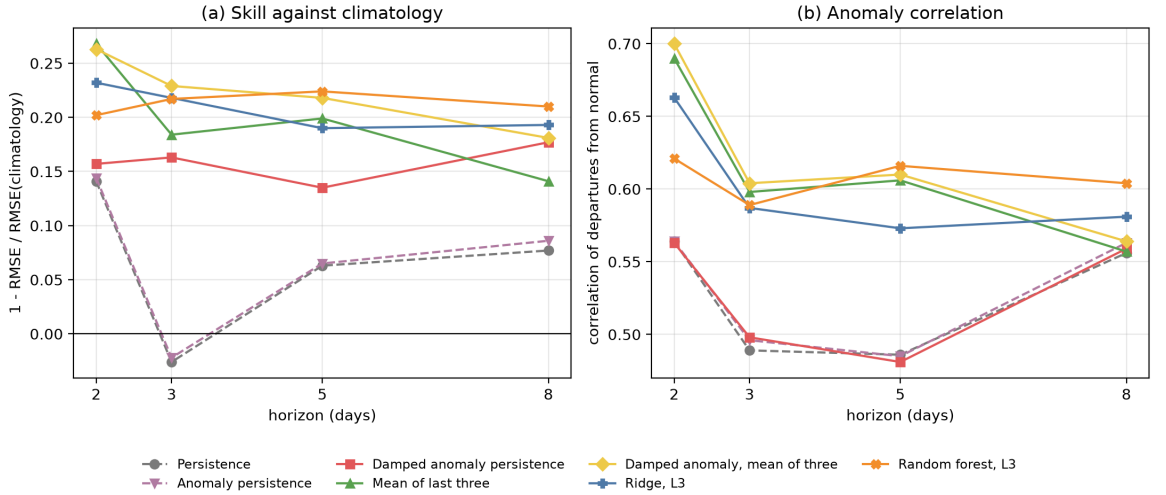


Figure 4.2: Cell (3,6), main run: (a) skill against seasonal climatology and (b) anomaly correlation (Equation 3.18), by horizon, for persistence, anomaly persistence, damped anomaly persistence (Equation 3.16), the mean of the last three retrievals, the damped anomaly of that mean (Equation 3.17), and ridge regression and the random forest on the L3 history.

4.6.2 Departures from normal

The anomaly correlation (Equation 3.18) of persistence was 0.564, 0.489, 0.486 and 0.556 at 2, 3, 5 and 8 days (Figure 4.2b). The three-retrieval mean reached 0.690, 0.598, 0.606 and 0.557, ridge regression 0.663, 0.587, 0.573 and 0.581, and the random forest 0.621, 0.589, 0.616 and 0.604. The learned models thus caught departures from normal better than persistence but by about as much as the three-retrieval mean did; these differences were not tested. Adding Sentinel-1 or weather changed the random forest's anomaly correlation little and mostly lowered it (at 3 days 0.594 and 0.575, against 0.589). Every forecast except climatology caught departures less well in the wet season; for

the random forest the anomaly correlation was 0.440 to 0.552 in the wet season and 0.630 to 0.684 in the dry season.

4.6.3 The soil-memory references

The fitted fraction ρ_h of damped anomaly persistence (Equation 3.16) lay between 0.47 and 0.57 in every fold at every horizon, with no trend from 2 to 8 days. If departures from normal simply decayed day by day, the fraction would fall as the horizon lengthened. Instead, about half of the departure on the issue date was lost at once and the remainder was retained over 8 days. This pattern is consistent with a slowly varying signal overlaid by short-lived variation such as retrieval noise, in line with the isolated retrievals of Table 4.16; without in-situ data the study cannot confirm that interpretation. For the damped anomaly of the three-retrieval mean (Equation 3.17) the fraction was 0.70 to 0.82. Ridge regression, fitted with its penalty chosen in each fold, arrives at a similar damping: the weights it gives the four recent retrievals sum to 0.72 to 0.83 on average (Section 4.3).

Table 4.18 sets these references beside the learned models. Its upper part gives the skill of each against persistence, and its lower part the skill of each learned model against DAM3, the strongest of the simple references at 3, 5 and 8 days. The lower part is the one to read: a negative entry means that a rule with one fitted number did better than the learned model.

Table 4.18: Cell (3,6): skill against persistence (Equation 3.9) of the soil-memory references and the learned models on the L3 history (upper part), and skill of each learned model against the damped anomaly of the three-retrieval mean, DAM3 (Equation 3.17), with the Diebold–Mariano p-value (lower part). DAM3 was defined after the main results were known.

Forecast	2 days	3 days	5 days	8 days
Skill against persistence				
Mean of last three	$+0.147$	$+0.204$	$+0.144$	$+0.069$
Damped anomaly persistence	$+0.018$	$+0.184$	$+0.077$	$+0.108$
DAM3	$+0.142$	$+0.249$	$+0.165$	$+0.112$
Ridge, L3	$+0.106$	$+0.238$	$+0.136$	$+0.125$
Random forest, L3	$+0.070$	$+0.237$	$+0.172$	$+0.144$
XGBoost, L3	-0.053	$+0.192$	$+0.139$	$+0.120$
Skill against DAM3 (Diebold–Mariano p)				
Ridge, L3	-0.042 (0.058)	-0.014 (0.31)	-0.035 (0.12)	$+0.015$ (0.38)
Random forest, L3	-0.083 (0.031)	-0.016 (0.48)	$+0.009$ (0.66)	$+0.036$ (0.075)
XGBoost, L3	-0.226 (0.014)	-0.076 (0.008)	-0.031 (0.33)	$+0.008$ (0.67)

Damped anomaly persistence beat persistence at 3, 5 and 8 days, with bootstrap intervals above zero (Table 4.18); anomaly persistence, with the fraction fixed at one, could not be distinguished from it ($+0.002$ to $+0.010$, every interval including zero). The damped anomaly of the three-retrieval mean (DAM3) is used here as a diagnostic benchmark, not as a proposed forecasting model. It scored $+0.142$, $+0.249$, $+0.165$ and $+0.112$ against persistence, above the three-retrieval mean at 3 days ($+0.056$, $p = 0.0003$) and 8 days ($+0.046$, $p = 0.016$). No learned model beat it with p below 0.05

at any horizon; this holds for every climatology tested in Section 4.6.5 except the benchmark's own and its purged variant, both built from training targets only. By the moving-block interval, too, no learned model lies above DAM3 (random forest at 8 days $+0.036$ $[-0.021, +0.080]$). Seven of the 24 intervals lie wholly below zero, five of them at 2 days; they include ridge regression at 2 and 5 days, where the Diebold–Mariano test gave $p = 0.058$ and 0.12. At 2 days DAM3 beat the random forest ($p = 0.031$) and XGBoost ($p = 0.014$), and at 3 days XGBoost ($p = 0.008$). The random forest with Sentinel-1, weather or both did no better: it lost to DAM3 at 2 days (p from 0.012 to 0.049, although with weather alone the interval includes zero) and differed from it by -0.027 to $+0.026$ at 3 to 8 days, with every $p \geq 0.24$. The increment that the learned models added over the three-retrieval mean at 3 and 8 days (Section 4.3) is thus largely what pulling the smoothed forecast part of the way back to the seasonal normal supplies. What the diagnostic shows is how difficult it is for a learned model to demonstrate information beyond a very simple seasonal adjustment of recent observations. Because DAM3 was chosen after the results were known and several references were compared, these p-values are not adjusted for that choice.

4.6.4 Shape of the forecasts and categories

The learned models' forecasts varied less than the retrievals they forecast: their amplitude ratio α in Equation 3.19 was 0.64 to 0.77, against 0.86 to 1.09 for persistence and 0.78 to 0.93 for the three-retrieval mean. A forecast that damps its departures lowers RMSE but understates how wet or dry the soil becomes. Scored by the Kling–Gupta efficiency, which penalises that, persistence ranked level with the learned models at 3 days and above them at 2, 5 and 8 days (Table 4.19), and the three-retrieval mean scored highest or close to it at every horizon. In the three categories, the learned models' Heidke skill scores (Equation 3.20) at 3 to 8 days (0.23 to 0.29 for the L3 models) were close to persistence's (0.25 to 0.30). They usually flagged fewer of the observed wet cases: 45 to 57% for the learned models against 55 to 62% for persistence, the one exception being ridge regression at 5 days (57% against 55%). The wet-category false-alarm ratio of ridge regression and the random forest was lower than persistence's at every horizon (29 to 41%, against 40 to 44%). Neither KGE nor the category scores were tested for significance.

Table 4.19: Cell (3,6): Kling–Gupta efficiency (Equation 3.19) and Heidke skill score (Equation 3.20) by horizon. DAM3 is the damped anomaly of the three-retrieval mean (Equation 3.17). Not tested for significance.

	KGE				Heidke skill score			
Forecast	**2 d**	**3 d**	**5 d**	**8 d**	**2 d**	**3 d**	**5 d**	**8 d**
Persistence	0.660	0.576	0.561	0.638	0.278	0.281	0.253	0.296
Mean of last three	0.688	0.684	0.645	0.627	0.349	0.339	0.317	0.270
DAM3	0.623	0.646	0.583	0.581	0.357	0.306	0.306	0.269
Ridge, L3	0.591	0.590	0.538	0.542	0.344	0.292	0.289	0.242
Random forest, L3	0.574	0.589	0.547	0.549	0.317	0.243	0.263	0.256
XGBoost, L3	0.527	0.582	0.527	0.555	0.256	0.226	0.228	0.227

4.6.5 Sensitivity to the definition of climatology

The climatology of the benchmark is one of several reasonable definitions (Section 3.7). Table 4.20 rebuilds it in seven ways, each from the training years only, and repeats the three comparisons that depend on it (`l3_climatology_sensitivity.py`); for the anomaly-based reference, DAM3 is rebuilt with the matching climatology. The first row reproduces the benchmark's climatology exactly. The third reproduces the anomaly climatology of Equation 3.15. The fourth and fifth use windows within the 4–8 weeks of the validation guidance (Gruber et al., 2020).

Table 4.20: Cell (3,6): the comparisons that depend on climatology, under seven definitions, all from training years only. Persistence: skill of persistence against climatology over the four horizons, and the smallest Diebold–Mariano p. Learned: skill of ridge regression, the random forest and XGBoost on the L3 history against climatology over the four horizons, and the largest p. RF vs DAM3: skill of the random forest on the L3 history against DAM3 (Equation 3.17) at 8 days, with p; this is the only comparison in which a learned model beat DAM3 under any of these definitions. The third row's p (0.073) differs from Table 4.18 (0.075) because the fraction is fitted here without the training pairs whose target falls in the test year.

	Persistence		**Learned**		**RF vs DAM3**
Climatology	**Skill**	**Min p**	**Skill**	**Max p**	**8 days (p)**
Training targets, ±7 d (as built)	−0.026 to +0.141	0.087	+0.096 to +0.232	0.015	+0.046 (0.039)
As built, purged	−0.026 to +0.141	0.088	+0.096 to +0.232	0.015	+0.046 (0.038)
All retrievals, ±7 d	−0.065 to +0.147	0.066	+0.102 to +0.238	0.007	+0.036 (0.073)
All retrievals, ±15 d (31 days)	−0.077 to +0.148	0.066	+0.103 to +0.239	0.007	+0.024 (0.182)
All retrievals, ±17 d (35 days)	−0.074 to +0.148	0.065	+0.103 to +0.238	0.006	+0.024 (0.183)
All retrievals, ±7 d, years weighted equally	−0.083 to +0.157	0.060	+0.112 to +0.246	0.007	+0.030 (0.101)
Harmonic fit, annual and semi-annual	−0.100 to +0.146	0.064	+0.101 to +0.236	0.019	+0.008 (0.621)

Among these seven, two of the three comparisons do not depend on the definition. Persistence and climatology differed with p below 0.05 under none of the seven, although persistence scored $+0.141$ to $+0.157$ at 2 days (p from 0.060 to 0.088). Every learned model on the L3 history beat climatology under all seven (largest $p = 0.019$). The third comparison does depend on it: under the benchmark's own climatology, built from the training targets with a ±7-day window, the random forest beat DAM3 at 8 days ($+0.046$, $p = 0.039$), as it did under the purged variant of that climatology, whereas under the five definitions built from all training-year retrievals no learned model did. That is one comparison among many and its p-value is not adjusted, but the statement that no learned model beats DAM3 holds only for climatologies built from all training-year retrievals.

An eighth definition was computed and set aside: the proposal's literal one, the mean on exactly the target day of year, with no window. On this record it averages a median of 2 retrievals, and 13.2% of test targets have none and fall back to the overall mean. It is too noisy to serve as a normal. Under it persistence beats climatology at every horizon ($p \leq 0.013$), DAM3 no longer beats persistence, and

the learned models beat DAM3 at 3, 5 and 8 days. The ± 7-day window was introduced to avoid this.

The climatology is weak where the record is shortest, though not only there. With two training years, in the fold testing 2018, the benchmark's climatology scored -0.329, -0.479 and -0.239 against persistence at 3, 5 and 8 days, with a median of 7 to 10 retrievals per window. Its weakness is not confined to short records, however: in 2024, with eight training years, it scored -0.298 at 2 days and -0.392 at 8 days, its lowest scores at those horizons apart from 2021 at 2 days (-0.327); at 2 and 5 days every forecast had its largest error of any fold in 2024.

4.7 Horizons and the Orbit

Table 4.21 extends the pair counts of Table 3.2 to further separations up to 24 days, with the error and correlation of persistence over all pairs at each separation. Pairs whose separation is a multiple of eight days come from the same slot of the repeat cycle and, where both values come from the same pass, from the same track; other separations join different swaths with different footprints. Because the series takes the morning value where it exists and the evening value otherwise, 19–26% of the pairs at 2 to 8 days join the two passes (19% at 8 days); those pairs are neither same-track nor exactly h days apart.

Persistence error does not grow monotonically with the gap. The 8-day RMSE, 0.0394, is below the 2-, 3- and 5-day values (0.0398, 0.0399 and 0.0405), and the 8-day correlation, 0.667, is above the 3- and 5-day values (0.646 and 0.632), though below the 2-day value (0.694). The 8-day pairs come from the same slot, while the 2-, 3- and 5-day pairs come from different slots. The lower 8-day error is consistent with persistence at the shorter horizons being charged with a change of footprint as well as with the change in soil moisture, but that mechanism is not established here, and a split by pass weakens it. Among pairs of two morning values, the 8-day RMSE (0.0401, 659 pairs) is below the 3-, 5- and 6-day values (0.0414, 0.0421 and 0.0458) but not below the 2-day value (0.0395); the pooled 8-day figure is pulled down by the pairs that join the two passes (0.0372 and 0.0353). In the evening-only series the ordering reverses: the 8-day RMSE (0.0362) is the highest of five (0.0355, 0.0330, 0.0344 and 0.0350 at 2, 3, 5 and 6 days). The differences are small, no interval was computed for them, and the off-pattern 7-day pairs have the same RMSE as the 8-day pairs (0.0394). In the pooled series the same ordering appears further out: 16-day pairs, same-slot, have a lower RMSE and a higher correlation (0.0440, 0.590) than 11-day pairs (0.0446, 0.564). It appears also at the original urban cell, on all retrievals regardless of quality, where the 8-day persistence RMSE (0.0504) is below the 2-, 3- and 5-day values (0.0522, 0.0517 and 0.0539). Memory decay is visible along the same-slot sequence of 8, 16 and 24 days: the RMSE rises from 0.0394 to 0.0440 and 0.0511, and the correlation falls from 0.667 to 0.590 and 0.449.

Whatever its cause, persistence error is about flat across the benchmark horizons of 2, 3, 5 and 8 days, with differences that are small and were not tested, so the benchmark horizons are not a sequence of steadily growing difficulty. That is one reason not to read the pattern of skill in Table 4.8, highest at 3 days, as a pattern of predictability alone.

Four separations of 8 days or less were not evaluated: 1, 4, 6 and 7 days. No two retrievals at this cell are exactly 4 days apart, so that separation does not arise at all. Exact 1- and 7-day pairs exist at this cell, 41 of each, because 44 of its retrievals fall off the regular pattern where adjacent swaths overlap;

Table 4.21: Recommended-quality retrieval pairs separated by exactly h days at cell (3,6), 2016–2024, with persistence RMSE (m^3/m^3) and correlation R over all pairs. One- and seven-day pairs do not arise from the regular slot pattern; they come from the off-pattern retrievals where adjacent swaths overlap.

h (days)	Pairs	RMSE	R	Slot	Benchmark horizon
1	41	0.0362	0.735	off-pattern	no: too few pairs
2	333	0.0398	0.694	different	yes
3	601	0.0399	0.646	different	yes
5	598	0.0405	0.632	different	yes
6	326	0.0430	0.645	different	no: run later as a check
7	41	0.0394	0.653	off-pattern	no: too few pairs
8	872	0.0394	0.667	same	yes
11	575	0.0446	0.564	different	no
16	834	0.0440	0.590	same	no
24	793	0.0511	0.449	same	no

at the original cell there were none. Forty-one pairs spread over nine years cannot meet the protocol's minimum of 120 training and 20 test samples in every fold, so the orbit rules out both horizons at this cell for lack of pairs. The 1-day pairs have the lowest persistence error in the table (0.0362), but that figure rests on those 41 pairs. Six-day pairs are more numerous, 326 of them; 6 days was not one of the benchmark's horizons, which were kept identical to the analysis presented to the advisor on 23 September 2026. The 5- and 8-day results do not stand in for it: persistence error over all 6-day pairs (0.0430) is higher than over the 5- or 8-day pairs (0.0405 and 0.0394). The benchmark horizons of 2, 3, 5 and 8 days are therefore this study's own choice from the separations with enough retrieval pairs (Section 3.5.1).

4.7.1 The 6-day horizon, run as a check

After the benchmark was complete, the same protocol was run at 6 days as a check (`results_l3_v3_h6_px3_6.csv`). Its three comparisons with the mean of three are not part of the family of twelve. There are 197 test samples in six folds, testing 2019 to 2024. Persistence has an RMSE of $0.0449\,\mathrm{m^3/m^3}$, the highest of the five horizons. Against it the mean of three scores $+0.227$ (interval $[+0.130,\ +0.271]$), ridge regression $+0.292$, the random forest $+0.286$ and XGBoost $+0.237$. Over the mean of three, ridge regression gains $+0.084$ ($[+0.050,\ +0.116]$, unadjusted $p = 0.004$), the random forest $+0.076$ ($[+0.023,\ +0.128]$, $p = 0.027$) and XGBoost $+0.013$ ($p = 0.75$). Adding Sentinel-1, weather or both changes the skill of ridge regression and the random forest by at most 0.01. With these three comparisons the family would hold fifteen: ridge regression's 8-day gain would still survive the adjustment (adjusted 0.029) and its 6-day gain would not (adjusted 0.056). The 6-day result thus resembles the 8-day one, a learned gain of a few hundredths over smoothing, and is reported as suggestive. Learned gains over the mean of three were therefore seen at 3, 6 and 8 days and not at 2 or 5 days, a pattern this study does not explain.

4.8 Retrieval Quality and the Former Saturation Ceiling

Earlier drafts of this report described a saturation ceiling: a recurring maximum of 0.4634 $\mathrm{m^3/m^3}$ on 28.6% of the morning retrievals at the original point. The province scan shows what that value is. Flag 13 marks a retrieval that was attempted but not successful, and in every cell of the scan, 98.1–100% of the flag-13 retrievals (99.86% on average) sit exactly at that cell's maximum value, the upper limit of a valid retrieval, which the product documentation gives as the soil's porosity (NSIDC, 2023). The former ceiling is therefore not a measurement of saturated soil but a value that SMAP marks as an unsuccessful retrieval, and the quality rule removes it.

At cell (3,6) the unfiltered maximum, 0.4692, is a flag-13 value. After filtering, the maximum is 0.4689, reached by a single retrieval (0.1%), so the recommended-quality record has no recurring bound. The analysis of the earlier unfiltered run, which treated samples at the ceiling as censored, no longer applies.

The rule has a cost. The retrievals it rejects at cell (3,6), 597 with flag 13 and 107 with flag 9, are concentrated in the wet season: 77% of them fall in May–October, with peaks in May (114)

and in September, October and November (101, 166 and 106). Within the same months they also follow more rain: in each of the nine months with at least ten of each, the reanalysis rainfall of the preceding 3, 7 and 30 days was higher before rejected retrievals than before accepted ones, by 7, 13 and 28 mm on average, and 178 of the 704 rejected retrievals carry the surface flag for precipitation above 1 mm/h, which none of the accepted ones does. The filtered record therefore under-represents the wettest conditions. Whether the soil was saturated on those days cannot be known from these data, since the rejected values are not measurements. The benchmark of Section 4.3 says correspondingly little about forecast skill in wet conditions, although they are the ones most relevant to the flood hazard that motivates the study.

4.9 Accuracy Relative to SMAP's Requirement

SMAP's baseline mission requirement is soil moisture in the top 5 cm with an error no greater than $0.04\,m^3/m^3$ (4% volumetric), one sigma, at three-day average intervals, over the global land area excluding snow and ice, frozen ground, mountainous terrain, open water, urban areas and vegetation with water content above $5\,kg/m^2$, averaged over the product's spatial resolution (Entekhabi et al., 2010). It is a design requirement for the product as a whole, assessed against ground measurements at validation sites (Chan et al., 2018); it is neither a measured error of the retrieval at any one cell nor a limit on that error. At SMAP's core validation sites the standard 36 km radiometer product reached an average ubRMSE of $0.039\,m^3/m^3$ from April 2015 to February 2016, on recommended-quality retrievals from the morning overpass (Colliander et al., 2017); no such site lies in Mainland Southeast Asia (Ganepola et al., 2026). The requirement's limit on vegetation water content is necessary rather than sufficient for accurate retrieval (Ganepola et al., 2026). Cell (3,6) meets that condition: its vegetation water content never exceeds $2.73\,kg/m^2$.

At cell (3,6) the forecasts on the L3 history with the lowest ubRMSE reach $0.0343\ m^3/m^3$ at 2 days (the mean of three), 0.0315 at 3 days (ridge), 0.0348 at 5 days (random forest) and 0.0340 at 8 days (random forest); with Sentinel-1 added, the random forest reaches 0.0314 at 3 days. The differences between models behind this ranking were not tested. Persistence has a ubRMSE of 0.0398 to 0.0420 on the same samples. The forecasts' values lie below $0.04\,m^3/m^3$, but the comparison establishes little: they measure the error of a forecast against the retrieval, whereas the requirement concerns the error of the retrieval against the ground. The figure serves here only as a familiar scale.

An earlier draft of this report, working from the unfiltered retrievals at the urban cell, argued that the benchmark operated at the noise level of its own reference. That claim is withdrawn. The $0.04\,m^3/m^3$ figure is a design requirement that the retrieval is meant to meet over the product as a whole, and not a measured error of the retrieval at this cell.

What remains true is narrower. The target carries its own retrieval error, of unknown size at this cell, and every error measured in this chapter combines the forecast's error with that retrieval error. The morning and evening retrievals of the same date differ with a standard deviation of $0.0332\ m^3/m^3$ (Table 4.3), of the same order as the forecast errors, but that spread mixes retrieval error with genuine change between the two passes and with their different footprints, so it does not measure the retrieval error either. Under one assumption it bounds a part of it. If nothing real changed between the two passes, the spread would be the difference of two independent errors. If those errors are also of equal size in the morning and the evening pass, the part of the retrieval error that is independent from

one retrieval to the next cannot have a standard deviation above $0.0332/\sqrt{2} \approx 0.023\,\mathrm{m^3/m^3}$. Noise of that size would add twice its variance to the squared error of persistence and four thirds of it to that of the three-retrieval mean, so averaging could remove at most about 0.00037 of mean squared error. The mean of three removes 0.00045, 0.00063, 0.00047 and 0.00021 at 2, 3, 5 and 8 days. Under that assumption independent noise could account for all of the smoothing gain at 8 days and about four fifths at 2 and 5 days, but only about three fifths at 3 days, and the rest would come from variation that the two passes share or from genuine short-lived change. The assumption of equal error was not tested, and the evening-only series has the lower persistence error (Section 4.7). Without it, one pass alone could carry an error of up to $0.033\,\mathrm{m^3/m^3}$, and averaging could then remove up to about 0.00073, more than the mean of three removes at any horizon. The strength of the three-retrieval mean is thus consistent with retrieval noise that averaging suppresses, and is not shown to be wholly explained by it; the study cannot separate the two without an independent reference. Consequently, an improvement in skill against L3 cannot be verified as an improvement against the true soil moisture without an independent reference, such as in-situ measurements or a third product for triple collocation (Gruber et al., 2020), and none was used in this study.

4.10 Product Intercomparison

Table 4.22 compares the daily series of SMAP L4, the ECMWF blend soil moisture (0–7 cm, Open-Meteo best match) and MERRA-2 surface wetness over all 3,288 days, at the original study point (for L4, the $\pm 0.15^\circ$ box around it). It is a product intercomparison, not a validation: none of the three is a reference truth, so agreement between them does not establish accuracy. The units differ—SMAP L4 and the ECMWF blend are volumetric, MERRA-2 wetness is a degree of saturation between 0 and 1—so the comparison is by correlation, both on the raw series and on the anomalies left after removing each series' smoothed day-of-year climatology. SMAP L3 at cell (3,6) is not part of the intercomparison.

Table 4.22: Product intercomparison, 2016–2024, $n = 3{,}288$ days. Pearson correlation of the raw daily series and of the anomalies from the smoothed day-of-year climatology.

Pair	r **raw**	r **anomaly**
SMAP L4 vs ECMWF blend (0–7 cm, Open-Meteo best match)	0.714	0.611
SMAP L4 vs MERRA-2 surface wetness	0.900	0.701
ECMWF blend vs MERRA-2	0.785	0.621
SMAP L4 surface vs SMAP L4 root zone	0.980	0.931

Every pair agrees less on anomalies than on the raw series, because part of the raw agreement is the shared seasonal cycle. SMAP L4 agrees most closely with MERRA-2 (0.900 raw, 0.701 anomaly).

A likely reason is a shared model: the MERRA-2 reanalysis describes the land surface with the same Catchment land-surface model as the L4 system (Reichle et al., 2019), so part of that agreement may be common model behaviour and not independent confirmation. The ECMWF blend departs most from both, with anomaly correlations of 0.611 with L4 and 0.621 with MERRA-2. The L4 surface layer tracks the L4 root zone closely (0.980 raw, 0.931 anomaly), consistent with the smoothness of the L4 surface series in Table 4.7.

Table 4.23 gives each product's response to rain: the mean day-to-day change on days with more than 10 mm of rainfall and on all other days, and the correlation between rainfall and that change. The rainfall is that of the ECMWF blend (Open-Meteo best match) at the original point, and the change is taken on the same day as the rain.

Table 4.23: Response to rain, 2016–2024. Mean day-to-day change on days with more than 10 mm of rainfall and on all other days, and the correlation between rainfall and the change. SMAP L4 and the ECMWF blend in m^3/m^3; MERRA-2 in units of wetness (0–1).

Product	**Days** > 10 **mm**	**Other days**	r**(rain, change)**
SMAP L4	$+0.0106$	-0.0020	0.387
ECMWF blend	$+0.0230$	-0.0043	0.471
MERRA-2 wetness	$+0.0175$	-0.0033	0.297

All three products wet up on days with more than 10 mm of rain and dry down on the others, so each responds to rain in the expected direction. The ECMWF blend responds most strongly, with a mean wetting on those days more than twice that of SMAP L4, and correlates best with rainfall (0.471), but its soil moisture and the rainfall come from the same reanalysis, which favours it in this test. SMAP L4's same-day response is the smallest in volumetric terms. MERRA-2's magnitudes are not comparable with the other two, because wetness is not a volumetric quantity; its correlation with rainfall is the lowest of the three (0.297). With no in-situ reference, these differences show how the products diverge, not which of them is closer to the soil. For the fifth research question the intercomparison therefore gives an order of agreement, not of accuracy. It also shows that the complete products cannot stand in for one another as targets: their anomalies share only 37 to 49% of their variance (the squares of the anomaly correlations of 0.611 to 0.701).

4.11 Summary of Findings

Table 4.24 brings the chapter together. For each of the five research questions of Section 1.4 it states the key finding, with the table that holds the evidence, the most likely reason for it, and what it implies for the objectives of the study and for the thesis. The reasons are those given in the sections above. Where they were not tested, the table says so, and Chapter 5 discusses each.

Table 4.24: The research questions, the key finding for each with its evidence, the most likely reason, and what the finding implies. Skill is the skill score S of Equation 3.9; all L3 results are for cell (3,6) and recommended-quality retrievals.

Question	Key finding and evidence	Likely reason	What it implies
RQ1: skill of standard models	Every forecast built on the retrieval history, except climatology, beat persistence at 3, 5 and 8 days ($+0.054$ to $+0.238$; Table 4.8). The mean of the last three retrievals, which has no parameters, supplied most of it ($+0.069$ to $+0.204$). Beyond that mean, ridge regression added $+0.060$ at 8 days (adjusted $p = 0.022$) and a suggestive $+0.042$ at 3 days; nothing was added at 2 or 5 days (Tables 4.9 and 4.10). On L4 no learned model beat persistence (-0.131 to -0.041; Table 4.6)	One retrieval differs from the next by short-lived variation, which an average damps. Its size is consistent with retrieval noise but is not shown to be noise alone (Section 4.9). L4 is very smooth, with a lag-one autocorrelation of 0.987	A benchmark now exists for one cell. A later model must be scored against the mean of three, not against persistence alone (Table 5.2)
RQ2: lead time and the orbit	Persistence error is about flat across the benchmark horizons of 2, 3, 5 and 8 days (RMSE 0.0394 to 0.0405 over all pairs; Table 4.21), and among them skill against persistence is highest at 3 days. Learned gains over the mean of three appeared at 3 and 8 days and, in a later check, at 6 days, and not at 2 or 5 days	No consistent evidence points to orbit geometry: within each pass the order of the errors is mixed or reversed. Why 5 days shows no gain is only partly explained: the history holds information there, but ridge regression's departures from the mean cost slightly more error than they saved	A change of skill across horizons is not a measure of predictability alone. Horizons should be chosen from the pair counts that the orbit gives
RQ3: information in each input	Adding Sentinel-1, weather or both changed the skill of ridge regression and the random forest by -0.025 to $+0.012$ (Table 4.15). No interval for ridge regression excludes zero, and where the random forest changed it became slightly worse. The largest gain the intervals still allow is $+0.040$. On L4 the current value had by far the largest SHAP attribution	Untested. The radar was averaged over the 9 km cell, not the sensed area, from a scene typically two days old and of either pass. Trailing weather describes a past that the retrieval already reflects. Every score is taken against a retrieval with its own error	The result is specific to these inputs as built. Inputs of another kind or construction should be tested before a larger model (Section 5.6)

continued on next page

Table 4.24 (continued)

Question	Key finding and evidence	Likely reason	What it implies
RQ4: where the baselines fail	No recommended retrieval exists in the urban-flagged cells, 76.2% of the province (Table 4.1). The quality rule removes mostly wet-season retrievals, and a usable one exists on 10% of October days. When more than 20 mm of rain fell in the forecast window, the retrieval rose by $0.017\,\mathrm{m}^3/\mathrm{m}^3$ on average and the forecasts moved by -0.002 to $+0.004$ (Section 4.5)	The product withholds its recommendation by design where the surroundings are more than a quarter urban or the rain is heavy. No input carries the rain still to come	The benchmark says least about the flood season. A thesis model needs a way to start a forecast when no usable retrieval exists, and an input for rain to come
RQ5: the products as targets	SMAP L4, the ECMWF blend and MERRA-2 are complete; L3 has a recommended value on 29.8% of days. The lag-one autocorrelation is 0.987 for L4 and 0.676 between consecutive L3 retrievals (Table 4.7). The anomalies of the three complete products correlate at 0.61 to 0.70 (Table 4.22)	L4 and MERRA-2 are outputs of the same land-surface model. The L4/L3 contrast also mixes footprint, time sampling and location, which the study cannot separate	A result reported for "SMAP" should state the product, footprint and time sampling. No product here is a reference truth, so a claim of accuracy needs an independent reference

Chapter 5

Discussion and Conclusions

This chapter interprets the results of Chapter 4, draws out what they imply for the thesis, and states the limitations. Unless stated otherwise, every statement about SMAP L3 refers to cell (3,6), to recommended-quality retrievals, and to the morning-first target defined in Chapter 3.

5.1 What the Benchmark Establishes

The study set out to measure how well standard machine-learning models forecast short-range soil moisture in Pathum Thani, with persistence as the reference. That measurement now exists for one non-urban SMAP cell, checked at two neighbouring cells, and for the SMAP L4 box around the original study point. Its answer is conditional: it differed between the settings examined, and those settings differ in the product used as the target, in the cell and in which retrievals are kept, often in several of these at once.

At this cell, the L3 retrieval is forecastable beyond persistence at 3, 5 and 8 days. With recommended-quality retrievals at a cell that SMAP does not flag as urban, every forecast built on the L3 history alone—the three-retrieval mean, damped persistence, and the ridge, random-forest and XGBoost models—beat persistence at 3, 5 and 8 days, with moving-block bootstrap intervals that exclude zero. At 2 days the result was weaker. Only the three-retrieval mean and ridge regression beat persistence with an interval above zero. Damped persistence, the random forest and XGBoost could not be distinguished from it at this sample size: their skills ($+0.008$, $+0.070$ and -0.053) all had intervals that included zero, XGBoost's being $[-0.199, +0.125]$, wide enough to hide a real gain or a real loss. Seasonal climatology had negative skill at 2, 5 and 8 days, though none of these intervals excluded zero, and could not be distinguished from persistence at 3 days. On the L3 history alone, XGBoost was the weakest learned model at 2, 3 and 8 days and level with ridge regression at 5 days ($+0.139$ against $+0.136$).

Most of the gain comes from smoothing. The mean of the last three retrievals (Equation 3.2) has no parameters, yet it scored $+0.147$, $+0.204$, $+0.144$ and $+0.069$ against persistence at 2, 3, 5 and 8 days. Learned models on the L3 history added a small further gain over that mean at 8 days. Ridge regression's, $+0.060$ (Diebold–Mariano $p = 0.002$), survives the Holm adjustment for the twelve comparisons both in the main run and in the purged run (Table 4.10), where it is unchanged; it was positive in six of the seven test years, and it also recurs, before adjustment, at both neighbouring cells ($p = 0.013$ and $p < 0.001$) and with the two-pass target ($+0.067$, $p = 0.0003$, which survives that run's own adjustment; Table 4.13). The random forest's $+0.081$ ($p = 0.0007$) survives the adjustment only in the main run: in the purged run it becomes $+0.071$ ($p = 0.014$, adjusted 0.12). Only ridge regression's 8-day gain is therefore established in both runs, within the twelve comparisons, a family fixed after the results were known. The verdict rests on that family: over all 48 comparisons with

the mean on the four feature sets, ridge regression's adjusted p would be about 0.10 (Section 3.8). Two later checks show how close to the threshold both gains sit (Section 4.3). With a penalty range widened afterwards, ridge regression's gain is $+0.057$ and its adjusted p is 0.051. And the random forest's gain passes or fails with its tuning choices, which are selected on a single held-out year, not with the removed pairs: on the purged pairs with the main run's tuning it is $+0.079$ ($p = 0.0009$). The 8-day gain of the learned models over smoothing is therefore best described as small, positive in every variant and, for ridge regression, in six of seven test years and at all 20 non-urban cells of the scan (Section 4.3.3), and close to the threshold of the adjusted test. A 6-day run made afterwards gave a similar gain for ridge regression ($+0.084$, unadjusted $p = 0.004$; Section 4.7.1). At 3 days ridge regression added $+0.042$ ($p = 0.009$, interval above zero), but that gain does not survive the adjustment and is not repeated at cell (4,6) or with the two-pass target; the random forest's $+0.040$ had $p = 0.076$ with the seed used and p from 0.037 to 0.076 over 21 seeds. The 3-day increment is therefore suggestive, not established. The learned models added nothing significant at 5 days (random forest $+0.032$, $p = 0.17$), and nothing at 2 days, where the three-retrieval mean was the best forecast and every learned model scored below it. Scored against persistence alone, the learned models would therefore be credited with skill that simple smoothing already supplies. Scored against the three-retrieval mean, their contribution is a few hundredths: established at 8 days and suggestive at 3 days. The meaningful question at this cell is therefore not whether a learned model beats persistence, which a parameter-free average already does, but whether it adds information beyond that simple smoothing baseline. Asking the stricter question makes the benchmark stronger, but no test year was held back untouched (Section 3.1): the three-retrieval mean, DAM3 and the horizons were chosen with the results of these test years in view. A later model shows more than smoothing only by passing the tests set out with Table 5.2, best on a test year that this study did not use.

A plausible reason why smoothing helps is that a single retrieval carries considerable short-term variation. On the 691 days with both a morning and an evening retrieval, the two correlate at only 0.739, and their difference has a standard deviation of $0.0332\,\mathrm{m^3/m^3}$, not far below the persistence RMSE of 0.040 to 0.042 in the test sets. Part of that difference may be genuine change between the two overpasses, and part may reflect their different footprints. Whatever part does not persist, however, averaging three retrievals damps. Under the assumption that the two passes carry independent errors of equal size, the same spread bounds how much of the gain such noise could explain: it could not exceed a standard deviation of about $0.023\,\mathrm{m^3/m^3}$, which could account for all of the smoothing gain at 8 days, about four fifths of it at 2 and 5 days and three fifths at 3 days. Without that assumption, which was not tested, one pass alone could carry enough noise to account for the gain at every horizon (Section 4.9). The gain from smoothing is therefore consistent with retrieval noise and is not shown to be wholly explained by it, and the study, without an independent reference, cannot separate noise from genuine short-term change.

This result sits beside the premise of Chapter 2, that soil-moisture memory makes the recent state a strong predictor of the next. That premise comes from a review and a model study of the soil column (Orth & Seneviratne, 2013; Seneviratne et al., 2010); studies of the layer that SMAP senses had already pointed to a shorter memory (McColl, Alemohammad, et al., 2017; Shellito et al., 2016). At this cell a single surface retrieval proved a weak predictor of the next one: consecutive retrievals correlate at 0.676, and persistence could not be distinguished from climatology. What persisted was the level that several retrievals share, which is what the mean of three estimates.

Why the learned gain differs across horizons. Three things were observed (Section 4.3). At 2 days the test set is the smallest, 203 samples over six folds, and ridge regression spreads its weights almost equally over four retrievals, so it has little to add to an average of three. At 8 days the mean of three is at its weakest against persistence ($+0.069$) and the learned models' gain over it is at its largest. And DAM3, which only damps the mean towards the season, also scores above the mean at 3 and 8 days ($+0.056$ and $+0.046$). Two readings are plausible and were not tested: that the mean of three keeps the whole of a recent departure from normal, which costs most at the longest horizon, and that the damping a learned model applies matters more as the horizon grows. One thing is only partly explained: why 5 days shows no gain while 3, 8 and, in the later check, 6 days do. A test built for nested models, computed afterwards (Section 4.3), finds that the retrieval history holds information beyond the mean of three at 5 days as at 3 and 8, and none that it can detect at 2 days. At 5 days ridge regression's departures from the mean pointed in the right direction on average but cost slightly more error than they saved. Why they paid at 3 days and not at 5, on the same number of test samples, is not known. Orbit geometry was examined as a cause and found no consistent support (Section 4.7).

On the real record much of the 3-day gain is the filtering of short-lived departures. The day-by-day forecasts (Section 4.5) support the reading that smoothing helps by damping short-lived variation. At 3 days, 59–65% of the gain over persistence of the three-retrieval mean, ridge regression and the random forest came from the one forecast in ten that touches an isolated retrieval; without those forecasts their skill was $+0.112$, $+0.155$ and $+0.151$, lower but still more than half of their skill with them. The share is descriptive and partly built into the label, which uses the retrieval after the issue date (Section 4.5). At 8 days the same share was 12–23%, but it is a net of opposite parts: forecasts issued from an isolated retrieval supplied 50% of the gain of the three-retrieval mean, 30% of ridge regression's and 20% of the random forest's, offset by losses where the target was isolated. The random forest's 8-day gain is thus largely independent of isolated retrievals; the three-retrieval mean's is not. Not every isolated retrieval is noise that is independent between passes: where the same date has a second pass, it supports the isolated value in half of the cases (17 of 34), which may reflect a brief real change or an error that both passes share. When the retrieval changed by $0.04\,\mathrm{m^3/m^3}$ or more over 3 days, the forecasts moved in the right direction but only about half as far on rises and a third as far on falls, a shortfall partly expected of any forecast scored against a target with short-lived variation, and when more than 20 mm of rain fell during the 3-day window they barely moved (-0.002 to $+0.004$, against an observed $+0.017$). The benchmark therefore measures two things: filtering departures that do not last, and following drying and recovery. It does not measure a forecast of the wetting that followed heavy reanalysis rainfall, which none of the inputs can supply; that the rain caused this wetting is plausible but was not tested. The two kinds of change must be kept apart (Section 4.5.5): isolated, short-lived changes in the retrievals, which the forecasts handle by smoothing, and wetting after heavy rain and in the wet season, which they do not anticipate. The second kind is also the one the quality filter removes most, so the benchmark under-represents the wettest conditions (Section 5.4).

No learned model clearly beat a one-parameter diagnostic. Re-scored with the yardsticks of Section 4.6, the benchmark's first conclusion holds: against seasonal climatology, which at this cell could not be distinguished from persistence, every learned model still gained $+0.10$ to $+0.23$. The second conclusion needs qualifying. The damped anomaly of the three-retrieval mean (Equation 3.17) keeps the seasonal normal and 70–82% of the recent average departure from it, with that fraction fitted on the training years, and no learned model beat it with p below 0.05 at any horizon when its

climatology was built from all training-year retrievals, whether with the ± 7-day window or with the 31- and 35-day windows that validation guidance recommends (Gruber et al., 2020) (Table 4.20). With the benchmark's own climatology of training targets, and with its purged variant, the random forest did beat it at 8 days ($+0.046$, $p = 0.039$ and 0.038, unadjusted): two of the seven definitions tested. The small increment that the learned models added over the three-retrieval mean at 3 and 8 days is therefore largely, though perhaps not entirely at 8 days, a pull towards the seasonal normal, which a single fitted fraction supplies. This rule is a diagnostic, post hoc benchmark and not another proposed forecasting model: it was defined after the main results were known, so the comparison is a check rather than a planned test, although it uses only training years and two standard ideas, smoothing and damping. What it shows is how difficult it is for the learned models to demonstrate information beyond a very simple seasonal adjustment of recent observations. For ridge regression part of the pull is observed directly: the weights it gives the four recent retrievals sum to 0.72 to 0.83, so it keeps between three quarters and four fifths of the recent level. That the rest is drawn towards the season, as in the diagnostic, is a reading of its other terms, whose coefficients are not reported (Section 4.3). For the random forest it remains an inference from similar scores. The fitted fraction for one retrieval, about 0.5 at every horizon from 2 to 8 days, points the same way, that part of each retrieval's departure from normal is short-lived, as retrieval noise would be; it rests, however, on the same property of the series as the strength of the three-retrieval mean, the weak correlation between successive retrievals, and is not independent evidence.

The ranking depends on the metric. The learned forecasts vary only 64–77% as much as the retrievals. That caution lowers RMSE, the basis of the skill score, but under the Kling–Gupta efficiency, which penalises it, persistence ranked level with or above the learned models, and in three categories of dry, normal and wet the learned models' Heidke scores at 3 to 8 days were close to persistence's while they flagged fewer of the observed wet cases. For a use in which missing a wet spell is costly, the lowest-RMSE forecast is not automatically the most useful one.

An error against the retrieval is not an accuracy against the soil. The best forecasts reached an ubRMSE of 0.0314 to $0.0348\,\mathrm{m^3/m^3}$ against the retrieval across the four horizons. These values are below $0.04\,\mathrm{m^3/m^3}$, SMAP's accuracy requirement (Entekhabi et al., 2010), but the two numbers measure different things. The requirement concerns the retrieval's error against the ground, over the product as a whole and outside excluded conditions; it is a design target, neither the measured retrieval error at this cell nor a limit on it. Being below it is therefore no evidence that the forecasts are accurate against the true soil moisture. An earlier version of this analysis claimed that the benchmark sat at the reference's own noise level. That claim is withdrawn. What remains true is that the target carries its own retrieval error, of unknown size here, and that the measured error of every forecast includes it.

On SMAP L4, no model beat persistence. Against SMAP L4 the best learned model, XGBoost, scored -0.131, -0.066 and -0.041 at 1, 3 and 7 days. At 1 and 3 days the interval lay below zero and p was below 0.05. At 7 days the interval included zero while the Diebold–Mariano test gave $p = 0.048$, unadjusted: a loss by the point estimate that is not established. SHAP, computed for the random forest at 7 days, ranked the current value first: its mean absolute SHAP value, 0.054, was about 4.5 times that of the next feature, the previous day's value (0.012). Although the current value thus had by far the largest attribution of any single feature, the learned models still scored below

persistence. SMAP L4 is produced by assimilating SMAP observations into a land-surface model, which supplies temporal continuity. Its lag-one autocorrelation over 2016–2024 is 0.987 at a daily step, against 0.676 between consecutive L3 retrievals at cell (3,6).

The L4/L3 contrast is confounded. One reading of the difference between the two outcomes is that processing level matters: a series smoothed by a land-surface model leaves little for a forecaster to add beyond persistence. The two benchmarks, however, differ in more than processing level. The L4 series is a box mean of $\pm 0.15^{\circ}$ around the original study point, 57% inside the province, and each daily value is the average of eight 3-hourly values. The L3 target is the value at one cell of the 9 km grid, observed at one instant, 38.1 km from the original point; since that value represents an area about 36 km across (Section 3.3.5), the two differ less in footprint than the descriptions suggest, and more in location. The L3 autocorrelation is measured between retrievals a median of 3 days apart, not between consecutive days. The two benchmarks also differ in horizons (1, 3 and 7 days against 2, 3, 5 and 8) and in how skill is aggregated: the L4 skill is a mean of per-fold skills over eight folds testing 2017–2024, whereas the L3 skill is computed once over the pooled samples of six folds (2 days, testing 2019–2024) or seven (testing 2018–2024). The contrast therefore mixes product level with footprint, time sampling and location, alongside these differences of design, and this study cannot apportion the difference among them. The L4 result is therefore kept separate from the L3 conclusions, and the two should not be interpreted as a direct product-against-product comparison of models. The pair does support a practical point: a result reported for "SMAP" can depend on which product, footprint and time sampling were used, and a study should state all three.

What the benchmark means for a user. The intended audience for a forecast of this kind is the province's irrigation, water-resource and disaster-preparedness offices. They are named as an audience, not as users: none of them used or assessed these forecasts, so nothing here is a validated benefit to a user. Table 5.1 sets what the benchmark supports beside what it cannot say. Two comparisons give the scale in the product's own unit. The mean of the last three retrievals lowered the RMSE of persistence by about 0.0085 m^3/m^3 at 3 days (from 0.0414 to 0.0329) and by about 0.0027 at 8 days (from 0.0398 to 0.0371). At 8 days ridge regression lowered the mean's RMSE by a further 0.0023 and the random forest by 0.0030 (Table 4.8). These gains are small beside the error that remains (ubRMSE 0.031–0.035 m^3/m^3, against the retrieval) and beside the unbiased RMSE of about 0.04 m^3/m^3 to which the enhanced product has been validated elsewhere (Chan et al., 2018). The target itself has a standard deviation of 0.049 m^3/m^3 and lies between 0.30 and 0.46 on nine days in ten, so the error that remains is about two thirds of the target's own spread. For a user, most of what the benchmark offers is the three-retrieval mean; whether a further gain of 0.002 to 0.003 m^3/m^3 would change a decision has not been tested, and testing it needs a decision threshold agreed with the user.

5.2 Implications for the Thesis

The benchmark to beat has two references and a stated test. Table 5.2 gives the reference scores at this cell. A thesis model must be compared with persistence *and* with the three-retrieval mean, on the same samples and folds, under the same tuning and testing protocol (Section 3.8), and with the Diebold–Mariano test applied against both references. A skill score against persistence alone

Table 5.1: What the benchmark supports for a user, and what it cannot say.

	Supported	**Not supported**
Place	Cell (3,6), whose value represents a footprint about 36 km across, roughly the size of the province, with 53–59% of that area inside it (Section 3.3.5); two overlapping neighbours gave the same broad pattern, and ridge regression's 8-day gain was positive at all 20 non-urban cells (Section 4.3.3)	Any field, farm or district; the urban-flagged cells, so 78.1% of the province cannot be benchmarked on recommended L3 retrievals
Season: data	The dry season has the denser record: a usable retrieval on 36–40% of days from January to April, and the latest one a median of 1 day old from November to April	The flood season: a usable retrieval on 10% of October days, the latest one at least 5 days old on 47% of days from September to November, and the wettest states removed by the quality filter
Season: skill	The mean of three, ridge regression and the random forest gained over persistence in both seasons at every horizon (Table 4.11)	Any difference in skill between the seasons: none was tested
Horizon	The smoothing gain over persistence at 2, 3, 5 and 8 days (at 8 days its lower bound is close to zero). A learned gain beyond smoothing at 8 days only: ridge regression's, within the twelve comparisons made	A learned gain beyond smoothing at 2, 3 or 5 days (at 3 days it is suggestive only); the 1-, 4- and 7-day horizons, which were not run; 6 days was run only as a later check
Lead time	Counted from the date of the latest retrieval, with the forecast issued at the end of that UTC day (Section 3.5)	The lead time a user would have. With the 50-hour latency listed for the product (Chan et al., 2018), about 0, 1, 3 and 6 days would remain of the four horizons; the delay was not measured here
Variable	Surface soil moisture (top $\sim$5 cm) as SMAP retrieves it	Root-zone wetness, crop water need, or accuracy against the ground, since no in-situ reference exists at the cell

would credit the model with what smoothing supplies. At 8 days damped persistence ($+0.089$ against persistence) scored above the three-retrieval mean ($+0.069$), though with $p = 0.38$, so at that horizon a thesis model should also be set against damped persistence. The re-scoring of Section 4.6 adds a stronger simple reference: the damped anomaly of the three-retrieval mean (Equation 3.17), which no learned model beat with p below 0.05 except under the benchmark's own climatology, built from training targets only (Section 4.6.5).

A thesis model passes the bar at a horizon only if three conditions hold. First, its gains over the three-retrieval mean and over DAM3 survive a Holm adjustment over every configuration tried at that horizon, those of this study and every model added later, so that the family grows as models are added; the twelve-comparison family of Table 4.9 belongs to this study alone. Second, a Diebold–Mariano test on the same samples shows it ahead of ridge regression on the L3 history, the single primary bar at each horizon. Ridge regression is chosen because it has no random seed, because its 8-day gain is the only one established in both the main and the purged run, within the twelve comparisons of this study, and because its weights can be read directly. The bar is ridge regression as tuned here, with its penalty capped at 100, the value chosen in 17 of its 27 fits on the L3 history (Section 3.8). A range widened to 10,000, tried afterwards, improved no score and lowered the 8-day gain to $+0.057$ (Section 4.3). The best of the twelve learned configurations, picked after the results were seen, is a description, not a bar. Third, because every test year of this study was used in choosing the references and the horizons (Section 3.1), the comparison should be repeated on a test period that this study did not use, such as 2025 onwards. One further year holds about 100 test samples at 8 days, and on a single year a gain is poorly determined: ridge regression's 8-day gain over the mean ranged from -0.092 to $+0.108$ across the seven test years here. A null on the new period should therefore be reported as not measurable, with its interval, and not as a failure. A point estimate above Table 5.2 is not, by itself, an improvement. The model should also be reported with at least one score that does not reward caution, such as the Kling–Gupta efficiency or the hit rate for wet conditions.

More model capacity brought no clear gain on the same history. On the L3 history alone, ridge regression, a linear model, matched the random forest at 3 days ($+0.238$ against $+0.237$) and scored higher at 2 days ($+0.106$ against $+0.070$), while the random forest scored higher at 5 and 8 days ($+0.172$ against $+0.136$, and $+0.144$ against $+0.125$). None of these differences between the two models was tested. XGBoost, the most flexible of the three, was the weakest at 2, 3 and 8 days. No learned model improved on the three-retrieval mean by more than $+0.081$, and neither Sentinel-1 nor weather added measurable additional information (Section 5.3). At this sample size and with these inputs, model capacity does not appear to be the binding constraint: where the random forest led the linear model, it did so by a few hundredths, and none of them clearly beat the one-parameter damped anomaly of the three-retrieval mean (Tables 4.18 and 4.20). Ridge regression's own weights show that its gain is a re-weighting of the same retrievals, read as a damping towards the season (Section 4.3), so where the forest only matched it, the forest's extra capacity found nothing that such an average lacks. XGBoost's scores are a lower bound for its kind of model at this sample size, because many of its tuning choices fell at the edge of their ranges (Section 3.8). Where progress would come from remains open. It may come from inputs of a different kind, meaning observations that carry information not already in the retrieval history, or from a better reference against which improvement can be measured. This study gives no sign that a bigger model trained on the same history would deliver it, but the differences between its models were not tested and no larger model was tried, so it does not rule that out.

Table 5.2: Reference scores and tests for a thesis model at cell (3,6): recommended-quality retrievals, morning-first target, pooled over walk-forward folds with test years up to 2024. Persistence RMSE in m^3/m^3; skill S follows Equation 3.9. Mean3 is the mean of the last three retrievals (Equation 3.2) and DAM3 its damped anomaly (Equation 3.17), defined after the main results. Ridge regression on the L3 history, with its penalty capped at 100, is the primary bar; its gain over mean3 is given with its Diebold–Mariano p-value and the Holm-adjusted value of Table 4.9. The best of twelve learned configurations (three models, four feature sets), selected after the results were seen, is descriptive only. The tests a thesis model must pass are stated in the text; a point estimate above this table is not an improvement.

	Persistence	S **against persistence**			**Ridge vs mean3**	**Best of twelve**	
Horizon (n)	**RMSE**	**Mean3**	**DAM3**	**Ridge, L3**	S (p; **Holm**)	**Model**	S
2 days (203)	0.0407	+0.147	+0.142	+0.106	−0.049 (0.18; 0.86)	ridge, L3+met	+0.112
3 days (434)	0.0414	+0.204	+0.249	+0.238	+0.042 (0.009; 0.092)	random forest, L3+S1	+0.241
5 days (434)	0.0420	+0.144	+0.165	+0.136	−0.010 (0.70; 1)	random forest, L3	+0.172
8 days (639)	0.0398	+0.069	+0.112	+0.125	+0.060 (0.002; 0.022)	random forest, L3	+0.144

Learned assimilation on L4 is difficult to justify. The thesis plan originally placed AI-assisted data assimilation at its centre. SMAP L4 is already the output of an assimilation system, and adding a second, learned assimilation layer to it would need a clear statement of what the first one failed to do. This study provides no evidence of such a deficiency, although the benchmark was not designed to reveal one. What it shows is that L4 is very smooth, with a lag-one autocorrelation of 0.987, and that neither learned model beat persistence on it. That direction is therefore on hold. Working from Level 3 places any assimilation closer to the observations, and the record leaves room for it: at this cell a recommended-quality retrieval exists on only 29.8% of days.

Improvement against the truth needs an independent reference. Because the target carries retrieval error of unknown size, a forecast that scores better against L3 may be closer to the true soil moisture. It may instead be closer to the retrieval's own pattern of error, and L3 alone cannot tell which. An independent reference is therefore a precondition for claiming that a method improves soil-moisture estimation, not merely a desirable validation step.

5.3 On Complementary Observations

Sentinel-1 was obtained and tested. Radiometrically terrain-corrected Sentinel-1 backscatter was read from Microsoft Planetary Computer: 846 scenes over the scan box for 2016–2024, none of which failed. At cell (3,6), 559 dual-polarisation scene-days cover at least half of the cell. Scenes were paired backward only, taking the latest scene on or before the issue date, and 978 of the 981

retrieval days had one. A two-sided window around the issue date would have paired more days, but only by using scenes acquired after the forecast is issued. That would leak future information into the predictors, so such windows were not used.

Neither Sentinel-1 nor weather added measurable additional information. For ridge regression and the random forest, adding Sentinel-1, weather or both to the L3 history changed skill by at most 0.02 at 2, 3 and 5 days, and by -0.025 to $+0.004$ at 8 days. For ridge regression the changes have no consistent sign, and every moving-block interval includes zero. For the random forest 11 of the 12 changes are negative, and adding weather lowered its skill at 8 days (-0.025, interval $[-0.053,\ -0.003]$) and marginally at 5 days (-0.011, $[-0.024,\ -0.0003]$): where the added inputs did anything, they made that model slightly worse. XGBoost's changes were larger, from -0.051 to $+0.078$, but at no horizon did they lift it to the level that ridge regression and the random forest reached on the L3 history alone. The highest 3-day score, $+0.241$ for the random forest with Sentinel-1, differs from the same model without it by 0.005. The finding is specific: there was no measurable additional information under the present data, quality filtering and evaluation protocol, that is, at this cell and these horizons, with this pairing and this spatial averaging. The intervals still allow a gain of a few hundredths of skill, at most $+0.040$ (ridge regression with weather at 5 days; Section 4.5), so a small contribution cannot be ruled out. The result does not show that Sentinel-1 or weather carries no information about soil moisture in general.

Possible reasons for the radar result. Five explanations are plausible. The study did not test them separately.

- *Physics.* Sentinel-1 operates at C-band. Compared with the L-band of SMAP, C-band senses a shallower surface layer and penetrates vegetation less well (Ganepola et al., 2026). Radar retrievals in the region are also affected by surface roughness, which frequent rainfall and farming alter between overpasses, and by water bodies and speckle (Ganepola et al., 2026).
- *Scale.* The backscatter was read at about 80 m and averaged over a 9 km cell. Field-scale contrast, one of the radar's main advantages, is averaged away, and the cell mean that remains may say little that the radiometer retrieval does not already say. In the other direction, the 9 km cell is much smaller than the area of about 36 km that the radiometer senses (Section 3.3.5), so the radar and the target do not describe the same ground.
- *Timing.* At the issue date the paired scene was typically 2 days old (median age), whereas the retrieval $\theta(t)$ is taken on the issue date itself. The scene therefore usually predates the most recent observation of the state it is meant to inform. Timing alone does not explain the result, however: with a scene from the same or the previous day, adding Sentinel-1 to the random forest changed 3-day skill by -0.004 $[-0.021,\ +0.012]$ (Section 4.5).
- *Mixed passes.* The feature takes the latest scene of either pass, ascending or descending, without saying which. Backscatter depends on the viewing geometry and the time of day as well as on soil moisture, and the report's own correlations are higher within the descending pass (0.38 to 0.39) than over all scenes (0.24 and 0.32).
- *The target's own error.* Every score is taken against a retrieval whose error at this cell is unknown. An input that tracks the soil, but not the retrieval's error, earns no credit against such a target.

A fused product already exists. NASA's SPL2SMAP_S combines the SMAP radiometer with Sentinel-1A and 1B at one to three kilometres (Das et al., 2019). It is the product that Jotisankasa et al. (2023) validated over Thailand. Any thesis proposing to combine L3 with Sentinel-1 must therefore state how it differs. Examples include forecasting rather than estimating the present state, using the radar at field scale rather than as a 9 km mean, or using it to inform the days on which no retrieval exists.

Weather. Rainfall, temperature and shortwave radiation from the ECMWF reanalysis (ERA5/ERA5-Land, Open-Meteo ERA5-Seamless) also added no measurable information. Radiation was absent from the earlier analysis and is now included. A plausible reason is that trailing weather describes the past, and the past is already expressed in the retrieval at the issue date. The real record is consistent with this: even after at least 20 mm of rain in the previous three days, adding weather to the random forest changed 3-day skill by only $+0.001$ $[-0.042, +0.047]$, and the forecasts did not anticipate the wetting that followed rain during the forecast window (Section 4.5). Rainfall still to come might supply what the trailing windows cannot, but that is a hypothesis this study did not test, and its own figures counsel caution: rainfall between retrievals correlates with the change in the retrieval at only 0.16, and after more than 30 mm the retrieval rose in only 66% of cases. These weak links may reflect the reanalysis rainfall, a model estimate for an area of about 27 km never compared with a gauge, or wetting that does not come from rain, such as irrigation, as much as the soil. The direct test, giving the models the reanalysis rainfall that fell in the forecast window, was not run. It would be an upper bound for rainfall of that kind, not for a perfect forecast: a gain would support going on, but a null would not close the question until the reanalysis rainfall has been compared with gauges or with another rainfall product for the sensed area.

5.4 Limitations

One cell. The main L3 results are for a single cell of the 9 km grid, (3,6), in the east of Thanyaburi district, whose value represents an area about 36 km across that reaches beyond the province: 53–59% of that area lies inside it, and about two fifths of it lies on cells that SMAP flags as urban (Section 3.3.5). Among the 12 cells at least 90% inside the province only two are not urban, and this one has the more recommended-quality days. Spatial variation of forecast error, which the advisor asked to be examined, cannot be assessed from one cell. It was examined afterwards for the reference forecasts and ridge regression at the 20 non-urban cells of the scan (Section 4.3.3): persistence was less accurate in the west, the mean of three beat it at every cell at 2, 3 and 5 days, and ridge regression's 8-day gain over the mean was positive at all 20. The random forest, XGBoost, Sentinel-1 and weather were not run there, and no urban-flagged cell can be scored on recommended retrievals, so the results still may not transfer to most of the province. Three further runs of the whole protocol bound this limitation without removing it. The adjacent non-urban cells to the north and south, (2,6) in Nong Suea district and (4,6) in Lam Luk Ka district, reproduced the broad pattern: the three-retrieval mean was the best forecast at 2 and 5 days, ridge regression and the random forest scored above it at 3 and 8 days, and Sentinel-1 and weather showed no consistent gain. The two cells share most of their footprint with cell (3,6), and their target series correlate with its series at 0.93 and 0.94, so they show that the result does not hinge on the exact grid position rather than that it holds elsewhere. The random forest's gain over the mean had p below 0.05 at 8 days ($+0.050$, $p = 0.027$, and $+0.063$, $p = 0.010$)

but not at 3 days ($+0.033$ and $+0.036$); these p-values are not adjusted (Table 4.13). Which model's 3-day gain over the mean fell below $p = 0.05$ varied: ridge regression's did at cell (2,6) ($+0.046$, $p = 0.003$), as at cell (3,6), but not at cell (4,6) ($+0.031$, $p = 0.090$), where neither model on the L3 history alone did. At the original urban cell (4,2), where no retrieval is of recommended quality and all retrievals were therefore used, persistence was less accurate (RMSE 0.049 to 0.055, against 0.040 to 0.042 at cell (3,6)), and the random forest added nothing over the three-retrieval mean at any horizon ($+0.001$, -0.007, $+0.018$ and $+0.003$ at 2, 3, 5 and 8 days; Diebold–Mariano p from 0.45 to 0.99), and neither did ridge regression or XGBoost (p from 0.22 to 0.67). A gain over smoothing therefore appeared only where retrievals were of recommended quality at a non-urban cell: a small increment at 8 days, which survives adjustment at cell (3,6) and has unadjusted p below 0.05 at both neighbours, a less consistent one at 3 days, none at 5 days, and none at 2 days. With not-recommended retrievals at the urban cell, no learned model's gain reached $p = 0.05$ at any horizon. The two settings differ in both the cell and the quality of the retrievals, and the urban series includes values at the retrieval's upper limit (flag 13 on 28% of its days), so these runs do not show which of these removes the increment; and because no urban-flagged cell carries a recommended-quality retrieval, the urban cells cannot be benchmarked on recommended retrievals at all. A run with all retrievals at cell (3,6), which holds 704 not-recommended retrievals, would hold the cell fixed, but it was not carried out. The three non-urban cells of the full protocol also lie in one column of the grid in the east of the province; at the other non-urban cells only the reference forecasts and ridge regression were scored.

Most of the province is outside the benchmark. Of the province's area, 76.2% lies in cells that SMAP flags as urban, and a further 1.9% lies in cell (5,3), which returned no retrieval at all; the non-urban cells cover 21.9%. The urban-flagged cells carry no recommended-quality retrieval over 2016–2024, including the cell of the original study point. The benchmark therefore describes one non-urban cell, checked at two of its neighbours, and not the urban-flagged cells that make up most of the province's area.

The wettest conditions are under-represented. The quality rule removes the flag-13 retrievals, which sit at the cell's maximum value, and the flag-9 retrievals. At cell (3,6) the rejected retrievals are flag 13 (597) and flag 9 (107). Of these, 77% fall in May–October, with peaks in May (114) and in September, October and November (101, 166 and 106), and within the same months they follow more rain than the accepted ones (Section 4.8). The filtered record therefore under-represents the wettest conditions; whether the soil was saturated on the rejected days cannot be known, because the rejected values are not measurements. These are the conditions of most interest for the flood hazard described in Chapter 1, and the benchmark says little about forecast skill under them.

The flood season is data-poor. The same rule leaves few usable retrievals when a forecast would matter most. A recommended-quality retrieval exists on 10% of October days, against 36–40% of days from January to April. From September to November the latest usable retrieval is at least 5 days old on 47% of days and at least 10 days old on 37%, and in 2022 there was almost no usable retrieval from August to late November (Figure 4.1), partly because the satellite was in safe mode from 7 August to 19 September. A forecast system built on this record could issue few forecasts in the peak flood season, and the forecasts it could issue would start from stale states. This is a property of the data, which no choice of model can change.

Land use and irrigation. Land cover was not joined, so the crops of cell (3,6), and whether its fields are irrigated or flooded on a crop calendar, are not known from the data used here. Irrigation and field flooding change surface wetness independently of rain. They are a possible, untested cause of the changes that the forecasts did not follow and of the weak link between rainfall and the retrieval, alongside retrieval noise and the rainfall data themselves.

Model settings. Many tuning choices fell at the edge of their ranges (Section 3.8), so wider ranges might have changed some scores, and XGBoost's most; its scores are a lower bound for that kind of model at this sample size. For ridge regression a wider penalty range was tried afterwards: it improved no score, and it lowered the 8-day gain over the mean of three from $+0.060$ to $+0.057$, enough to move its adjusted p from 0.022 to 0.051 (Section 4.3). Settings are chosen on one held-out year of 25 to 113 pairs, and both 8-day verdicts are sensitive to that choice.

No in-situ reference. The International Soil Moisture Network holds no station in Thailand, and Dr. Natthachet Tangdamrongsub (personal communication, 18 September 2026) confirmed that no publicly accessible soil-moisture network currently exists in the country. The Hydro-Informatics Institute operates ten stations, used by Jotisankasa et al. (2023). The three nearest to the original study point lie 114–128 km away in Saraburi, Chai Nat and Prachinburi, and no data request was sent during the study. The regional picture is the same. Ganepola et al. (2026) find no SMAP core or candidate validation site in Mainland Southeast Asia, and they recommend at least 13 stations within one 36 km grid cell for robust validation of coarse products. The comparison of SMAP L4 with the ECMWF blend (Open-Meteo best match) and MERRA-2 is a product intercomparison, not validation, and SMAP L3 at cell (3,6) was not part of it. Forecasts were therefore scored against the retrieval itself, whose error at this cell is unknown, and improvements could not be verified against the true soil moisture. Indirect checks that need no in-situ data, such as triple collocation with two other independent products (Gruber et al., 2020), were not attempted.

Horizons chosen within the orbit's limits. The horizons of 2, 3, 5 and 8 days were chosen from the separations with enough retrieval pairs (Section 3.5.1) and approved by the advisor on 6 October 2026. One and 7 days were excluded at this cell because exact 1- and 7-day pairs number only 41 each, too few for the walk-forward protocol, and 4-day pairs do not occur. Six-day pairs are numerous enough (326) but 6 days was not made a benchmark horizon, so that the four horizons stayed consistent with the analysis presented on 23 September 2026. It was run afterwards as a check, outside the family of twelve comparisons (Section 4.7.1); its persistence RMSE in the test samples (0.0449) is the highest of the five horizons. The horizons also differ in geometry. Four in five of the 8-day pairs come from the same track, and the 2-, 3- and 5-day pairs from different tracks, so a comparison of skill across horizons compares lead times and orbit geometries together. The lower pooled persistence error at 8 days (0.0394 against 0.0398 to 0.0405) might suggest an effect of same-track pairing, but there is no consistent evidence for one: the pooled figure is pulled down by pairs that join the morning and evening passes, which are not same-track, among morning pairs the 8-day error is not below the 2-day error, and among evening pairs it is the highest (Section 4.7). The differences are small and were not tested. What is supported is that persistence error does not increase from 2 to 8 days, so the horizons are not a sequence of steadily growing difficulty.

Small samples. The test sets hold 203 samples at 2 days (6 folds), 434 at 3 and 5 days, and 639 at 8 days (7 folds each). The 2-day result rests on the fewest. With 6 or 7 folds, the per-fold Wilcoxon test cannot return a p-value below 0.031 or 0.016 respectively. It serves as a check that results are consistent across folds, not as strong evidence.

One random seed. The random forest and XGBoost are built with random numbers, and the benchmark fitted each once, with one seed. Refitting them with 20 other seeds, on the same folds and with the tuning choices of the main run held fixed (`l3_seed_sensitivity.py`), moved the random forest's skill against persistence over a range of 0.004 to 0.009 at a given horizon, and XGBoost's over 0.016 to 0.033; Table 4.10 gives the corresponding ranges of the gains over the three-retrieval mean. The clearest result did not change: the random forest's 8-day gain over the three-retrieval mean had p below 0.05 for all 21 seeds (0.0004 to 0.0013). XGBoost's 8-day gain also stayed below 0.05 for all 21, but over a wide range (0.003 to 0.043), and the random forest's 5-day gain fell below it for none. Results near that level did change. The random forest's 3-day gain over the three-retrieval mean ranged from $+0.040$ to $+0.047$, with p from 0.037 to 0.076, below 0.05 for 5 of the 21 seeds, about one in four. With Sentinel-1 the same gain had p from 0.021 to 0.051. Which learned configuration is best at 3 days, and whether the random forest's 3-day gain falls below $p = 0.05$, therefore depend on the seed. Ridge regression and the reference forecasts have no seed. The tuning step was not repeated, so these ranges are a lower bound; the robustness runs at the other cells were fitted with one seed as well.

Sentinel-1 means over part of a cell. The backscatter is averaged scene by scene. Where a cell straddles two consecutive frames of one pass, the two parts are not combined: the frame that covers at least half of the cell is used and the other is dropped. At cell (3,6) this affects 29 scene-days in 2017 and 2018, where the mean covers 88% of the cell. At the neighbouring cell (4,6) it affects nearly every ascending pass, 37% of its scene-days, where the mean covers 68% of the cell. A check that combined the two frames changed the random forest's skill with Sentinel-1 by at most 0.007 at cell (3,6) and 0.008 at cell (4,6), comparable to the spread between random seeds, and left the conclusion that Sentinel-1 adds no measurable information unchanged. The check is not part of the committed pipeline.

Climatology from a short record. Climatology was defined by the study, with a $\pm$7-day window, rather than taken from a convention, and it rests on 2 to 8 training years where validation guidance asks for at least 5 to 10 (Gruber et al., 2020). It was weakest in the first fold at 3 and 5 days; at 2 and 8 days its weakest years were 2021 and 2024. The comparisons that depend on it were therefore repeated under six other definitions (Table 4.20); two of three held under all of them, and the third, that no learned model beats the damped anomaly of the three-retrieval mean, held under five. An eighth definition, the proposal's literal day-of-year mean with no window, was too noisy to use (a median of 2 retrievals per day of year) and reverses these comparisons (Section 4.6.5).

Training pairs at the year boundary. The main run keeps the 0 to 3 training pairs per fold whose target falls in the test year, and the purged run removes them (Section 3.8, Table 4.10). It changed the skill of climatology, the three-retrieval mean and damped persistence by at most 0.001, and that of ridge regression by at most 0.003. The random forest and XGBoost changed by up to

0.009 and 0.016 because the models were refitted. Changes of that size in skill also arise from the random seed alone, but the change in the random forest's 8-day p-value against the three-retrieval mean does not: over 21 seeds with the boundary pairs included it stayed between 0.0004 and 0.0013 (paragraph *One random seed*), whereas without them it is 0.014. Taking the purged run apart shows where the change comes from (Section 4.3): with the purged pairs and the main run's tuning choices the gain is $+0.079$ ($p = 0.0009$), and with the main run's pairs and the purged run's tuning choices it is $+0.070$ ($p = 0.016$). The random forest's 8-day result therefore depends on its tuning choices, selected on one held-out year, and not on the removed pairs themselves. Of 324 recorded tuning choices 62 differ, but 48 of those are random-forest depth limits that tie in validation error and are picked arbitrarily; 14 changed in substance (5 for ridge regression, 6 for XGBoost, 3 for the random forest). The changes are spread across the test years rather than concentrated in early January. The random forest's 8-day gain over the three-retrieval mean became $+0.071$ ($p = 0.014$) instead of $+0.081$ ($p = 0.0007$), which no longer survives the Holm adjustment of Table 4.9 (adjusted 0.12); ridge regression's 3- and 8-day gains were unchanged ($p = 0.0094$ and 0.0020), and its 8-day gain still survives it (adjusted 0.024). The only Diebold–Mariano p to cross 0.05 was that of XGBoost with weather at 8 days against persistence (p from 0.012 to 0.069; its bootstrap interval changed with it). The per-fold Wilcoxon p of the random forest with weather at 8 days moved from 0.078 to 0.047, and one interval against the three-retrieval mean, the random forest with Sentinel-1 at 2 days, moved its upper bound from $+0.0002$ to -0.0003. The L4 benchmark uses the same rule, with up to h such pairs per fold, and was not re-run, nor were the robustness runs at the other cells.

A confounded L4/L3 contrast. The L4 series is a $\pm 0.15^\circ$ box mean of daily-averaged 3-hourly values around the original study point. The L3 target is one 9 km cell observed at one instant, 38.1 km away, forecast at different horizons. The skill scores are also summarised differently: the L4 skill in Table 4.6 is the mean of the per-fold skills, with its interval and Diebold–Mariano test taken over the pooled test samples, whereas the L3 skill is computed once over the pooled samples. The folds differ too: eight on L4, testing 2017–2024, against six on L3 at 2 days (testing 2019–2024) and seven at the other horizons (testing 2018–2024). The contrast between the two benchmarks therefore mixes product level with footprint, time sampling and location, and the two benchmarks also differ in horizons, folds and skill aggregation, so the contrast cannot be read as an effect of processing level alone.

Target definition. The L3 target is the morning retrieval, or the evening one where the morning one is missing. The choice affects the character of the series. The lag-one autocorrelation between consecutive retrievals is 0.676 for the morning-first series, 0.735 for the evening-only series and 0.745 for the mean of the two passes. On the 691 days with both passes, the morning value exceeds the evening one by $0.0067\,\mathrm{m^3/m^3}$ on average, with a standard deviation of $0.0332\,\mathrm{m^3/m^3}$. The benchmark may therefore depend on this choice. It also means that the separation is exact only in file dates: 18–22% of test samples pair a morning value with an evening one, or the reverse, and are half a day less or more than h apart. Those pairs carry the morning–evening offset as a bias that no feature describes (a mean change of -0.008 to $-0.019\,\mathrm{m^3/m^3}$ from morning to evening and $+0.009$ to $+0.012$ from evening to morning). A repeat of the benchmark at cell (3,6) under the same protocol, with the mean of the two passes as the target (the mean of both retrievals where both exist, otherwise the one available), addresses this dependence. Averaging the two passes lowered persistence error to 0.034–$0.035\,\mathrm{m^3/m^3}$ and left the broad pattern unchanged (Section 4.3.2, Table 4.13): the three-retrieval mean was still the best forecast at 2 days, no learned model's gain over it reached $p = 0.05$ at

5 days, and at 8 days the gains of ridge regression ($+0.067$, $p = 0.0003$) and the random forest ($+0.102$, $p < 0.001$) both survive the Holm adjustment of that run. The 3-day gain with p below 0.05, however, moved from ridge regression, whose gain with this target was $+0.024$ ($p = 0.15$), to the random forest ($+0.061$, $p = 0.008$, adjusted 0.068). The broad pattern is therefore not specific to the morning-first target, but which learned model's 3-day gain falls below $p = 0.05$ depends on the target. Only the mean of the two passes was tested; an evening-only target was not run.

Explainability only for L4. SHAP was computed only for the L4 random forest, at 7 days; XGBoost, the best L4 model, was not analysed, so research gap G3 (Table 2.3) is only partly closed. On L3, the contribution of inputs was measured only by the information-gain ladder. The ladder shows whether an input group helps, but not how a model uses the retrieval history; only ridge regression's weights show that (Section 4.3).

Datasets not used. ESA WorldCover was not joined, and GLDAS could not be obtained, so MERRA-2 took its role as the land-surface-model comparator.

5.5 Conclusions

The first conclusion summarises the benchmark. Conclusions 3, 4, 5 and 9 answer RQ1; 12 answers RQ2; 7, with the SHAP result in 9, answers RQ3; 2, 6, 8, 11 and 14 answer RQ4; and 10 and 13 answer RQ5.

1. A persistence-referenced, significance-tested benchmark for short-range surface soil-moisture forecasting at a selected SMAP L3 cell in Pathum Thani over 2016–2024 has been established, supported by a scripted pipeline that acquires every dataset without Google Earth Engine. It covers SMAP L3 at one non-urban cell of the 9 km grid, (3,6), whose value represents an area about 36 km across, with two neighbouring cells as robustness checks, at 2, 3, 5 and 8 days, and, separately, SMAP L4 for a $\pm 0.15^{\circ}$ box around the original study point at 1, 3 and 7 days. It is not a forecasting result for the province: it does not cover the urban-flagged cells that make up 76.2% of the province's area.

2. Retrieval quality limits where SMAP L3 can be used. Only 27.2% of the retrievals in the scan box are recommended. The urban-flagged cells, which cover 76.2% of the province, carry none, and this includes the cell of the original study point; a further 1.9% of the province lies in a cell that returned no retrieval. Of the province's seven districts, recommended retrievals exist over parts of four (Nong Suea, Thanyaburi, Lam Luk Ka and Lat Lum Kaeo) and over none of Khlong Luang, Sam Khok or Mueang Pathum Thani. A cell is flagged urban when more than a quarter of its surroundings, about 33 to 36 km across, is urban, and the product then withholds its recommendation by design. The recurring maximum earlier read as a saturation ceiling is a value at the upper limit of a valid retrieval that SMAP marks as unsuccessful.

3. At cell (3,6), SMAP L3 is forecastable beyond persistence at 3, 5 and 8 days. Every forecast built on the L3 history alone—the three-retrieval mean, damped persistence, ridge regression,

the random forest and XGBoost—beat persistence with bootstrap intervals above zero; one of those bounds, the mean of three's at 8 days ($+0.006$), is close enough to zero to depend on the bootstrap seed.

4. Most of that skill comes from smoothing, so the meaningful question is whether learning adds information beyond a simple smoothing baseline. The mean of the last three retrievals scored $+0.147$, $+0.204$, $+0.144$ and $+0.069$ at 2, 3, 5 and 8 days. Ridge regression added a small increment over it at 8 days ($+0.060$, $p = 0.002$) that survives adjustment for the twelve comparisons in the main run and in the purged run, without the training pairs whose target falls in the test year (Table 4.10); it was positive in six of the seven test years. The random forest's $+0.081$ ($p = 0.0007$) survives it in the main run only; in the purged run it is $+0.071$ ($p = 0.014$, adjusted 0.12), a change that comes from its tuning choices and not from the removed pairs. The family of twelve was fixed after the results were known, and with a penalty range widened afterwards ridge regression's 8-day gain is $+0.057$ with an adjusted p of 0.051. The 8-day gain of the learned models is thus small, positive in every variant and close to the threshold of the adjusted test. Over a family of all 48 comparisons with the mean, ridge regression's adjusted p would be about 0.10. Ridge regression's $+0.042$ at 3 days ($p = 0.009$) does not survive the adjustment and is suggestive. The learned models added nothing significant at 5 days and nothing at 2 days. A test built for nested models, computed afterwards, rejects the mean of three as the correct model at 3, 5 and 8 days (one-sided $p \leq 0.0003$) and not at 2 days: the history holds information beyond the mean at those horizons, which at 5 days ridge regression did not turn into a more accurate forecast. Ridge regression's weights show its increment to be a re-weighting of the same retrievals, read here as a damping towards the season. Ridge regression scored higher than the random forest at 2 days and matched it at 3 days; the random forest scored higher at 5 and 8 days. None of these differences between the two models was tested. A diagnostic benchmark defined after these results, the three-retrieval mean damped towards the seasonal normal by one fitted fraction, which is not a proposed forecasting model, was not beaten with p below 0.05 by any learned model at any horizon when its climatology used all training-year retrievals; with the benchmark's own climatology, built from training targets only, the random forest beat it at 8 days ($p = 0.039$, unadjusted; two of the seven definitions tested).

5. The conclusions hold against seasonal climatology as well as against persistence, which at this cell were statistically indistinguishable. They depend on the metric: the learned forecasts vary less than the retrievals, so under the Kling–Gupta efficiency persistence ranked level with or above them, and they flagged fewer of the observed wet cases.

6. The two neighbouring non-urban cells and the mean of the two passes as the target reproduced the broad pattern: the three-retrieval mean was the best forecast at 2 days, no learned model's gain over it reached $p = 0.05$ at 5 days, and the 8-day gains of ridge regression and the random forest over it had p below 0.05 before adjustment in all four runs with recommended retrievals at a non-urban cell (ridge regression: Diebold–Mariano $p = 0.002$ at cell (3,6), 0.013 at cell (2,6), $p < 0.001$ at cell (4,6) and $p = 0.0003$ with the two-pass target; random forest: $p = 0.0007$, 0.027, 0.010 and $p < 0.001$; Table 4.13). The neighbouring cells share most of their footprint with cell (3,6), and their series correlate with its series at 0.93 to 0.94, so these runs are consistency checks on largely shared data, not independent confirmations. Which model's 3-day gain over the mean fell below $p = 0.05$ varied, and none survives adjustment: ridge regression at cells (3,6) and (2,6), the random forest with the two-pass target, and neither on the L3 history alone at cell (4,6). For the random forest these 3-day verdicts also depend on

its random seed. At the original urban cell, where no retrieval is of recommended quality and all were therefore used, no learned model's gain over the three-retrieval mean reached $p = 0.05$ at any horizon (Diebold–Mariano p from 0.22 to 0.99). A gain over smoothing with p below 0.05 thus appeared only with recommended retrievals at a non-urban cell, but the cell and the quality of the retrievals change together between these runs, and the urban series includes values at the retrieval's upper limit, so the study does not show which of these removes the increment. In a check made afterwards on the retrieval history alone, ridge regression's 8-day gain over the three-retrieval mean was positive at all 20 non-urban cells of the scan (unadjusted p below 0.05 at 15, including 5 of the 8 western cells that share little with cell (3,6)), and the three-retrieval mean beat persistence at every cell at 2, 3 and 5 days.

7. Neither Sentinel-1 backscatter nor weather (rainfall, temperature and shortwave radiation) added measurable additional information beyond the retrieval history under the present data, quality filtering and evaluation protocol. The intervals still allow a gain of a few hundredths of skill, and several features of the inputs as built, untested here, may have hidden information. That is not a finding that either is uninformative about soil moisture in general. The result held also in the situations where they should help most: with a radar scene from the same or the previous day, and after heavy rain: in those situations every interval of the change, computed by resampling forecasts independently, included zero. Over all forecasts, adding weather lowered the random forest's skill at 8 days (-0.025) and marginally at 5 days (-0.011), by intervals that are not adjusted and of which the 5-day one reaches to within 0.0003 of zero.

8. On the real record, much of the 3-day gain over persistence is concentrated in forecasts touching an isolated, short-lived departure: 59–65% of the reduction in squared error, a share partly built into how such retrievals are identified; more than half of the 3-day skill remained without them. At 8 days the share is smaller and depends on the forecast: forecasts issued from an isolated retrieval supplied 20% of the random forest's gain and 50% of the three-retrieval mean's. The forecasts followed rises between retrievals about halfway and falls about a third of the way, a description rather than a measure of missed signal, and did not anticipate the wetting that followed heavy reanalysis rainfall.

9. On SMAP L4 neither learned model, the random forest or XGBoost, beat persistence at 1, 3 or 7 days (best skill, as a mean of per-fold skills, -0.131, -0.066 and -0.041). For the random forest at 7 days, the current value had the largest mean absolute SHAP value, about 4.5 times that of the next feature, the previous day's value. L4 is a $\pm 0.15^\circ$ box mean of daily-averaged 3-hourly values and L3 one 9 km cell at one instant, 38.1 km from the centre of the L4 box. The contrast between them therefore mixes product level with footprint, time sampling and location, and the two benchmarks also differ in horizons and in skill aggregation (a mean of per-fold skills over eight folds on L4, a single pooled score over six or seven folds on L3). The L4 result is kept separate from the L3 conclusions and is not a direct product-against-product comparison. It shows that a result reported for "SMAP" can differ with the product and the setting used, but it isolates no single cause.

10. The candidate targets differ sharply in completeness, footprint and memory (RQ5). SMAP L4 (the $\pm 0.15^\circ$ box), the ECMWF blend (Open-Meteo best match of the IFS, ERA5 and ERA5-Land) and MERRA-2 (daily series for the original study point) are complete over all 3,288 days, whereas SMAP L3 at cell (3,6), one cell of the 9 km grid observed at one instant, has a recommended-quality retrieval on only 29.8% of them. SMAP L4 is very persistent, with a lag-one autocorrelation of 0.987 at a daily step, against 0.676 between consecutive L3 retrievals,

which are a median of 3 days apart. On L4 no learned model beat persistence; on L3 at cell (3,6) every forecast built on the retrieval history alone beat it at 3, 5 and 8 days. MERRA-2 surface wetness is about as persistent as L4 (0.989 at lag 1), whereas the ECMWF blend loses its memory faster (0.772 at lag 7). With the ECMWF blend as the target, the L4 protocol gave $+0.016$, $+0.041$ and $+0.048$ at 1, 3 and 7 days, of which only the 3-day value excluded zero; this is reported as suggestive only. As the previous item states, the L4/L3 contrast also reflects footprint, time sampling and location, not processing level alone.

11. The benchmark says least about the wettest conditions. At cell (3,6), 77% of the retrievals rejected by the quality rule fall in May–October, and in every month compared they follow more rain than the accepted ones, so the wettest states are under-represented. A usable retrieval exists on 10% of October days, and from September to November the latest one is at least 5 days old on 47% of days. Split by season, the three-retrieval mean, ridge regression and the random forest gained over persistence in both seasons at every horizon. The largest seasonal differences were at 2 days, where the three-retrieval mean and ridge regression gained more in the dry season ($+0.174$ and $+0.121$, against $+0.100$ and $+0.078$), and at 8 days, where the random forest gained more in the wet season ($+0.203$ against $+0.106$); none of these differences was tested for significance. On L4, one untuned random forest lost to persistence in all 21 combinations of horizon with four antecedent-rainfall bands and three wetness terciles (skill -0.017 to -0.567): none of the 21 tested conditions favoured it.

12. The orbit constrains the horizons, and persistence error does not increase from 2 to 8 days. Exact 1- and 7-day pairs are too few to evaluate, and 4-day pairs do not occur; the horizons of 2, 3, 5 and 8 days were chosen from the separations with enough retrieval pairs and approved by the advisor, and 6 days, with 326 pairs, was not made a benchmark horizon, to stay consistent with the earlier analysis; run afterwards as a check, it gave ridge regression a gain of $+0.084$ over the three-retrieval mean (unadjusted $p = 0.004$). Over all recommended-quality pairs, 8-day pairs, four in five of them from the same track, have a lower persistence RMSE (0.0394) than 2-, 3- and 5-day pairs from different tracks (0.0398 to 0.0405), but there is no consistent evidence of an orbit-geometry effect: the pooled 8-day figure is pulled down by pairs that join the two passes, which are not same-track, and within each pass the ordering is mixed or reversed. The differences in persistence error are small and were not tested. Among the four benchmark horizons, skill against persistence was highest at 3 days; the study does not explain that peak, and it should not be read as a measure of predictability alone.

13. In the product intercomparison at the original study point, SMAP L4 agreed most closely with MERRA-2 ($r = 0.900$ on the raw series, 0.701 on anomalies) and least with the ECMWF blend (0.714 and 0.611). This is an intercomparison, not a validation: with no in-situ reference, it shows how the products diverge, not which of them is closer to the soil. Products built on similar models can also agree because they share errors.

14. The best forecasts reached an ubRMSE of 0.031 to $0.035\,\mathrm{m^3/m^3}$ against the retrieval. This is not comparable with SMAP's $0.04\,\mathrm{m^3/m^3}$ requirement, which concerns the retrieval's error against the ground over the product as a whole, and it is no evidence of accuracy against the true soil moisture. Without an independent reference, improvement cannot be verified against the true soil moisture.

5.6 Research Gaps Revisited and Future Research

Chapter 2 set out ten research gaps (Table 2.3). This section returns to them with the results in hand. It first states which gaps remain open after this study. It then derives, from each finding or limitation, a research topic that is specific enough to be carried out: a question, a first experiment with data that exist or can be obtained, and the result that would count as an answer. Table 5.3 gives the topics in brief, and the paragraphs that follow give the reasoning and the literature behind each. Every topic that produces a forecast is to be judged under the protocol of Section 3.8, against the references and tests of Table 5.2.

Which gaps remain open. Two gaps are closed at the sites studied: persistence is now the reference against which skill is stated (G2), and a local baseline exists against which a more elaborate model can be justified (G4). Two are answered as far as one study can answer them: the apparent saturation ceiling is a flagged value (G8), and the horizons were taken from the orbit's sampling (G9). The other six remain open, each in a narrower form than before.

- **G1.** The benchmark exists only where SMAP recommends its retrievals, which is 21.9% of the province's area, and in full at only three of the 20 non-urban cells (Topic 5).
- **G3.** On L3, the way a model uses the retrieval history is known for ridge regression alone, from its weights (Topic 7).
- **G5.** Failure in the wettest conditions could not be characterised, because the quality rule removes them from the record (Topics 1 and 2).
- **G6.** L3 and L4 were not forecast under one protocol, at one place and on the same dates, so the contrast between them is confounded (Topic 2).
- **G7 and G10.** No independent reference exists against which the retrieval, or an improvement on it, can be checked (Topic 3).

The study also leaves two questions that Table 2.3 does not contain. Neither Sentinel-1 nor weather, as built here, added information, so which observation would add it is open (Topics 1 and 4). And the forecast is for the top 5 cm over an area about 36 km across, whereas the decisions it is meant to inform are taken for a root zone and a field (Topic 6).

Table 5.3: Future research topics, the finding or limitation of this study that each starts from, the question with its first experiment, and the literature that supports it. The gap of Table 2.3 that a topic addresses is given in brackets.

Topic	Finding or limitation	Question and first experiment	Supporting literature
1. Rain in the forecast window (G5)	Wetting after more than 20 mm of rain in the forecast window was not forecast, and weather up to the issue date added nothing	Does rainfall between the issue and target dates improve the forecast at 3 to 8 days? First add the reanalysis rainfall of the window to the ladder, an upper bound for rainfall of that kind; if a gain appears, repeat with archived weather forecasts as issued	Orth and Seneviratne (2013); L. Li et al. (2024); McColl, Alemohammad, et al. (2017)
2. Forecasts without a usable retrieval; L3 against L4 under one protocol (G5, G6)	A usable retrieval exists on 10% of October days, and the record has 23 gaps of ten days or more. L4 is complete but was benchmarked at another place and on other horizons	Can the retrieval be estimated on the days it is missing well enough to start a forecast? First sample L4 at cell (3,6) on the L3 retrieval dates and repeat the L3 benchmark on it. Then withhold recommended retrievals, estimate them from L4 and weather, and score the estimates and the forecasts started from them	Fang et al. (2017); Fang and Shen (2020); Reichle et al. (2019); Ganepola et al. (2026)
3. The retrieval's own error (G7, G10)	Every error is measured against a retrieval whose error is unknown, and the bound of 0.023 m^3/m^3 rests on untested assumptions	How large is the random error of the L3 retrieval at the cell, and how much of the smoothing gain does it explain? Estimate it by triple collocation with two products whose errors are independent of it and of each other, and set it beside the bound. Request the existing station data	Gruber et al. (2020); Colliander et al. (2017); Jotisankasa et al. (2023); Ganepola et al. (2026)
4. Radar at the scale and geometry of the target	Sentinel-1 as a 9 km mean of the latest scene of either pass added nothing, yet changes between scenes of one pass correlate with changes in the retrieval at 0.51 on 28 pairs	Does radar add information when it is built to match the target? In order of cost: a mean over the sensed area; separate features for the two orbit tracks; the change since the previous scene of the same track; L-band data from NISAR	Das et al. (2019); Hosseinizadeh et al. (2025); Jotisankasa et al. (2023); Ganepola et al. (2026)
5. The rest of the province (G1, G9)	The full protocol ran at 3 of the 20 non-urban cells, and 76.2% of the province lies in urban-flagged cells, which carry no recommended retrieval	Does the pattern hold with all three models and the added inputs at the other 17 cells, above all the eight western ones, and at 6 days? Can a finer product give a usable target inside the urban-flagged area? First count its usable values in each district	Das et al. (2019); Liu et al. (2020); Ganepola et al. (2026)

continued on next page

Table 5.3 (continued)

Topic	Finding or limitation	Question and first experiment	Supporting literature
6. From the surface layer to irrigation advice	The target is the top 5 cm over about 36 km, and the benchmark supports no statement about the root zone or a field	Can root-zone wetness for the irrigated fields of the cell be forecast 3 to 8 days ahead with skill over its own persistence? It needs a root-zone target, crop and land-cover records, a water balance with explicit terms, and a decision threshold agreed with users	Carranza et al. (2021); Wu et al. (2025); Xi et al. (2025); Fu et al. (2023)
7. Model capacity, after an input has shown a gain (G3, G4)	Ridge regression matched the random forest at 3 days, XGBoost was the weakest, and no learned model clearly beat a rule with one fitted number	Once an input from Topics 1, 2 or 4 shows a gain with ridge regression, does a sequence model or a physics-guided model enlarge it? Judged by the three conditions of Section 5.2, on a test period from 2025	Y. Wang et al. (2024); Pan et al. (2025); Xi et al. (2025); L. Li et al. (2024)

Topic 1. Rain in the forecast window. This topic starts from the one failure that the benchmark located precisely. When more than 20 mm of reanalysis rain fell during a 3-day forecast window, the retrieval rose by $+0.017\,\mathrm{m^3/m^3}$ on average and the forecasts moved by -0.002 to $+0.004$ (Section 4.5), and weather up to the issue date added no measurable information (Section 4.4). Trailing weather describes a past that the retrieval at the issue date already reflects, so the missing input is the rain still to come. The literature supports looking there. In a model study of Swiss catchments, Orth and Seneviratne (2013) found that the initial soil moisture alone gave skill up to about one week, and that adding the initial snow and seasonal weather forecasts doubled that lead time. L. Li et al. (2024) fed the soil-moisture forecast of a numerical weather model into a learned model, and the combination beat the learned model alone at 79.5% of stations at a 16-day lead. And the layer that SMAP senses keeps a median of only 14% of the rain that falls on land after three days (McColl, Alemohammad, et al., 2017), which suggests that at horizons of 3 to 8 days the rain inside the window bears more on the target than the rain before the issue date. Whether it helps at this cell is untested, and the weak link found here between rainfall and the retrieval, a correlation of 0.16, makes the outcome uncertain. The first experiment therefore uses data already in hand: add the reanalysis rainfall that fell between the issue and target dates as one more rung of the ladder. It is an upper bound for rainfall of that kind, not for a perfect forecast. A gain, shown by a paired interval that excludes zero (Section 3.9), would justify the second step, the same rung built from archived weather forecasts as they were issued and not from observed rain. A null would not close the question until the reanalysis rainfall has been compared with gauges or with another rainfall product for the sensed area.

Topic 2. Forecasts without a usable retrieval, and L3 against L4 under one protocol. A recommended-quality retrieval exists on 10% of October days, the record holds 23 gaps of ten days or more, and in 2022 there was almost none from August to late November (Section 4.5). A forecast system built on this record is silent for much of the flood season. SMAP L4 has a value every day, but this study benchmarked it at another place, on other horizons and with another summary of skill, so it cannot say what L4 would offer at cell (3,6). Two experiments follow. The first closes gap G6: sample L4 at cell (3,6) on the dates of the L3 retrievals, and repeat the L3 benchmark on that series,

at the same horizons and folds. With the place, the dates and the protocol held fixed, what remains of the contrast between the two products is their processing. The second asks whether the retrieval can be estimated on the days it is missing. Withhold a share of the recommended retrievals, estimate them from L4 and the weather, and score the estimates against the withheld values; then compare forecasts started from the estimate with forecasts started from the last real retrieval, which from September to November is at least five days old on 47% of days. Fang et al. (2017) showed that a network can reproduce SMAP L3 from weather and a land-surface model, with a test RMSE of 0.027 at the median pixel of the continental United States, and Fang and Shen (2020) forecast L3 from the most recent SMAP observations with a median RMSE of 0.022 at one day and 0.024 at two and three days. Gap filling with region-specific validation is also the first pillar of the regional roadmap (Ganepola et al., 2026). One caution applies. The wettest days carry no measurement in L3, since the rejected values are not retrievals, so an estimate can be scored only where recommended retrievals were withheld. For the wettest states it depends on the reference of Topic 3. NISAR is a further source for the days without a retrieval (Topic 4).

Topic 3. The retrieval's own error. Every error in this report is measured against a retrieval whose own error at the cell is unknown. The one bound the study could give, a standard deviation of at most 0.023 m^3/m^3 for the part of the error that is independent from one retrieval to the next, rests on assumptions that were not tested (Section 4.9). The question is how large that error is, and how much of the gain from smoothing it explains. It can be approached without ground data. Triple collocation estimates the random error of each of three products, provided that their errors are not correlated with one another (Gruber et al., 2020). The first experiment is to apply it to the anomalies of the L3 retrieval and of two such products, for example a reanalysis and a radar-based retrieval, and to set the estimated error beside the bound of 0.023. Ganepola et al. (2026) list it among the indirect checks suited to the region, with the conversion of soil moisture to rainfall for comparison with rain gauges. Ground data would settle more. At SMAP's core validation sites the standard radiometer product reached an average ubRMSE of 0.039 m^3/m^3 (Colliander et al., 2017), but none of those sites is in Mainland Southeast Asia, and the one published comparison of a SMAP product with buried sensors in Thailand found R^2 of 0.13 to 0.58 at seven stations after monthly averaging (Jotisankasa et al., 2023). A formal request for the data of those stations, which the Hydro-Informatics Institute operates, was not sent during this study and remains possible. The first research-grade site in the region, three sensors at 5 cm within one 200 m cell at Kasetsart University's Suwan Farm in Nakhon Ratchasima, is due by mid-2027 and will join the Institute's telemetry network (Ganepola et al., 2026). It lies outside the province, so it would bear on SMAP's retrieval error in general and not at cell (3,6); the same roadmap recommends at least 13 stations within one 36 km cell for a robust check.

Topic 4. Radar at the scale and geometry of the target. Sentinel-1 entered the benchmark as the mean backscatter of a 9 km cell from the latest scene of either pass, and in that form it added nothing (Section 4.4). The study's own figures suggest that the form, and not the sensor, may be the reason. The 9 km cell is much smaller than the area that the radiometer senses. The correlation between backscatter and the retrieval is higher within the descending pass (0.38 to 0.39) than over all scenes (0.24 and 0.32). And the changes between consecutive descending scenes correlate with the changes in the retrieval at 0.51, though on only 28 pairs (Section 4.5.4). Four experiments follow, in order of cost: the mean of the backscatter over the sensed area in place of the 9 km cell; separate features for the two orbit tracks, 172 and 62, in place of one mixed feature; the change since the previous

scene of the same track in place of the level; and L-band data from NISAR, whose signal penetrates vegetation better than Sentinel-1's C-band (Ganepola et al., 2026), which would test whether the absence of a gain is specific to C-band. Change between scenes is an established way to read soil moisture from Sentinel-1: the labels on which Hosseinizadeh et al. (2025) trained their network were derived by change detection. Each experiment is one more rung of the ladder and is judged by the paired interval of the change in skill. A thesis that combines L3 with Sentinel-1 must also say how it differs from NASA's fused product, which gives soil moisture at 3 and 1 km with an unbiased RMSE of about 0.05 m^3/m^3 (Das et al., 2019) and has been compared with stations in Thailand (Jotisankasa et al., 2023): by forecasting instead of estimating the present state, by using the radar at field scale, or by informing the days on which no retrieval exists.

Topic 5. The rest of the province. The full protocol ran at three of the 20 non-urban cells, and the later check at all 20 used the retrieval history alone (Section 4.3.3). Two questions remain. The first is whether the pattern holds with the random forest, XGBoost, Sentinel-1 and weather at the other 17 cells. The eight western cells matter most, because their series correlate with that of cell (3,6) at only 0.45 to 0.55, whereas neighbouring cells share most of their footprint; the 20 cells are closer to two places than to twenty. The experiment needs weather for 17 more cells, the Sentinel-1 means being already extracted, and the 6-day horizon, run here only as a check, can be made a horizon of the benchmark at the same time. The second question is larger. Of the province's area, 76.2% lies in urban-flagged cells, which carry no recommended retrieval, and no forecast can be scored there on L3. A finer product is the only route. The SMAP/Sentinel-1 product at 3 and 1 km (Das et al., 2019) is one candidate, and statistical downscaling, for which Liu et al. (2020) found the random forest the best of six algorithms, is another. The first step is a count like that of Section 4.1: how many usable values the finer product holds in each of the seven districts, and under which quality flags. Ganepola et al. (2026) put the operational need at a resolution finer than 200 m, so this topic also bears on whether satellite soil moisture can serve the province's districts at all.

Topic 6. From the surface layer to irrigation advice. The benchmark supplies a protocol and the references that a later model must beat. It does not supply a variable on which irrigation can be decided: irrigation depends on the water available to roots, while the target here is the top 5 cm as SMAP sees it over an area about 36 km across (Table 5.1). The research question is whether root-zone wetness for the irrigated fields of the cell can be forecast 3 to 8 days ahead with skill over its own persistence. It needs five additions. The first is a root-zone target, or a tested link from the surface to the root zone. Carranza et al. (2021) estimated root-zone soil moisture with a random forest about as well as a process model with data assimilation, and the surface value was the forest's most important input, which makes such a link plausible. The SMAP L4 root-zone series is already downloaded, although its close agreement with the L4 surface layer (anomaly correlation 0.931, Table 4.22) comes from the same land-surface model and does not show that the L3 surface value tracks the root zone; Wu et al. (2025) forecast that product 1 to 14 days ahead. The second is land-cover and crop-calendar records, so that a forecast can be tied to fields that are actually irrigated and to the stage of the crop; with irrigation records, they would also show whether managed water explains part of the change that rainfall does not. The third is radar at field scale, since one cell describes far more than a farm (Topic 4). The fourth is a simple water balance, using the reference evapotranspiration ET_0 already retrieved for cell (3,6), so that rainfall, evaporation and irrigation enter the forecast as explicit terms. Xi et al. (2025) fed water-balance terms into a recurrent network and beat both the process model and the network alone, and Fu et al. (2023) combined a water balance with a seasonal time-series model

below the surface layer. The fifth is a decision threshold agreed with the people who would act on the forecast, so that skill can be scored as decisions, by hit rate and false-alarm ratio, and not as RMSE. One feature of the record helps: the dry season, when irrigation stress arises, is when usable retrievals are most frequent (36–40% of days from January to April, Section 4.5).

Topic 7. Model capacity, after an input has shown a gain. On the retrieval history alone, more model capacity brought no clear gain: ridge regression matched the random forest at 3 days, XGBoost was the weakest, and no learned model clearly beat a rule with one fitted number (Section 5.2). No larger model was tried, so the study does not rule one out. The literature gives a similar picture where a comparable test was made. In the comparison of Y. Wang et al. (2024) at 5 cm depth, the LSTM led the random forest by about 0.02 in R^2 at 1, 3 and 7 days (0.943, 0.895 and 0.816 against 0.924, 0.874 and 0.797). In that of Pan et al. (2025), a graph and convolutional network reached an RMSE of 0.018 to 0.038 m^3/m^3 at 1 to 7 days where the random forest reached 0.021 to 0.042. The order of work of Figure 3.1 therefore stands: a larger model is the last step, taken once an input from Topics 1, 2 or 4 has shown a gain with ridge regression. The question is then whether a sequence model or a physics-guided one (L. Li et al., 2024; Xi et al., 2025) turns that gain into a larger one. Such a model passes only under the three conditions of Section 5.2: gains over the mean of three and over DAM3 (Equation 3.17) that survive adjustment, a head-to-head test against ridge regression, and a test period that this study did not use, from 2025. It should report a score that does not reward caution beside RMSE, and it should state what it uses, by weights or by attribution, so that gap G3 narrows further. Two directions of the original thesis plan are qualified by the results. AI-assisted data assimilation built on SMAP L4 is on hold (Section 5.2). And a graph over neighbouring 9 km cells would pass largely the same measurement from node to node, since neighbouring cells correlate at 0.93 to 0.94; a graph has something to carry only between the eastern and western groups of cells, or between nodes at field scale.

Order of work. Topics 1 and 3 come first. Each needs one experiment with data that are in hand or freely available, and their answers decide what the others are worth. If rain in the forecast window adds nothing and the retrieval's error proves as large as the gain from smoothing, the limit lies in the target, and Topics 2 and 5, which change the target, come before any change of model. If rain adds a gain, Topic 7 has an input to work with.

REFERENCES

Ahmad, S., Kalra, A., & Stephen, H. (2010). Estimating soil moisture using remote sensing data: A machine learning approach. *Advances in Water Resources, 33*(1), 69–80. https://doi.org/10.1016/j.advwatres.2009.10.008

Akkala, A., Boubrahimi, S. F., Hamdi, S. M., Hosseinzadeh, P., & Nassar, A. (2025). Spatio-temporal graph neural networks for streamflow prediction in the Upper Colorado Basin. *Hydrology, 12*(3), Article 60. https://doi.org/10.3390/hydrology12030060

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Carranza, C., Nolet, C., Pezij, M., & van der Ploeg, M. (2021). Root zone soil moisture estimation with random forest. *Journal of Hydrology, 593*, Article 125840. https://doi.org/10.1016/j.jhydrol.2020.125840

Chan, S. K., Bindlish, R., O'Neill, P., Jackson, T., Njoku, E., Dunbar, S., Chaubell, J., Piepmeier, J., Yueh, S., Entekhabi, D., Colliander, A., Chen, F., Cosh, M. H., Caldwell, T., Walker, J., Berg, A., McNairn, H., Thibeault, M., Martínez-Fernández, J., … Kerr, Y. (2018). Development and assessment of the SMAP enhanced passive soil moisture product. *Remote Sensing of Environment, 204*, 931–941. https://doi.org/10.1016/j.rse.2017.08.025

Chavoshi, A., Dashtian, H., Bakhshian, S., Young, M. H., & Niyogi, D. (2025). PINN-SM: A physics-informed neural networks model for vadose zone soil moisture profile prediction. *Journal of Geophysical Research: Machine Learning and Computation, 2*(4), Article e2024JH000547. https://doi.org/10.1029/2024JH000547

Chen, P.-Y., Chen, C.-C., Kang, C., Liu, J.-W., & Li, Y.-H. (2025). Soil water content prediction across seasons using random forest based on precipitation-related data. *Computers and Electronics in Agriculture, 230*, Article 109802. https://doi.org/10.1016/j.compag.2024.109802

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). Association for Computing Machinery. https://doi.org/10.1145/2939672.2939785

Clark, T. E., & West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics, 138*(1), 291–311. https://doi.org/10.1016/j.jeconom.2006.05.023

Colliander, A., Jackson, T. J., Bindlish, R., Chan, S., Das, N., Kim, S. B., Cosh, M. H., Dunbar, R. S., Dang, L., Pashaian, L., Asanuma, J., Aida, K., Berg, A., Rowlandson, T., Bosch, D., Caldwell, T., Caylor, K., Goodrich, D., al Jassar, H., … Yueh, S. (2017). Validation of SMAP surface soil moisture products with core validation sites. *Remote Sensing of Environment, 191*, 215–231. https://doi.org/10.1016/j.rse.2017.01.021

Cong, Y., Khanna, S., Meng, C., Liu, P., Rozi, E., He, Y., Burke, M., Lobell, D. B., & Ermon, S. (2022). SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery. *Advances in Neural Information Processing Systems, 35*. https://arxiv.org/abs/2207.08051

Das, N. N., Entekhabi, D., Dunbar, R. S., Chaubell, M. J., Colliander, A., Yueh, S., Jagdhuber, T., Chen, F., Crow, W., O'Neill, P. E., Walker, J. P., Berg, A., Bosch, D. D., Caldwell, T., Cosh, M. H., Collins, C. H., Lopez-Baeza, E., & Thibeault, M. (2019). The SMAP and Copernicus Sentinel 1A/B microwave active-passive high resolution surface soil moisture product. *Remote Sensing of Environment, 233*, Article 111380. https://doi.org/10.1016/j.rse.2019.111380

Delworth, T. L., & Manabe, S. (1988). The influence of potential evaporation on the variabilities of simulated soil wetness and climate. *Journal of Climate, 1*(5), 523–547. https://doi.org/10.1175/1520-0442(1988)001<0523:TIOPEO>2.0.CO;2

Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics, 13*(3), 253–263. https://doi.org/10.1080/07350015.1995.10524599

Dorigo, W., Wagner, W., Albergel, C., Albrecht, F., Balsamo, G., Brocca, L., Chung, D., Ertl, M., Forkel, M., Gruber, A., Haas, E., Hamer, P. D., Hirschi, M., Ikonen, J., de Jeu, R., Kidd, R., Lahoz, W., Liu, Y. Y., Miralles, D., … Lecomte, P. (2017). ESA CCI Soil Moisture for improved Earth system understanding: State-of-the-art and future directions. *Remote Sensing of Environment, 203*, 185–215. https://doi.org/10.1016/j.rse.2017.07.001

Dorigo, W. A., Wagner, W., Hohensinn, R., Hahn, S., Paulik, C., Xaver, A., Gruber, A., Drusch, M., Mecklenburg, S., van Oevelen, P., Robock, A., & Jackson, T. (2011). The International Soil Moisture Network: A data hosting facility for global in situ soil moisture measurements. *Hydrology and Earth System Sciences, 15*(5), 1675–1698. https://doi.org/10.5194/hess-15-1675-2011

Entekhabi, D., Njoku, E. G., O'Neill, P. E., Kellogg, K. H., Crow, W. T., Edelstein, W. N., Entin, J. K., Goodman, S. D., Jackson, T. J., Johnson, J., Kimball, J., Piepmeier, J. R., Koster, R. D., Martin, N., McDonald, K. C., Moghaddam, M., Moran, S., Reichle, R., Shi, J. C., … Van Zyl, J. (2010). The Soil Moisture Active Passive (SMAP) mission. *Proceedings of the IEEE, 98*(5), 704–716. https://doi.org/10.1109/JPROC.2010.2043918

Fang, K., & Shen, C. (2020). Near-real-time forecast of satellite-based soil moisture using long short-term memory with an adaptive data integration kernel. *Journal of Hydrometeorology, 21*(3), 399–413. https://doi.org/10.1175/JHM-D-19-0169.1

Fang, K., Shen, C., Kifer, D., & Yang, X. (2017). Prolongation of SMAP to spatiotemporally seamless coverage of continental U.S. using a deep learning neural network. *Geophysical Research Letters, 44*(21), 11030–11039. https://doi.org/10.1002/2017GL075619

Fu, R., Xie, L., Liu, T., Zheng, B., Zhang, Y., & Hu, S. (2023). A soil moisture prediction model, based on depth and water balance equation: A case study of the Xilingol League grassland. *International Journal of Environmental Research and Public Health, 20*(2), Article 1374.

https://doi.org/10.3390/ijerph20021374

Ganepola, G. A. C., Tangdamrongsub, N., Torsri, K., Jotisankasa, A., Meechaiya, C., Yueh, S., Dong, J., & Senanayake, I. P. (2026). Roadmap for leveraging satellite data to advance soil moisture monitoring in Mainland Southeast Asia. *Vadose Zone Journal, 25*(3), Article e70107. https://doi.org/10.1002/vzj2.70107

Gelaro, R., McCarty, W., Suárez, M. J., Todling, R., Molod, A., Takacs, L., Randles, C. A., Darmenov, A., Bosilovich, M. G., Reichle, R., Wargan, K., Coy, L., Cullather, R., Draper, C., Akella, S., Buchard, V., Conaty, A., da Silva, A. M., Gu, W., … Zhao, B. (2017). The Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2). *Journal of Climate, 30*(14), 5419–5454. https://doi.org/10.1175/JCLI-D-16-0758.1

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment, 202*, 18–27. https://doi.org/10.1016/j.rse.2017.06.031

Gruber, A., De Lannoy, G., Albergel, C., Al-Yaari, A., Brocca, L., Calvet, J.-C., Colliander, A., Cosh, M., Crow, W., Dorigo, W., Draper, C., Hirschi, M., Kerr, Y., Konings, A., Lahoz, W., McColl, K., Montzka, C., Muñoz-Sabater, J., Peng, J., … Wagner, W. (2020). Validation practices for satellite soil moisture retrievals: What are (the) errors? *Remote Sensing of Environment, 244*, Article 111806. https://doi.org/10.1016/j.rse.2020.111806

Gupta, H. V., Kling, H., Yilmaz, K. K., & Martinez, G. F. (2009). Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. *Journal of Hydrology, 377*(1–2), 80–91. https://doi.org/10.1016/j.jhydrol.2009.08.003

Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting, 13*(2), 281–291. https://doi.org/10.1016/S0169-2070(96)00719-4

Hengl, T., Mendes de Jesus, J., Heuvelink, G. B. M., Ruiperez Gonzalez, M., Kilibarda, M., Blagotić, A., Shangguan, W., Wright, M. N., Geng, X., Bauer-Marschallinger, B., Guevara, M. A., Vargas, R., MacMillan, R. A., Batjes, N. H., Leenaars, J. G. B., Ribeiro, E., Wheeler, I., Mantel, S., & Kempen, B. (2017). SoilGrids250m: Global gridded soil information based on machine learning. *PLOS ONE, 12*(2), Article e0169748. https://doi.org/10.1371/journal.pone.0169748

Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J., Bonavita, M., … Thépaut, J.-N. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society, 146*(730), 1999–2049. https://doi.org/10.1002/qj.3803

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation, 9*(8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics, 12*(1), 55–67. https://doi.org/10.1080/00401706.1970.10488634

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics, 6*, 65–70.

Hosseinizadeh, A., Sheng, Z., & Liu, Y. (2025). High-resolution spatiotemporal mapping of surface soil moisture using ConvLSTM model and Sentinel-1 data. *Water, 17*(22), Article 3300. https://doi.org/10.3390/w17223300

Jakubik, J., Roy, S., Phillips, C. E., Fraccaro, P., Godwin, D., Zadrozny, B., Szwarcman, D., Gomes, C., Nyirjesy, G., Edwards, B., Kimura, D., Simumba, N., Chu, L., Mukkavilli, S. K., Lambhate, D., Das, K., Bangalore, R., Oliveira, D., Muszynski, M., … Ramachandran, R. (2023). *Foundation models for generalist geospatial artificial intelligence*. arXiv. https://arxiv.org/abs/2310.18660

Jotisankasa, A., Torsri, K., Supavetch, S., Sirirodwattanakool, K., Thonglert, N., Sawangwattanaphaibun, R., Faikrua, A., Peangta, P., & Akaranee, J. (2023). Investigating correlations and the validation of SMAP-Sentinel L2 and in situ soil moisture in Thailand. *Sensors, 23*(21), Article 8828. https://doi.org/10.3390/s23218828

Kang, J., Jin, R., Li, X., Zhang, Y., & Zhu, Z. (2018). Spatial upscaling of sparse soil moisture observations based on ridge regression. *Remote Sensing, 10*(2), Article 192. https://doi.org/10.3390/rs10020192

Kerr, Y. H., Waldteufel, P., Wigneron, J.-P., Delwart, S., Cabot, F., Boutin, J., Escorihuela, M.-J., Font, J., Reul, N., Gruhier, C., Juglea, S. E., Drinkwater, M. R., Hahne, A., Martín-Neira, M., & Mecklenburg, S. (2010). The SMOS mission: New tool for monitoring key elements of the global water cycle. *Proceedings of the IEEE, 98*(5), 666–687. https://doi.org/10.1109/JPROC.2010.2043032

Komori, D., Nakamura, S., Kiguchi, M., Nishijima, A., Yamazaki, D., Suzuki, S., Kawasaki, A., Oki, K., & Oki, T. (2012). Characteristics of the 2011 Chao Phraya River flood in Central Thailand. *Hydrological Research Letters, 6*, 41–46. https://doi.org/10.3178/hrl.6.41

Kratzert, F., Klotz, D., Brenner, C., Schulz, K., & Herrnegger, M. (2018). Rainfall–runoff modelling using long short-term memory (LSTM) networks. *Hydrology and Earth System Sciences, 22*(11), 6005–6022. https://doi.org/10.5194/hess-22-6005-2018

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics, 17*(3), 1217–1241. https://doi.org/10.1214/aos/1176347265

Lamichhane, M., Mehan, S., & Mankin, K. R. (2025). Soil moisture prediction using remote sensing and machine learning algorithms: A review on progress, challenges, and opportunities. *Remote Sensing, 17*(14), Article 2397. https://doi.org/10.3390/rs17142397

Li, L., Dai, Y., Shangguan, W., Wei, N., Wei, Z., & Gupta, S. (2022). Multistep forecasting of soil

moisture using spatiotemporal deep encoder–decoder networks. *Journal of Hydrometeorology, 23*(3), 337–350. https://doi.org/10.1175/JHM-D-21-0131.1

Li, L., Dai, Y., Wei, Z., Shangguan, W., Wei, N., Zhang, Y., Li, Q., & Li, X.-X. (2024). Enhancing deep learning soil moisture forecasting models by integrating physics-based models. *Advances in Atmospheric Sciences, 41*(7), 1326–1341. https://doi.org/10.1007/s00376-023-3181-8

Li, Q., Li, Z., Shangguan, W., Wang, X., Li, L., & Yu, F. (2022). Improving soil moisture prediction using a novel encoder-decoder model with residual learning. *Computers and Electronics in Agriculture, 195*, Article 106816. https://doi.org/10.1016/j.compag.2022.106816

Liu, Y., Jing, W., Wang, Q., & Xia, X. (2020). Generating high-resolution daily soil moisture by using spatial downscaling techniques: A comparison of six machine learning algorithms. *Advances in Water Resources, 141*, Article 103601. https://doi.org/10.1016/j.advwatres.2020.103601

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*, 4765–4774.

McColl, K. A., Alemohammad, S. H., Akbar, R., Konings, A. G., Yueh, S., & Entekhabi, D. (2017). The global distribution and dynamics of surface soil moisture. *Nature Geoscience, 10*(2), 100–104. https://doi.org/10.1038/ngeo2868

McColl, K. A., Wang, W., Peng, B., Akbar, R., Short Gianotti, D. J., Lu, H., Pan, M., & Entekhabi, D. (2017). Global characterization of surface soil moisture drydowns. *Geophysical Research Letters, 44*(8), 3682–3690. https://doi.org/10.1002/2017GL072819

Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D. G., Piles, M., Rodríguez-Fernández, N. J., Zsoter, E., Buontempo, C., & Thépaut, J.-N. (2021). ERA5-Land: A state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data, 13*(9), 4349–4383. https://doi.org/10.5194/essd-13-4349-2021

Murphy, A. H. (1988). Skill scores based on the mean square error and their relationships to the correlation coefficient. *Monthly Weather Review, 116*(12), 2417–2424. https://doi.org/10.1175/1520-0493(1988)116<2417:SSBOTM>2.0.CO;2

Murphy, A. H. (1992). Climatology, persistence, and their linear combination as standards of reference in skill scores. *Weather and Forecasting, 7*(4), 692–698. https://doi.org/10.1175/1520-0434(1992)007<0692:CPATLC>2.0.CO;2

National Snow and Ice Data Center. (2023). *SMAP Enhanced L3 Radiometer Global and Polar Grid Daily 9 km EASE-Grid Soil Moisture, Version 6: User guide*. NASA National Snow and Ice Data Center Distributed Active Archive Center. https://nsidc.org/sites/default/files/documents/user-guide/spl3smp_e-v006-userguide.pdf

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and auto-

correlation consistent covariance matrix. *Econometrica, 55*(3), 703–708. https://doi.org/10.2307/1913610

O'Neill, P., Bindlish, R., Chan, S., Chaubell, J., Colliander, A., Njoku, E., & Jackson, T. (2021). *SMAP algorithm theoretical basis document: Level 2 & 3 soil moisture (passive) data products, Revision G* (JPL D-66480). Jet Propulsion Laboratory.

O'Neill, P. E., Chan, S., Njoku, E. G., Jackson, T., Bindlish, R., Chaubell, J., & Colliander, A. (2023). *SMAP Enhanced L3 Radiometer Global and Polar Grid Daily 9 km EASE-Grid Soil Moisture, Version 6* (SPL3SMP_E) [Data set]. NASA National Snow and Ice Data Center Distributed Active Archive Center. https://doi.org/10.5067/M20OXIZHY3RJ

ORNL DAAC. (2017). *Terrestrial Ecology Subsetting & Visualization Services (TESViS) Global Subsets Tool*. ORNL Distributed Active Archive Center. https://doi.org/10.3334/ORNLDAAC/1379

Orth, R., & Seneviratne, S. I. (2013). Predictability of soil moisture and streamflow on subseasonal timescales: A case study. *Journal of Geophysical Research: Atmospheres, 118*(19), 10963–10979. https://doi.org/10.1002/jgrd.50846

Pan, Z., Xu, L., & Chen, N. (2025). Combining graph neural network and convolutional LSTM network for multistep soil moisture spatiotemporal prediction. *Journal of Hydrology, 651*, Article 132572. https://doi.org/10.1016/j.jhydrol.2024.132572

Reichle, R. H., Liu, Q., Koster, R. D., Crow, W. T., De Lannoy, G. J. M., Kimball, J. S., Ardizzone, J. V., Bosch, D., Colliander, A., Cosh, M., Kolassa, J., Mahanama, S. P., Prueger, J., Starks, P., & Walker, J. P. (2019). Version 4 of the SMAP Level-4 soil moisture algorithm and data product. *Journal of Advances in Modeling Earth Systems, 11*(10), 3106–3130. https://doi.org/10.1029/2019MS001729

Ren, Y., Ling, F., & Wang, Y. (2023). Research on provincial-level soil moisture prediction based on extreme gradient boosting model. *Agriculture, 13*(5), Article 927. https://doi.org/10.3390/agriculture13050927

Rodell, M., Houser, P. R., Jambor, U., Gottschalck, J., Mitchell, K., Meng, C.-J., Arsenault, K., Cosgrove, B., Radakovich, J., Bosilovich, M., Entin, J. K., Walker, J. P., Lohmann, D., & Toll, D. (2004). The Global Land Data Assimilation System. *Bulletin of the American Meteorological Society, 85*(3), 381–394. https://doi.org/10.1175/BAMS-85-3-381

Seneviratne, S. I., Corti, T., Davin, E. L., Hirschi, M., Jaeger, E. B., Lehner, I., Orlowsky, B., & Teuling, A. J. (2010). Investigating soil moisture–climate interactions in a changing climate: A review. *Earth-Science Reviews, 99*(3–4), 125–161. https://doi.org/10.1016/j.earscirev.2010.02.004

Shellito, P. J., Small, E. E., Colliander, A., Bindlish, R., Cosh, M. H., Berg, A. A., Bosch, D. D., Caldwell, T. G., Goodrich, D. C., McNairn, H., Prueger, J. H., Starks, P. J., van der Velde, R., & Walker, J. P. (2016). SMAP soil moisture drying more rapid than observed in situ

following rainfall events. *Geophysical Research Letters, 43*(15), 8068–8075. https://doi.org/10.1002/2016GL069946

Torres, R., Snoeij, P., Geudtner, D., Bibby, D., Davidson, M., Attema, E., Potin, P., Rommen, B., Floury, N., Brown, M., Navas Traver, I., Deghaye, P., Duesmann, B., Rosich, B., Miranda, N., Bruno, C., L'Abbate, M., Croci, R., Pietropaolo, A., … Rostan, F. (2012). GMES Sentinel-1 mission. *Remote Sensing of Environment, 120*, 9–24. https://doi.org/10.1016/j.rse.2011.05.028

Vyas, A., & Bandyopadhyay, S. (2020). *Dynamic structure learning through graph neural network for forecasting soil moisture in precision agriculture*. arXiv. https://arxiv.org/abs/2012.03506

Wang, L., Shi, L., Reimers, C., Wang, Y., He, L., Wang, Y., Reichstein, M., & Jiang, S. (2025). A self-supervised deep learning model for enhanced generalization in soil moisture prediction. *Journal of Hydrology, 662*, Article 133974. https://doi.org/10.1016/j.jhydrol.2025.133974

Wang, W., Wei, Y., Hao, L., Wei, Z., & Zhao, T. (2025). Soil moisture forecasting in wireless sensor networks via spatiotemporal graph convolutional networks. *Vadose Zone Journal, 24*(1), Article e70000. https://doi.org/10.1002/vzj2.70000

Wang, Y., Shi, L., Hu, Y., Hu, X., Song, W., & Wang, L. (2024). A comprehensive study of deep learning for soil moisture prediction. *Hydrology and Earth System Sciences, 28*(4), 917–943. https://doi.org/10.5194/hess-28-917-2024

Wang, Y., & Zha, Y. (2024). Comparison of transformer, LSTM and coupled algorithms for soil moisture prediction in shallow-groundwater-level areas with interpretability analysis. *Agricultural Water Management, 305*, Article 109120. https://doi.org/10.1016/j.agwat.2024.109120

Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin, 1*(6), 80–83. https://doi.org/10.2307/3001968

Wu, T., Xu, L., Lv, Y., Cai, R., Pan, Z., Zhang, X., Zhang, X., & Chen, N. (2025). Integrating causal inference with ConvLSTM networks for spatiotemporal forecasting of root zone soil moisture. *Journal of Hydrology, 659*, Article 133246. https://doi.org/10.1016/j.jhydrol.2025.133246

Xi, X., Zhuang, Q., & Liu, X. (2025). A hybrid physics-guided deep learning modeling framework for predicting surface soil moisture. *Journal of Geophysical Research: Machine Learning and Computation, 2*(3), Article e2025JH000682. https://doi.org/10.1029/2025JH000682

Yao, X., Fu, B., Lü, Y., Sun, F., Wang, S., & Liu, M. (2013). Comparison of four spatial interpolation methods for estimating soil moisture in a complex terrain catchment. *PLOS ONE, 8*(1), Article e54660. https://doi.org/10.1371/journal.pone.0054660

Zheng, W., Zheng, K., Gao, L., Zhangzhong, L., Lan, R., Xu, L., & Yu, J. (2024). GRU–Transformer: A novel hybrid model for predicting soil moisture content in root zones. *Agronomy, 14*(3), Article 432. https://doi.org/10.3390/agronomy14030432